Structural Entropy Dark Energy: a fixed-parameter, growth-gated holographic dark-energy model without a cosmological constant

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Abstract

We present Structural Entropy Dark Energy (SEDE), in which the cosmological constant is replaced by the thermal energy of the cosmic apparent horizon, $\rho_{\rm DE} = T_{\rm AH}\ s_{\rm grav}\ f_{\rm sat}(z)$, switched on by the growth of cosmic structure — a dark sector with **zero fitted parameters and three stated prescriptions**: the amplitude is fixed by flatness, the gate shape by a halo-binding prescription ($\gamma \approx 1.5$), the H-scaling by the volume-law postulate ($\lambda = 1 - \Delta/2 = 1/2$), and the sound speed by the closure $c_{\rm s}^2 = 1$. Nothing is tuned to the data; each prescription is a stated choice, declared before the fit, and the one discrete postulate is a volume-law (Barrow $\Delta = 1$) horizon entropy, the quantity our forecasts target. To be explicit about scope, SEDE modifies *only* the dark-energy density in an otherwise standard-GR Friedmann background (holographic DE, not modified gravity), with $\rho_{\rm DE} = T_{\rm AH}\ s_{\rm grav}\ f_{\rm sat}$ imposed as an exact ansatz whose equation of state is a *derived* consequence of energy conservation, not an equilibrium $\rho + p = Ts$ relation. In a marginalised, CAMB-in-the-loop joint analysis of DESI DR2 BAO, Pantheon+/DES-5YR/Union3 supernovae, cosmic chronometers, $f\sigma 8$, and Planck data, SEDE is mildly-to-moderately preferred over ΛCDM (ΔDIC ≈ -3.5 to -4.7; false-preference-calibrated p < 0.0015, 0/2000 mocks, $\sim$2–3σ across pipelines, $\sim$2.3σ without the contested SH0ES prior), and its *derived* equation of state crosses $w = -1$ in DESI's evolving-dark-energy quadrant, $(w_0, w_{\rm a}) \approx (-0.98, -0.11)$. A cubic Kinetic-Gravity-Braiding completion (FPAB, "fixed-point activated braiding") supplies the perturbation sector: $c_{\rm T} = 1$, zero slip, a $\sim$5% sub-horizon force enhance-

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ment, a +2–3% $f\sigma 8$ boost — the model **raises** growth ($\sigma 8 \approx 0.811$ vs 0.800) and makes **no** claim to ease the S8 tension — and $a - 7.6\%$ low-ℓ ISW suppression. This is a moderate, calibrated preference, not an established result: the SH0ES-free full-likelihood companion reads it as *statistically consistent with* ΛCDM *with a mild pull*, and **if DESI DR3 confirms the DR2 central values, SEDE is disfavoured alongside ΛCDM, not rescued by proximity.** The decisive falsifier is the discrete entropy index, $\Delta = 1$ versus 0, measurable by DESI DR3 + Euclid at a Fisher-forecast $\sigma(\Delta) \approx 0.09$.

Keywords: dark energy theory, dark energy experiments, cosmological perturbation theory, modified gravity

1. Introduction

The standard cosmological model, ΛCDM, fits an extraordinary range of data with a cosmological constant Λ whose value confronts two long-standing puzzles. The **cosmological-constant problem**: if Λ is the energy of the quantum vacuum, naïve effective field theory overshoots the observed value by $\sim 10^{120}$. The **coincidence problem**: the dark-energy and matter densities are comparable today, $\rho_{\rm DE} \sim \rho_{\rm m}$, despite scaling differently with expansion — so we appear to live at a special epoch. Recent DESI DR1 and DR2 results, moreover, give a mild preference for *evolving* dark energy (w crossing −1) over a constant Λ [1, 2, 3], reopening the question of whether Λ is fundamental at all and prompting a wave of dynamical-DE model-building — including holographic and non-equilibrium-thermodynamic constructions [4, 5] — within which SEDE's distinction is a *zero-extra-parameter* w(z) rather than a fitted one.

We approach this through an accounting question. As cosmic structure forms, collapsing halos release gravitational binding energy, and dark energy activates over the *same* epoch that structure formation approaches saturation. Is that a coincidence, or is the dark sector tied to structure formation by an energy ledger? It is tempting to guess that the binding energy released by structure *is* the dark energy — but a one-line budget settles it: a virialised system has $|E_{\rm bind}|/Mc^2 \sim \sigma^2/c^2$, and even clusters have $\sigma^2/c^2 \sim 10^{-5}$ (galaxies $\sim 10^{-7}$), so the binding-energy density is $\sim 10^6$ times too small to *be* the observed $\rho_{\rm DE}$. The link, if there is one, cannot be an energy supply.

The resolution we propose is thermodynamic. We identify dark energy with the conjugate thermal energy density of the cosmic horizon —

$\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$, the conjugate energy of the apparent-horizon entropy (motivated by the horizon thermodynamics of [6, 7]) — whose *scale* is the horizon's own, $T_{\rm AH}$ s $\sim M_{\rm P}^2 H^2 \sim \rho_{\rm crit}$, rather than the QFT vacuum (so the vacuum-energy-scale problem is recast, not inherited) and not structure. What structure supplies is not energy but **entropy bookkeeping**: by the horizon first law $\delta Q = T$ dS, the binding-entropy ledger associated with halo virialisation fixes the *redshift dependence* of the effective activation fraction $f_{\rm sat}(z) \in$ [0,1], while its amplitude and the $f_{\rm sat}(0) = 1$ normalisation are fixed separately by the horizon-scale normalisation. Energy is conserved throughout — total covariant conservation is automatic, and the dark-energy fluid is self-conserved — with the binding energy playing a purely entropic, shape-setting role (§2). The coincidence is then a consequence of the mechanism, not a tuning: dark energy turns on with a gate whose redshift dependence tracks structure formation as it approaches saturation, i.e. near "now."

The idea that dark energy is tied to the growth of structure is not new: the coincidence-via-onset framing traces to backreaction/timescape cosmology, where apparent acceleration switches on as the void/collapse fraction matures [8, 9, 10] (against which mainstream "why-now" solutions are the tracker/scaling class [11]); the *structure-sourced entropic* version originates with Gough's information dark energy [12, 13, 14]; and a contemporaneous phenomenological cousin ties $\rho_{\rm DE}(z)$ to nonlinear structure and voids [15]. We credit these priorities explicitly. What is specific to SEDE is the *mechanism* (a gravitational horizon-entropy realisation, $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$, driven by the gravitational growth factor rather than the baryonic star-formation history) and a dark sector with no fitted parameter — fixed by stated prescriptions and one discrete postulate; the relationship is quantified in §7.

The model is, to a useful approximation, ΛCDM with a cosmological constant that fades in as structure forms: schematically $\Lambda_{\rm eff}(z) \propto f_{\rm sat}(z)$ (the exact running term is $3\Omega_{\rm DE}0$ H H_0 $f_{\rm sat}$, §2), with $f_{\rm sat}$ running from ≈ 0 in the early universe to 1 today. By Λ-free we mean specifically that the late-time acceleration is *not* a constant stress-energy term ($p = -\rho = const.$) but a self-conserved horizon fluid whose present amplitude is fixed by flatness ($\Omega_{\rm DE}0 = 1 - \Omega_{\rm m} - \Omega_{\rm r}$) — exactly as flat ΛCDM fixes $\Omega_\Lambda 0$ — and whose redshift dependence is set by the horizon/growth prescription rather than held constant; it does *not* mean the absence of a dark-energy normalisation. Its distinctive feature is that the dark sector adds no fitted parameter — every input is fixed by a stated theoretical choice (§3), so SEDE has the same cosmological parameter count as the ΛCDM baseline used in this analysis (the five of the compressed late-time pipeline of §4). The price is a single postulate about the quantum-gravitational structure of the horizon (§8), which we make explicit, plus the accompanying fixed choices (the halo-binding-entropy derivation of γ (Prescription 4C), the smooth-DE closure $c_s^2 = 1$), which we flag as such rather than as derivations.

This paper is organised around the energy ledger. §2 states the model and establishes energy conservation as its backbone, fixing the (entropic, not energetic) role of binding energy; §3 fixes the dark-sector inputs — the structure coupling γ from the halo-binding-entropy derivation (Prescription 4C), then w(z), λ, and the sound speed c_s^2 — separating the horizon-set *magnitude* from the structure-set *shape*. §4 describes the data and the CAMB-in-the-loop methodology; §5 presents the results, including a false-preference calibration and out-of-sample tests; §6 gives the falsifiable predictions; §7 the relation to dark matter and other sectors. §8 then supplies the quantum-gravity underpinning — the magnitude scale and the maximal deformation — and isolates the one irreducible postulate; §9 discusses the open problems and §10 concludes. Throughout we maintain that SEDE is a *falsifiable proposal*, preferred at a moderate level, not an established result.

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2. The SEDE model and energy conservation

SEDE is general relativity with a single extra fluid: a horizon-fluid dark-energy density identified with the conjugate thermal energy density of the cosmic apparent horizon. We stress the scope at the outset: the horizon entropy sources only ρ_{DE} and never the Friedmann equation itself, so the construction is *holographic* dark energy — not entropic or modified gravity — and the Barrow exponent Δ multiplies a component that f_{sat} gates to a negligible fraction at early times, leaving H(z), the sound horizon r_{drag}, and hence the modified-gravity Barrow/BBN bounds untouched. (In Barrow *modified-gravity* cosmology, by contrast, the same first law is applied to derive the background itself, giving $H^{2-\Delta} \propto \rho$ [16]; that is deliberately *not* the construction here — the Barrow deformation enters ρ_{DE} alone.) We stress that this is a *substantive* restriction rather than a stylistic one, and that the alternative is disfavoured rather than merely different: feeding the same deformation into the field equations makes the scalar sector modify gravity directly, and a ghost-free, gradient-stable completion with $c_{\mathrm{T}} = 1$ then forces the effective coupling upward — an enhancement of structure growth larger than the present data permit. The asymmetry between deforming $\rho_{\rm DE}$ and deforming the geometry is therefore load-bearing: it is the branch of the construction that survives the growth constraints, not simply the branch we chose to develop. Correspondingly, $\rho_{\rm DE} = T_{\rm AH}\ s_{\rm grav}\ f_{\rm sat}$ is imposed as an *exact effective relation* that fixes the dark-energy density's amplitude and its H-scaling; the equation of state is then a derived, kinematic consequence of energy conservation on that standard background (it crosses $w = -1$ by *sourcing*), not the equilibrium enthalpy relation $\rho + p = Ts$, which does not hold for this open, structure-fed sector. The whole construction is governed by one bookkeeping rule — that energy is conserved when this fluid's redshift dependence is gated by structure growth. We first state the model (the conjugate horizon-fluid ansatz, the structure gate, the Friedmann equation, and how Λ is replaced), and then make energy conservation the backbone: it is what fixes the role of structure's gravitational binding energy as *entropic*, not energetic, and what shows the $w = -1$ crossing to be sourcing rather than a phantom field.

The conjugate horizon-fluid ansatz. At the cosmological apparent horizon $R_{\rm AH} = 1/H$ (flat FRW), the dynamical Cai–Kim temperature is $T_{\rm AH} = (H/2\pi)(1 - \epsilon/2)$, $\epsilon{\equiv}{-}\dot{H}/H^2$. For the dark-energy sector, which tends to a de Sitter attractor ($\epsilon \to 0$), we adopt the Gibbons–Hawking temperature $T_{\rm AH} = H/2\pi$ — the $\epsilon \to 0$ limit, i.e. the appropriate *leading* temperature for a slowly-evolving horizon fluid. This is an approximation, made deliberately: the dynamical $(1-\epsilon/2)$ correction is largest in the matter era, exactly where $f_{\rm sat} \to 0$ and dark energy is negligible, and including it in full on the volume-law fluid is in fact *disfavoured* — it yields an unstable, phantom-everywhere background with no $w = -1$ crossing (§6). So $T_{\rm AH} = H/2\pi$ is the physically-appropriate choice for this sector and it is what enters the calibrated E(z). We emphasise that this selection is made on the physical argument alone and does not depend on the outcome: the $(1 - \epsilon/2)$ correction is parametrically negligible precisely where this fluid carries any weight, so the $\epsilon \to 0$ limit would be the right leading choice for a slowly-evolving horizon fluid *even had the dynamical variant produced an acceptable background*. That the dynamical variant additionally fails is a consistency check on the choice, not its justification. We are nonetheless explicit about the epistemic status of this step: the Gibbons–Hawking form is a convention, and one whose adoption is corroborated by the failure of the dynamical alternative, so quantities downstream of it — notably the $w = -1$ crossing redshift (§6) — are convention-conditional outputs of the adopted temperature, not convention-free predictions. We *identify* the dark-energy density with the conjugate horizon-fluid energy density,

$$\rho_{\rm DE} = T_{AH}\, s_{\rm grav}\, f_{\rm sat}(z), \tag{1}$$

where $s_{\rm grav}$ is the gravitational entropy *density* and $f_{\rm sat}$ is an effective horizon-entropy activation gate, normalised to $f_{\rm sat}(0) = 1$ — i.e. a value *relative to today*, not a literal fraction bounded by 1 (it reaches ≈ 1.23 in the future, below); its redshift dependence is set by the growth-weighted halo prescription of §3.1, not derived from the Bousso bound itself. *Range and future behaviour (defining the convention explicitly):* for $z \geq 0$ — the regime of every data fit here — the linear growth obeys $0 \leq D(z) \leq 1$, so $0 \leq f_{\rm sat} \leq 1$. $f_{\rm sat}$ is not clipped; in the future (z < 0) linear growth *saturates* (D freezes at $D_\infty \approx 1.44$ as dark energy suppresses growth — it does not run away), so $f_{\rm sat}$ saturates to a finite de Sitter value $f_{\rm sat}(D_\infty) \approx 1.23$ (modestly above 1, never reaching the formal x $\to \infty$ ceiling $1/(1-e^{-\gamma}) \approx 1.29$). The background tends to a de Sitter attractor (H $\to H_\infty$, $\rho_{\rm DE} \to$ const, w $\to -1$; A.7, Result 12) — bounded and well-defined for all z, with the ≤ 1 statement specific to $z \geq 0$. (The gate is normalised to its present value; it is *not* the literal fraction of horizon entropy deposited by collapsed structures, which is far smaller — $\sim 10^{-7}$; Appendix Appendix F.) Clausius horizon thermodynamics motivates the relation and supplies the conservation logic, but the absolute identification is the defining SEDE ansatz (Appendix Appendix A.1). To be precise about a distinction used throughout: the relation $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav} \cdot f_{\rm sat}$ is *imposed as an exact relation* of the effective description — that is what lets the reservoir argument (§3.1) and the free-energy step (Appendix Appendix A.6) use it — while "postulate" refers to its lack of a *microscopic* derivation (the horizon-entropy density is not derived from a quantum-gravity state count), not to any looseness in how it is applied. We take $s_{\rm grav}$ to be the coarse-grained entanglement entropy in the apparent-horizon volume, $S_{\rm grav} \propto V_{\rm AH} \propto R_{\rm AH}^3$ — equivalently a *constant* entropy density $s_{\rm grav} = s_0$ (in contrast to the Bekenstein–Hawking area-law density $s_{\rm AH} \propto H$). With $T_{\rm AH} \propto H$,

$$\rho_{\rm DE} = T_{AH}\, s_0\, f_{\rm sat} \;\propto\; H\, f_{\rm sat}, \qquad s_0 = \frac{\Omega_{\rm DE0}\, \rho_{\rm crit,0}}{T_{AH,0}} \text{ (fixed by flatness)}, \tag{2}$$

giving $\rho_{\rm DE} \propto H\ f_{\rm sat}$ with no free parameter. This volume law is exactly the maximal Barrow deformation ($S \propto A^{1+\Delta/2} = A^{3/2} = R^3$ at $\Delta = 1$, $\lambda =$

$1 - \Delta/2 = 1/2$); we keep Δ only as a *test* parameter quantifying deviations from the volume law (§8, and the data measurement of Δ), not as a model input.

Beyond the Barrow characterisation, there is a *plausibility ordering* for the volume law that needs no entropy maximisation (we stress at the outset that it is an ordering, not a selection principle). Expanding the vacuum response to the cosmic expansion scalar $\Theta = 3H$ as a derivative series $\rho_{\mathrm{DE}} \propto \Theta^{2-\Delta}$, the constant term ($\Delta = 2$) is a cosmological constant, the quadratic term ($\Delta = 0$) is the area law, and the linear term ($\Delta = 1$) is the unique *time-reversal-odd* — hence dissipative, entropy-producing — response. This is also why the linear power is not the covariant-vacuum "red flag" it can appear to be: a *covariant* vacuum energy is invariant under H $\rightarrow -$H and so admits only even powers (the running-vacuum program's $\rho_{\mathrm{DE}} \propto H^2 + H^4$ [17]); SEDE's odd, linear term is not a covariant vacuum energy but the T-odd dissipative response of the apparent horizon, so the broken H $\rightarrow -$H invariance is the *signature* of entropy production, not a covariance violation. Because SEDE admits no bare Λ, the leading surviving response is the linear one, so $\Delta = 1$ is the leading dissipative response *provided* the higher, time-reversal-even Θ^2 (area) term is subdominant — a plausibility ordering that ranks $\Delta = 1$ above $\Delta = 0$, not a selection principle (dimensionally, ranking Θ^1 above Θ^2 requires the Θ^2 coefficient to be sub-Planckian, an input we do not derive; $\Delta = 1$ remains a postulate tested by the data, §5.6, §8.4). Equivalently, $\rho_{\mathrm{DE}} \propto H\ f_{\mathrm{sat}}$ is a structure-gated bulk viscosity, $\zeta = \rho_{\mathrm{DE}}$ (d ln f_{sat}/d ln a)/(9H) > 0 (second-law-respecting), so the $w = -1$ crossing is a viscous–dilution balance rather than a phantom kinetic term. The *ungated* H-linear form is itself not new: entropic cosmology derives an H-linear driving term from a volume (Tsallis–Cirto) horizon entropy, explicitly analogized to bulk viscosity [18, 19] (whereas the area-law entropic-force term is H^2, not H [20, 17]); the *same* H-linear law is reached from an entirely different route — QCD topological (Veneziano contact-term) vacuum energy, $\rho \propto H \cdot \Lambda^3$ [21, 22, 23], the hidden-sector version of which the scale-sector companion uses to source $\mu^3 H$ — so that $\rho \propto H$ arises from horizon thermodynamics and from QCD contact terms alike is a convergence, not a coincidence. What is specific to SEDE is the structure gate f_{sat}, the zero-fitted-parameter closure, and the derived w(z) — not the $\rho_{\mathrm{DE}} \propto H$ scaling per se. Ghost DE leaves the amplitude α free and Barrow holographic DE leaves the exponent Δ free; SEDE fixes both ($\Delta = 1$ from the capacity theorem of the count companion, the amplitude by flatness) and multiplies by the derived gate, so it is the ungated, free-amplitude limit of SEDE that reproduces those models, not the reverse. A density linear in H cannot descend from a local conservative action, so the mechanism is intrinsically dissipative; its covariant completion is a braided (kinetic-gravity-braiding) or Schwinger–Keldysh dissipative-fluid theory (§7). We flag the status honestly: a *single* bulk-viscous fluid is in fact gradient-unstable across the crossing (the principal-symbol speed uses the total enthalpy $(1+w)\rho_{\rm DE}$, which changes sign), so stability requires the positive kinetic structure that braiding supplies — demonstrated for a computed KGB member but conditional on a shift-charge tuning; the completion is not yet closed (§7, and the dissipative-stability analysis in the reproduction repo).

The structure gate. Structure formation supplies the *redshift dependence* of the activation gate at a rate set by the collapsed-halo abundance (Result 3, Appendix Appendix A), yielding a closed form for the normalised activation fraction,

$$f_{\rm sat}(z) = \frac{1 - e^{-\gamma D^2(z)}}{1 - e^{-\gamma}}, \qquad D(z) \text{ the linear growth factor,} \tag{3}$$

where D(z) is the model's self-consistent growth factor (computed with SEDE's own E(z), which in turn depends on $f_{\rm sat}$ — the background is solved as the joint fixed point; §4.4, Appendix Appendix B). with $f_{\rm sat}(0) = 1$ imposed as the present-day normalisation convention and $f_{\rm sat} \to 0$ at high z. The single shape parameter γ is fixed by a halo-binding-entropy derivation (§3.1, Prescription 4C), not fitted. One property of the gate deserves plain statement: because its argument is the normalised growth *ratio* $x = D^2(z) = \sigma 8^2(z)/\sigma 8^2(0)$, the gate is independent of the absolute structure amplitude — a universe with four times the primordial power ($A_{\rm s}$) has the identical $f_{\rm sat}(z)$. What activates dark energy is the relative *maturation* of the growth history, not the accumulated amount of structure; the phrase "structure activates dark energy" is to be read in this shape (not amount) sense throughout.

The Friedmann equation. Inserting $\rho_{\rm DE}$ into the (unmodified) Friedmann constraint gives a single fixed-point equation for the dimensionless expansion rate $E = H/H_0$:

$$\boxed{E^2(z) = \Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_{\rm DE0}\, f_{\rm sat}(z)\, E(z)} \tag{4}$$

(for $\lambda = 1/2$, so $E^{2\lambda} = E$). The boxed closure equation is the entire background model. It is ΛCDM with the constant Λ replaced by the structure-gated, horizon-coupled term $\Omega_{\rm DE}0\ f_{\rm sat}(z)$ E(z); at z = 0 with $f_{\rm sat}$ = 1 it reproduces the observed $\Omega_{\rm DE}0 = 1 - \Omega_{\rm m}$.

Replacement of Λ in the field equations. SEDE is *holographic dark energy* (the programme founded by Li [24], with the density set by an entropy bound at a cosmological length), not modified gravity: the Einstein tensor and Newton's constant are untouched, and the deformation enters only a smooth ($c_{\rm s}^2 = 1$; §3.4) perfect fluid on the right-hand side,

$$G_{\mu\nu} = 8\pi G\,(T_{\mu\nu}^{(\rm matter)} + T_{\mu\nu}^{(\rm DE)}), \qquad \rho_{\rm DE} = \tfrac{3}{8\pi G}\,\Omega_{\rm DE0}\,HH_0\,f_{\rm sat}, \quad p_{\rm DE} = w(z)\rho_{\rm DE}. \tag{5}$$

Equivalently, Λ becomes a *running* scalar $\Lambda_{\rm eff}(z)$ = 3 $\Omega_{\rm DE}0$ H H_0 $f_{\rm sat}$ that equals the observed Λ today and is negligible in the early universe (Figure 1). The choice of right-hand side (fluid) over left-hand side (geometry) is not cosmetic: a Barrow-*modified-gravity* Friedmann equation would rescale the early-universe expansion by $(M_{\rm P}/H)^{\Delta} \sim 10^{61}$, a Big-Bang-nucleosynthesis catastrophe (§8, Result 11). Keeping the deformation in the dark-energy fluid leaves the early universe exactly standard.

Energy conservation. Because SEDE is general relativity with a fluid source, the Bianchi identity $\nabla^{\mu} G_{\mu\nu} = 0$ forces the total stress–energy to be covariantly conserved, $\nabla^{\mu}(T_{\mu\nu}^{(\rm matter)} + T_{\mu\nu}^{(\rm DE)}) = 0$, identically — there is no violation. At the background level matter dilutes standardly, $\rho_{\rm m} \propto a^{-3}$, and dark energy is a self-conserved fluid whose density evolves entirely through its own pressure work,

$$\dot{\rho}_{\rm DE} + 3H\,(1 + w_{\rm DE})\,\rho_{\rm DE} = 0, \qquad w_{\rm DE}(a) = -1 - \tfrac{1}{3}\,\frac{d\ln\rho_{\rm DE}}{d\ln a} \tag{6}$$

(the w(z) of §3.2 and Fig. 4). At the background level this "self-conservation" is definitional — for any prescribed $\rho_{\rm DE}(a)$ one *defines* w(a) by the second equality to satisfy continuity — so the non-trivial content is not background conservation but the existence of a local action whose equations of motion produce this w(a) without an external matter coupling; that is what the FPAB completion (§7) supplies, and it is trajectory-level, not yet closed. There is no literal energy transfer from matter to dark energy: such a transfer of order $\rho_{\rm crit}$ over a Hubble time would spoil $\rho_{\rm m} \propto a^{-3}$ and with it BBN and the CMB. This is the crucial accounting point of the model. The magnitude of $\rho_{\rm DE}$ is anchored to the horizon gravitational scale $\sim M_{\rm P}^2\,H^2 \sim \rho_{\rm crit}$ through the CKN bound and the flatness normalisation (§8.2), not to energy borrowed from structure: the gravitational binding energy released as halos virialise

($E_{\mathrm{bind}} \propto M^{5/3}$) is $\sim \sigma^2/c^2 \sim 10^{-6}$ of the collapsed rest-mass energy, six orders of magnitude too small to *be* the dark energy. This makes the genuinely new content precise: with the area law ($\Delta = 0$) the ratio $\rho_{\mathrm{DE}}/\rho_{\mathrm{crit}} \propto H^{-\Delta}$ is a fixed fraction that merely tracks — GR's own horizon energy repackaged (Jacobson) — whereas the Barrow deformation $\Delta > 0$ makes it grow as H falls, turning a tracking term into dark energy that comes to dominate; the new physics is *entirely* the deformation ("is the energy new?" $\iff$ "is $\Delta \neq 0$?"). What structure formation supplies is therefore not energy but **entropy bookkeeping**: by the apparent-horizon first law (A.1), the deposited binding entropy $\Delta S_{\mathrm{AH}} = E_{\mathrm{bind}}/T_{\mathrm{AH}}$ fixes the *weight* p = 5/3 that sets the shape γ of the gate f_{sat} (§3.1) — it determines *when* and *how sharply* dark energy turns on, while the overall scale, and the normalisation $f_{\mathrm{sat}}(0) = 1$, come from the horizon (§3, §8). Read this way, SEDE's crossing of $w = -1$ is not a phantom field — there is no negative kinetic term or gradient instability — but the pressure work of a horizon-sourced fluid whose density rises as f_{sat} grows (Result 13: 1 + w is the horizon entropy-production rate). (A single canonical or k-essence field cannot cross $w = -1$ without its kinetic term $\rho + p = 2X \cdot P_{\mathrm{X}}$ passing through zero — the Vikman no-go — so a fundamental-field description would require a ghostly quintom; SEDE avoids this because its phantom is *effective*: $\rho_{\mathrm{DE}} = T_{\mathrm{AH}} \cdot s_{\mathrm{grav}}$ is built from covariant horizon scalars, not a fundamental scalar.) Finally, since f_{sat} is a functional of the *background* growth ($c_{\mathrm{s}}^2 = 1$, §3.4), the coupling lives at the background level only and introduces no local energy non-conservation in the perturbations.

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3. Fixed inputs: zero fitted parameters, three stated prescriptions

In flat ΛCDM the dark-energy amplitude $\Omega_{\Lambda}0 = 1 - \Omega_{\mathrm{m}} - \Omega_{\mathrm{r}}$ is itself fixed by the flatness closure, not an independent fit; SEDE keeps exactly this — no independent dark-energy amplitude beyond flatness — and differs only in that the *redshift dependence* is set by the horizon/growth prescription rather than held constant. What that prescription fixes is the triplet (γ, w(z)-shape, λ) plus the sound speed c_{s}^2 — and each is fixed by a stated theoretical choice (not fitted to cosmological data), so SEDE adds no free dark-energy parameter. We are explicit about the status of each choice below (some follow from flatness or the model construction; others, notably γ and c_{s}^2, are fixed prescriptions). The choices separate along the two scales of §2. The

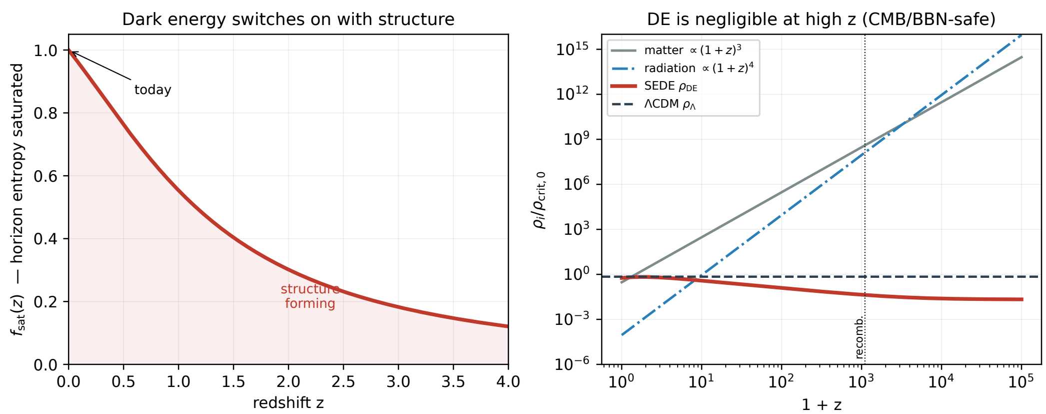

Figure 1: The mechanism. *Left:* the effective activation fraction (structure-growth gate) $f_{\rm sat}(z)$ rises toward 1 today as structure forms, activating the dark-energy gate — the effective running term $\Lambda_{\rm eff} = 3\Omega_{\rm DE}0$ H H_0 $f_{\rm sat}$ *fades in* (computed from the model's self-consistent linear growth factor D(z), §2). *Right:* the cosmic density inventory; SEDE's $\rho_{\rm DE}$ is orders of magnitude below matter and radiation at high z and falls below $\Lambda CDM's$ constant ρ_Λ, so the early universe (BBN, recombination) is standard — the deformation acts only on the late-time dark-energy fluid.

magnitude of dark energy is the horizon scale, $\sim M_{\rm P}^2 H^2 \sim \rho_{\rm crit}$ (the CKN bound normalised by flatness, §8.2) — not a fitted dark-sector parameter. What the choices control is the **shape**: *when* and *how sharply* that energy turns on. We therefore lead with the structure coupling γ — the shape, set by the binding energy — and then give w_0, λ, and c_s^2 (Figure 2).

3.1. The structure coupling $\gamma \approx 1.5$ from a halo-binding-energy derivation (Prescription 4C)

γ sets the sharpness of the structure gate, $\gamma = d \ln \Sigma_{\rm S}/d \ln \sigma 8^2$ with $\Sigma_{\rm S}$ the entropy-weighted collapsed mass, $\Sigma_{\rm S}{=}\int M^p$ (dn/d ln M) d ln M — the derivative is with respect to the variance $\sigma 8^2$ because the gate's argument is $x = D^2 = \sigma 8^2(z)/\sigma 8^2(0)$ (§3.4, A.2); equivalently $\gamma{=}\frac{1}{2}$ d ln $\Sigma_{\rm S}/d$ ln $\sigma 8$. The weight exponent p is fixed by *where* the structure-formation entropy is deposited — and the reservoir is fixed *before any data enter*, by the model's own §2 identity. Because SEDE identifies $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$ (§2), $f_{\rm sat}$ is by construction the cosmic-*horizon* entropy activation fraction (A.1), so the binding energy released by virialisation is accounted in the horizon ledger at its temperature $T_{\rm AH} \propto H$ — *not* the halo's internal virial temperature

$T_{\rm vir} \propto M^{2/3}$. With $T_{\rm AH}$ mass-independent at a fixed epoch, the Clausius relation gives $\Delta S_{\rm AH} = E_{\rm bind}/T_{\rm AH} \propto E_{\rm bind}$, and the virial binding energy $E_{\rm bind} \propto M^{5/3}$ then fixes p = 5/3. This reservoir choice follows from the §2 identity, and we state its history plainly rather than calling it forced: an initial treatment used $T_{\rm vir}$ (giving p = 1), a reservoir error corrected for consistency with §2, not a tuning — but the correction is a discrete modelling revision, made after the p = 1 mis-step. We flag the epistemic risk plainly: the revision post-dates the p = 1 attempt and lands on the data-consistent value, so — absent a demonstrated microphysical transport channel depositing the binding heat on the horizon (which we do not claim; below) — p = 5/3 is a prescription whose consistency with the data is a check, not a uniquely forced output. What makes it more than post-hoc is that the reservoir (horizon, $T_{\rm AH}$, not virial $T_{\rm vir}$) is fixed by the §2 relation independently of any fit; the residual freedom is whether that relation is the right accounting, not the value of p given it.

γ has a transparent closed form. A self-similar reduction gives $\gamma = (p - 1)\langle 1/\alpha\rangle\Sigma$ ($\alpha \equiv -d$ *ln* σ/d *ln* M *the effective spectral slope,* $\langle\cdot\rangle\Sigma$ the entropy-weighted average), which reproduces the full mass-function integral to 0.1% (A.3) — so the dark-sector input is *only* p = 5/3, with $\langle 1/\alpha\rangle$ ordinary halo physics. The *numerical* γ then follows from a standard, pre-specified halo model and inherits its systematics: a Sheth–Tormen mass function over 10^{10}–$10^{16}\ M\odot$ with an EH98 transfer gives $\gamma = 1.50$ (Prescription 4C), and varying the mass function (Press–Schechter, Tinker08, Despali16, Watson13), mass range, and transfer function spans $\gamma \in [1.14, 1.60]$ (median 1.50), dominated by the mass-range low-cutoff and robust (1.48–1.57) to the mass-function and transfer choice. We therefore claim precisely: the reservoir — hence the weight p = 5/3 — is fixed by the model's own construction; the numerical $\gamma \approx 1.5$ carries standard halo-model systematics of order ± 0.2, none tuned to cosmological data. That it lands near the diagnostic-fit value is then a *consistency check*, not the motivation. (We do not claim a demonstrated microphysical transport channel depositing the binding heat onto the horizon; the prescription is the accounting, and the smooth-DE closure of §3.4 ensures it induces no local dark-energy perturbation.) Binding energy thus sets the gate *shape*, not the magnitude (§2, §10 ledger).

3.2. *The equation of state w(z) from the fixed background, conservation, and flatness*

The model has a single physical equation of state — that of the canonical fixed-point background (the boxed Friedmann equation of §2), obtained from energy conservation,

$$w(z) = -1 - \tfrac{1}{3}\,\frac{d\ln\rho_{\rm DE}}{d\ln a}, \tag{7}$$

with $\rho_{\rm DE} = \Omega_{\rm DE}0\ \rho_{\rm crit},0\ \mathrm{E}\ f_{\rm sat}$ and the present-day normalisation fixed by spatial flatness ($\Omega_{\rm DE}0 = 1 - \Omega_{\rm m} - \Omega_{\rm r},\ f_{\rm sat}(0) = 1$). Evaluated on the self-consistent background this gives $w_0 \approx -1.0$, crossing -1 at low redshift ($z \approx 0.2$) and dipping to ≈ -1.08 by $z \approx 2$ — DESI DR2's thawing/crossing quadrant. So w(z) is fixed by $\Omega_{\rm m}$ (through the background) with no free EOS parameter; this is the w(z) used throughout (§5, Fig. 4). *(Convention: the literal present-day value is* $w(z = 0) = -0.996$*; the reported pair* (w_0, $w_{\rm a}$) $\approx$ *(*-0.98*,* -0.11*) is the CPL projection of this same w(z) over 0 <* z *< 1 — the range used by the locked* ***`results/predictions.json`****. The CPL* $w_{\rm a}$ *steepens if fit over a wider range, e.g.* $-\,0.17$ *over 0 <* z *< 2, so the fit window is quoted with the pair.)* Three w_0 values appear in this paper for *different* objects; to avoid confusion:

object	scaling	role	w_0
canonical SEDE	$\rho_{\rm DE} \propto H\ f_{\rm sat}$	the actual model	≈ -1.0, crosses -1
$\lambda = 1$ "temperature-factor" variant	—	closed-form structural estimate (side note)	≈ -0.85
area-law branch	$\rho_{\rm DE} \propto H^2\ f_{\rm sat}$	disfavoured diagnostic alternative (§5.6, §8.1)	≈ -0.49

Only the first row is the model's prediction; the -0.85 is a sanity check on the $\Omega_{\rm m}$-dependence (closed form $w_0 = (4\Omega_{\rm m}/3 - 1)/(1 - \Omega_{\rm m})$) and the -0.49 is the $\Delta = 0$ alternative the data disfavour.

3.3. The H-coupling $\lambda = 0.5$ from the horizon's volume entropy

λ fixes the *magnitude*'s scaling with H, with no free parameter: the gravitational entropy is taken to be the coarse-grained entanglement entropy in the apparent-horizon volume, $S_{\rm grav} \propto V_{\rm AH} \propto R^3_{\rm AH}$, i.e. a *constant* horizon entropy density. The conjugate horizon-fluid ansatz ($\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$, $T_{\rm AH} \propto H$) then gives $\rho_{\rm DE} \propto H \equiv H^{2\lambda}$ with $\lambda = 1/2$ — no deformation parameter. Equivalently, in the Barrow language $S = (A/A_0)^{1+\Delta/2}$ this is the maximal deformation $\Delta = 1$ ($s_{\rm AH} \propto H^{1-\Delta}$, $\lambda = 1 - \Delta/2$): the volume law *is* $\Delta = 1$ ($A^{3/2} = R^3 = V$). We keep Δ only as a falsifiable test of the volume law (its data measurement, §5/§6), not as a model input; §8 motivates the volume/space-filling state (capping $\Delta \leq 1$ and postulating saturation). (Verified Δ-free $\equiv \Delta = 1$ bit-for-bit.)

3.4. The perturbation sector: from a smooth-fluid closure to the FPAB-SEDE cubic-KGB action

SEDE's dark energy is *background-growth-gated* (its $f_{\rm sat}$ depends on the growth history), which might suggest it clusters with structure. The minimal treatment adopts a **smooth-dark-energy closure**: $f_{\rm sat}$ is a functional of the background growth factor D(z) — a function of cosmic time only — so the structure coupling is background-level / nonlocal / coarse-grained, not local, $\rho_{\rm DE}$ is homogeneous by construction, and the dark energy has the standard metric-driven response with $c_s^2 = 1$. That closure is a *modelling* choice, not a theorem, and it is the sub-horizon (`c_s²=1`) limit of the completed theory rather than the theory itself.

The completed perturbation sector (FPAB-SEDE). The background $\rho_X = \mu^3 H\, g(a)$ (with g the growth-shaped gate, §2) admits a local, luminal **cubic Kinetic-Gravity-Braiding** completion, $\mathcal{L} = M_{\rm P}^2 R/2 + K(X,\phi) - G_3(X,\phi)\,\Box\phi$, whose finite (16-coefficient) action reproduces the background to $\max|\delta H/H| \approx 2 \times 10^{-4}$ over $0 \leq z \leq 99$; we call this completed effective theory **FPAB-SEDE** (fixed-point activated braiding). This is a trajectory-level completion — an instance of designer / EFT-of-dark-energy reconstruction [25, 26, 27] rather than a symmetry-derived Lagrangian — delivered in the stable α-basis [28] to mochi-class [29, 30], with braiding $\alpha_B = b\,\Omega_X$ the standard density-tracking ansatz and the cubic-KGB no-ghost/no-gradient structure of [31, 32]. Because it carries no G_4/G_5 term, $\alpha_M = \alpha_T = 0$ and $c_T = 1$ exactly (GW170817-safe). This is a *derived* statement, not merely an EFT prior: an exact all-orders-in-the-tensor-perturbation expansion of the FPAB action shows that K and G_3 contribute nothing to the quadratic tensor action for arbitrary functional forms, so the entire tensor sector is that of GR and the GW and EM luminosity distances coincide ($\Xi_0 = 1$ exactly) — see the tensor-sector and standard-siren predictions in §6 (script `src/experiments/tensor_sector/fpab_tensor_sector.py`). Its perturbations are solved *dynamically* (Mochi-CLASS, not a PPF or an imposed sound speed): ghost- and gradient-stable, with

$$c_s^2 \in [0.018,\, 0.18], \qquad \gamma_{\rm slip} = 1, \qquad \mu_\infty(0) = 1.05, \tag{8}$$

i.e. **unit gravitational slip and a $\sim$5% near-horizon scalar-force enhancement** that turns on across $k_{50} = 0.70\,aH$ to $k_{90} = 2.1\,aH$ — a scale-dependent $\mu(k,z)$, *not* a smooth $c_s^2 = 1$ fluid. The earlier $c_s^2 = 1$ "smooth dark energy" statement is thus the sub-horizon limit of this result, and the marginalised fit of §5 (run in that PPF limit) is valid at the background/distance level while the growth-side observables are revised to the FPAB values below and treated as forecasts pending a full perturbative refit (§5, §6).

input	value	status — fixed by
λ (H-coupling)	0.5	postulate — volume-law horizon entropy $S_{\rm grav} \propto V_{\rm AH}$ ($\equiv$ Barrow $\Delta = 1$), §3.3, §8
w(z) (EOS)	$w_0 \approx -1.0$, crosses -1	derived — spatial flatness + the boxed closure equation (§2), §3.2 (no free EOS parameter)
$\gamma/gate$ q	≈ 1.50; gate $q = \sigma_R^2(a)$, R = 8 $h^{-1}Mpc$	fixed prescription — halo-binding weighting (§3.1); the model-owned filtered variance replaces the abstract D^2

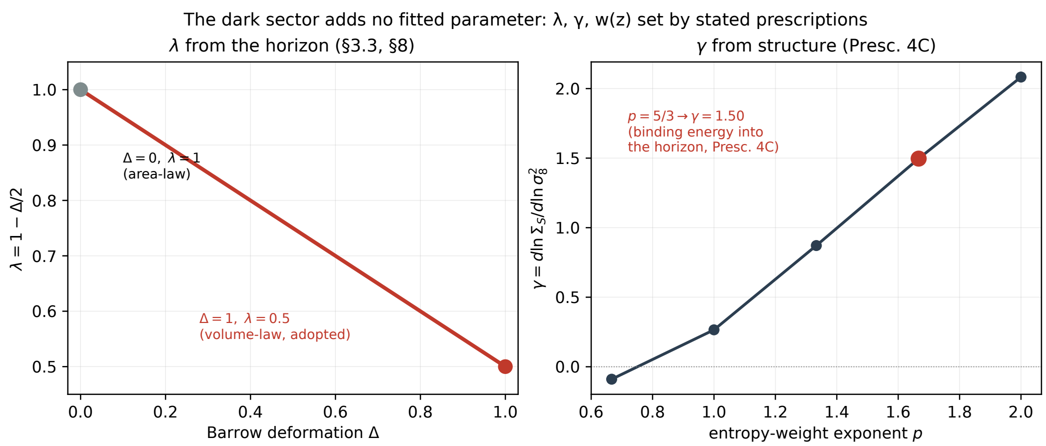

Figure 2: The two fixed dark-sector couplings. *Left:* the H-coupling $\lambda = 1 - \Delta/2$ as a function of the Barrow deformation; the volume-law endpoint $\Delta = 1$ (the postulate of §8) fixes $\lambda = 0.5$. *Right:* the structure coupling $\gamma = d\ \ln\ \Sigma_S/d\ \ln\ \sigma 8^2$ (variance convention, §3.1) as a function of the entropy-weight exponent p; the binding-energy-into-the-horizon argument (Prescription 4C, A.3) fixes p $= 5/3$, giving $\gamma = 1.50$ — close to the value favoured in diagnostic fits. Neither is a fitted parameter.

input	value	status — fixed by
perturbations	$c_s^2 \in [0.018, 0.18]$, $\gamma_{\text{slip}} = 1$, $\mu_\infty = 1.05$	**computed** — FPAB-SEDE cubic-KGB action + Mochi-CLASS (§3.4); $c_s^2 = 1$ is the sub-horizon limit

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4. Data and analysis methodology

4.1. Data

We use a standard late-time-plus-compressed-CMB compilation, with both models fit to the identical data vector: - **BAO:** DESI DR2 (galaxy, QSO, and $Ly\alpha$ tracers, z = 0.3–2.33), as $D_{\rm M}/r_{\rm d}$ and $D_{\rm H}/r_{\rm d}$. - **Supernovae:** Pantheon+ as the baseline, with DES-5YR and Union3 used independently for the robustness cross-check of §5.4. - **Cosmic chronometers:** the Moresco et al. H(z) compilation (model-independent expansion rates). - **Redshift-space distortions:** the Gold-2018 [33] $f\sigma8(z)$ growth compilation (a standard, vetted set chosen for comparability with prior growth analyses; combining it with DESI DR2 BAO follows common late-time practice, and the audit below corrects a few mis-entered points), with the data audit of Appendix Appendix C (a small number of mis-entered points corrected to their published values). - **CMB:** the compressed Planck distance priors (shift parameter R, acoustic scale $l_{\rm A}$, $\omega_{\rm b}$), with the full $plik_{\rm lite}$ likelihood used as a cross-check. - **Local distance ladder:** the SH0ES H_0 prior. - **Out-of-sample (§5.3):** the pre-DESI eBOSS DR16 full-shape BAO, held entirely out of the main fit.

4.2. CAMB-in-the-loop: a physical sound horizon

The single most important methodological point is that the sound horizon is not a free parameter. Earlier exploratory fits that floated $r_{\rm d}$ found large apparent preferences for SEDE; those did not survive a physical sound horizon. Here, at every point in the chain, $r_{\rm drag}$, $r_{\rm star}$, and θ^* are computed self-consistently from the background by CAMB ("CAMB-in-the-loop"), so the BAO and CMB distances are tied to the same early-universe physics for both models. This is what makes the ΔDIC of §5 a like-for-like comparison rather than an artifact of a free standard ruler. The CMB enters as the compressed (R, $l_{\rm A}$) distances computed on the SEDE background through the same CAMB call. *Scope of the compressed likelihood, and its full-likelihood validation:* the compressed priors capture the background distance to last scattering (and hence the early-dark-energy fraction that drives the Δ leverage of §5.6), but not the full acoustic-peak/polarisation or lensing/ISW information. We therefore validate the headline against the full primary CMB likelihood: a CAMB-in-the-loop fit on the full Planck $plik_{\rm lite}$ TTTEEE + $low\ell$ TT/EE, with the five cosmological parameters and the Planck nuisance *all free*, gives $\Delta\chi^2(\text{SEDE} - \Lambda\text{CDM}) = -0.43$ — the preference sitting in the high-ℓ peak structure (-0.37), *negative* (mildly favouring SEDE), not the large positive $\Delta\chi^2$ that the holographic-DE early–late tension would produce (§5.1, §9). This directly tests the acoustic-peak channel the compression omits, and we have confirmed it against the full non-lite plik with all 15 foregrounds sampled ($\Delta\chi^2 = -0.33$, consistent with the lite −0.43; §9). *Stated scope of what remains:* only the ACT DR6 *primary* multifrequency likelihood is not run (ACT *lensing*, the DE-sensitive channel, is, evaluated at the primary best-fit; §5.1). It is formally deferred under a pre-registered trigger (pre-registration §Appendix E): the `act_dr6_mflike` likelihood and its multi-GB data products are not installed, and with three concordant CMB datasets — $plik_{\rm lite}$, full non-lite plik, and ACT DR6 lensing — all at $|\Delta\chi^2| < 1$ (these are not the same estimator: the $plik_{\rm lite}$/*non*-lite numbers are *profiled* over the cosmological and nuisance parameters, whereas the ACT-lensing value is a *single-point* evaluation at the primary best-fit — so their numerical closeness is corroborative but not a like-for-like concordance, and the full *marginalised* CMB analysis is the companion paper's, §5.1), a fourth primary likelihood is not expected to overturn them; the committed (falsifiable) expectation is $|\Delta\chi^2(\mathrm{SEDE}-\Lambda\mathrm{CDM})| < 1$, a large positive value being a genuine failure. The %-level metric effects (§4.4) are reported but not used in the headline fit.

4.3. Inference and model comparison

Parameters are sampled by MCMC, marginalising over $\Omega_{\rm m}$, H_0 (with the SH0ES prior), $\omega_{\rm b}$, the *amplitude*/$\sigma 8$, and $r_{\rm d}$-determining physics — the same five for SEDE and ΛCDM, since SEDE's dark sector is fixed (§3). Model comparison uses the deviance information criterion (DIC), which penalises the effective number of parameters $p_{\rm D}$. The marginalised chains give $p_{\rm D} = 4.8$ (SEDE) and 4.9 (ΛCDM) — nearly equal ($\Delta p_{\rm D} \approx 0.1$), as expected since the two share the same five fitted parameters — so ΔDIC reduces essentially to the χ^2 difference: ΔDIC = −4.69 (Barrow preferred); the chains' $\Delta\chi^2_{\rm min} = -4.66$ agrees with the $optimiser's - 4.68$ to sampling precision. With k = 5 fitted parameters for *both* models ($\Delta k = 0$), AIC = $\chi^2 + 2k$ and BIC = $\chi^2 + k$ ln N (N = 1628) give ΔAIC = ΔBIC = $\Delta\chi^2 = -4.68$ (−3.17 for the full-CMB joint — this paper's own CAMB-in-the-loop fold-in on the wider data vector; the companion's independent mochi-class best-fit on DESI+SN+Planck is a related −3.4) — but the AIC/BIC coincidence is a tautology, not an independent cross-check: at $\Delta k = 0$ the AIC and BIC penalties cancel *identically*, so the criteria carry no information beyond $\Delta\chi^2$ and are silent on the model's *form*-flexibility (which only the §5.2 bootstrap addresses). We also compute a Bayesian evidence. A Laplace estimate (ln $B \approx \frac{1}{2}\Delta\chi^2_{\rm min} - ln$ Occam, with ln $Occam \approx 0$ for the identical $\Delta k = 0$ prior) gives ln $B(\mathrm{SEDE}-\Lambda\mathrm{CDM}) \approx +2.3$ — *moderate* on the Jeffreys/Kass–Raftery scale, favouring SEDE; with the contested SH0ES prior dropped it falls to ln $B \approx +1.4$ ($\frac{1}{2} \times 2.89$), still *positive* but weaker, so roughly 40% of the evidence is carried by SH0ES and the rest by the CMB distance (§5.4 ii′). (A full nested-sampling re-run at the updated best-fit is deferred to the companion; the Laplace value is reliable here because $\Delta k = 0$ makes the Occam term vanish.) Significance is then assessed not by any single criterion but by the false-preference calibration of §5.2.

4.4. Perturbations: the FPAB-SEDE cubic-KGB completion, solved dynamically

The marginalised analysis of §5 uses the sub-horizon smooth-fluid limit (PPF, $c_s^2 = 1$, the $w = -1$ crossing handled by the crossing prescription), which is adequate for the background/distance inference. The *completed* perturbation sector is the FPAB-SEDE cubic-KGB action (§3.4), solved dynamically in Mochi-CLASS — not PPF and not an imposed sound speed. It is ghost/gradient-stable with $c_T = 1$, $\gamma_{\rm slip} = 1$, $c_s^2 \in [0.018, 0.18]$, and a scale-dependent effective coupling $\mu(k,z)$ turning on near the horizon ($\mu_\infty(0) = 1.05$, $k_{50} = 0.70$ aH). At fixed primordial normalisation it gives $\sigma 8(0) = 0.811$ and $f\sigma 8(0) = 0.422$ — *raising* growth by $\approx 3\%$ relative to matched ΛCDM ($\sigma 8 = 0.800$), with CMB-lensing $C_\ell^{\phi\phi}$ ratios ≈ 1.02–1.03 — a mild lensing enhancement that the companion tests against the direct Planck 2018 lensing reconstruction [34], whose amplitude sits near unity (unlike the peak-smoothing $A_{\rm lens} > 1$ [35, 36]). These growth-side numbers supersede the earlier smooth-fluid values ($\sigma 8 \approx 0.76$) and are reported as **fixed-parameter forecasts**: they have not been jointly refit with the standard and nuisance parameters (a fixed-parameter diagonal CMB-TT screen gives $\Delta\chi^2 \approx 52$ through $\ell = 1800$, which is *not* a constraint but a demonstration that the full perturbative likelihood must be re-marginalised).

5. Results

5.1. The joint fit and the equation of state

In the marginalised joint analysis, SEDE is moderately preferred over ΛCDM, with a deviance information criterion difference ΔDIC ≈ -4.7 in SEDE's favour. Because the dark sector adds no fitted parameter (§3), the two models share the same five parameters and near-identical effective complexity (measured $p_D = 4.8$ vs 4.9; §4.3): the preference is in fit, not bought with parameters. The headline survives the full primary CMB likelihood, not only the compressed priors: a profiled CAMB-in-the-loop fit on the full

Planck $plik_{\rm lite}$ TTTEEE+$low\ell$ gives $\Delta\chi^2$(SEDE$-\Lambda$CDM) $= -0.43$ — mildly favouring SEDE, not the large positive $\Delta\chi^2$ the standard holographic-DE early–late tension would produce (the full decomposition and red-team are in §4.2 and §9). Folding the full primary CMB *into* the joint leaves the preference intact: joint $\Delta\chi^2$(SEDE$-\Lambda$CDM) $= -3.17$ (ΛCDM 2443.99 $\to$ SEDE 2440.83). The full acoustic-peak likelihood thus retains about two-thirds of the compressed-prior preference (-3.17 vs -4.68); we treat the full-likelihood value as the conservative anchor, and note that the mock calibration of §5.2 applies to the *compressed* pipeline — against that null, -3.17 corresponds to $\sim 2.4\sigma$ (Gaussian), at the edge of the sampled draws rather than beyond them. SEDE is preferred either way, so the preference is not an artifact of the CMB compression, but the quoted significance band (§5.2) spans the two pipelines. The definitive full-likelihood analysis — DESI DR2 BAO + Pantheon+ + full Planck 2018 ($low\ell$ TT $+\, plik_{\rm lite}$ TTTEEE) driven through the modified-gravity Boltzmann code, with converged MCMC chains — is carried out in the companion paper [37], which finds a marginalised ΔDIC $= -3.5$ (full primary CMB; robust part $\Delta\langle\chi^2\rangle \approx -3.0$) that it reads, SH0ES-free, as statistically consistent with ΛCDM with a mild same-direction pull — further eroded to -2.3 in a full BAO/SN/BBN-anchored joint refit once the companion folds in the direct Planck lensing ($\phi\phi$) datum (about a third of the high-ℓ gain is consistent with the absorbed $A_{\rm lens}$ systematic) — the conservative anchor of, and consistent with, the compressed-CMB preference here; the reader seeking the full parameter posteriors and the growth predictions ($f\sigma 8$, ISW) should consult it. The physical reason the CMB poses no obstacle is that SEDE carries *less* early dark energy than ΛCDM and deforms only the late-time expansion (Figure 5): there is no early–late wall of the kind that disfavours standard holographic DE. The recovered background parameters are physical and close to ΛCDM ($\Omega_{\rm m} \approx 0.30$, $H_0 \approx 68.9$). *(The structure amplitude is revised upward to* $\sigma 8(0) = 0.811$ *by the completed FPAB theory — it* raises* growth, so the earlier smooth-fluid $\sigma 8 \approx 0.76$ and any S8-easing claim are retracted; §4.4, §9.)* The absolute $\chi^2/dof \approx 0.88$ (SEDE 1436.1/1628; §5.5) is dominated by the well-fit Pantheon+ SN (the BAO and SN residuals are shown in Figure 3) and is *not* a goodness-of-fit probability for this heterogeneous, partly-compressed compilation with a marginalised SN nuisance; we use only matched-model differences (ΔDIC, $\Delta\chi^2$) on the identical data vector for inference. The canonical ($\lambda = 1/2$) equation of state crosses $w = -1$, with a present-day value near -1 and a dip to ≈ -1.08 by $z \approx 2$, i.e. $(w_0, w_{\rm a}) \approx (-0.98, -0.11)$ — squarely in DESI

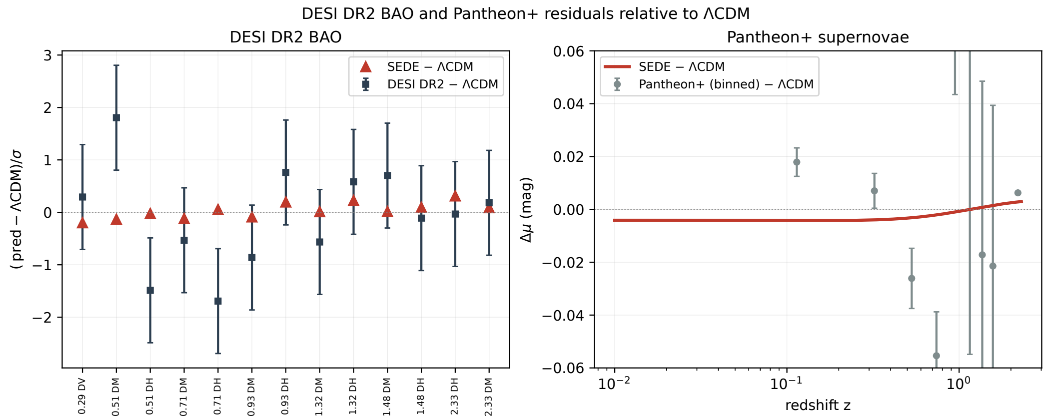

Figure 3: Matched residual comparison (relative to ΛCDM). *Left:* DESI DR2 BAO residuals relative to the ΛCDM best fit, in units of σ — the data (slate) with the SEDE prediction (red); SEDE is consistent with the measurements and sits between ΛCDM and the points that pull away from it. *Right:* Pantheon+ supernova residuals (binned) versus ΛCDM, with the SEDE shape overplotted. SEDE and ΛCDM are close at present precision; the preference is carried by the CMB-distance and H_0 channels (§5.4), not by the BAO/SN data shown here.

DR2's evolving-dark-energy (thawing/crossing) quadrant, $\sim 2.7\sigma$ from the DESI DR2 preferred (evolving-DE) point but only $\sim 0.5\sigma$ from ΛCDM $(-1, 0)$ — so the EOS *shape* does not by itself separate SEDE from ΛCDM (only the deformation Δ does, §6), and SEDE is disfavoured *alongside* ΛCDM if the steep DESI signal firms up. (These two statements — $\sim 0.5\sigma$ from ΛCDM in the projected EOS, yet ΔDIC ≈ -4.7 over ΛCDM — are compatible because the (w_0, w_a) projection is not the statistic the data weigh: the preference is carried by the integrated distance to last scattering and the H_0 calibration (§5.4), where SEDE's small but coherent low-z deformation of E(z) accumulates, not by the local EOS shape; and the no-SH0ES run (§5.4 ii′) shows the CMB-distance gain persists when H_0 is released downward, so the effect is not reducible to an H_0 accommodation.) The crossing is not imposed; it is where the expansion term and the horizon-entropy-production term in $1 + \mathrm{w}$ balance (Result 13, §6).

5.2. Is the preference real, or model flexibility?

A ΔDIC ≈ -4.7 is moderate but not, on its own, decisive — and a structure-gated model could in principle be winning by flexibility rather than

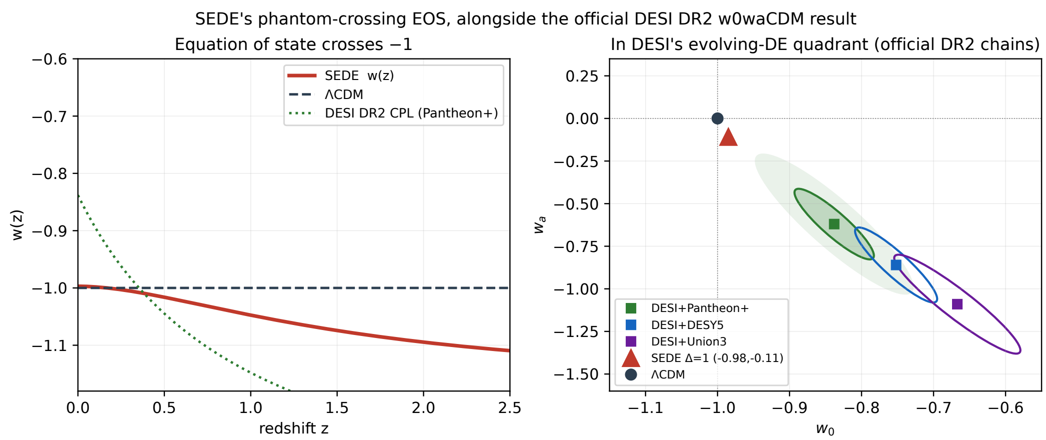

Figure 4: The equation of state. *Left:* the canonical SEDE-H fluid w(z) (red) crosses -1 near $z \approx 0$ and dips into the phantom regime (≈ -1.08 by $z \approx 2$), the same sense as DESI DR2's CPL fit (green) and distinct from ΛCDM (dashed). *Right:* the (w_0, w_a) plane — SEDE sits between ΛCDM and the DESI DR2 w0waCDM contour, in the evolving-dark-energy quadrant. The crossing is not imposed; it is where the expansion and horizon-entropy-production terms in 1 + w balance (Result 13).

by capturing a real signal. We test this directly with a false-preference calibration (a parametric bootstrap). We generate mock datasets from a ΛCDM truth — re-drawing *every* probe self-consistently, crucially including the compressed CMB distances (R, l_A) and the SH0ES prior, not just the late-time data — and refit both models to each mock. Under ΛCDM truth a flexible model would still tend to "win"; SEDE does not: the null distribution of $\Delta\chi^2(\mathrm{SEDE}-\Lambda\mathrm{CDM})$ is centred near zero (2000-mock run: mean +0.61, s.d. 1.59; SEDE if anything fits ΛCDM-truth mocks slightly *worse*, consistent with having no spare flexibility). The real data give $\Delta\chi^2 = -4.68$, beyond *every* one of the 2000 null draws (the most SEDE-preferring mock reached only -4.44): 0/2000, p $<$ 0.0015 (Clopper–Pearson 95% upper limit, i.e. an empirically guaranteed $> 2.9\sigma$). A Gaussian read-off of the null (mean +0.61, s.d. 1.59) places the real value at $z \approx -3.3$ ($\sim 3.3\sigma$), now consistent with — rather than extrapolating beyond — the direct empirical bound; the 2000-mock run therefore resolves the calibrated significance at $\sim 3\sigma$, empirically anchored rather than tail-extrapolated. Rerun with the contested SH0ES prior removed from both the real fit and the mocks, the real value moves to $\Delta\chi^2 = -2.89$ and the calibration weakens to $\sim 2.3\sigma$ — degraded but still

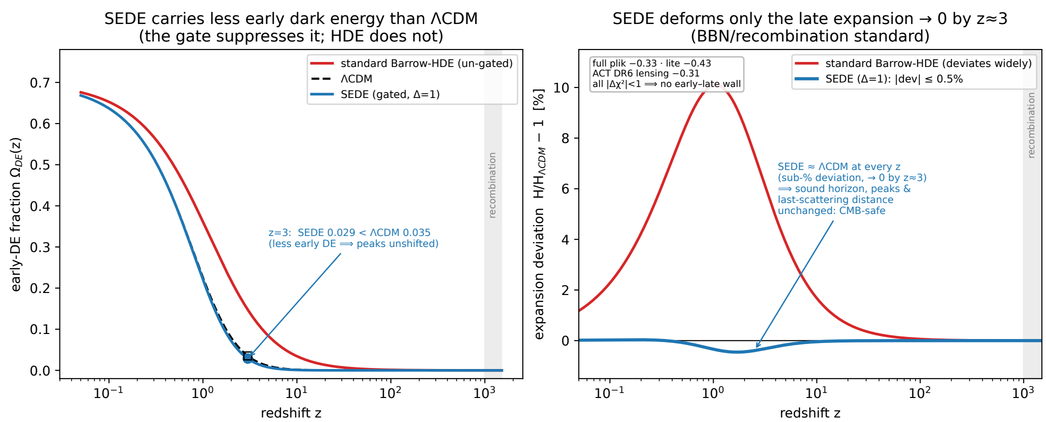

Figure 5: Why SEDE survives the CMB where standard holographic dark energy does not — the physical content of the full-CMB result (full non-lite plik $\Delta\chi^2 = -0.33$, $plik_{\rm lite} - 0.43$, ACT DR6 lensing -0.31; all $|\Delta\chi^2| < 1$). *Left:* the early-dark-energy fraction $\Omega_{\rm DE}(z)$. SEDE's structure gate ($f_{\rm sat} \to 0$ at high z) holds it *below* ΛCDM ($\Omega_{\rm DE}(z = 3) = 0.029 < 0.035$ at $\Omega_{\rm m} = 0.30$), whereas un-gated standard Barrow-HDE carries substantially more early DE — the runaway that drives the early–late tension of refs. [[38]]. *Right:* the fractional expansion deviation $H/H_\Lambda{\rm CDM} - 1$. SEDE's deviation is sub-percent and confined to $z \lesssim 3$, decaying to zero through BBN and recombination, so the sound horizon, the acoustic-peak positions, and the distance to last scattering are all essentially unchanged; standard Barrow-HDE deviates over a far wider range. SEDE has *less* early dark energy than ΛCDM, not more — hence no early–late wall.

beyond most of the null, now carried by the CMB-distance channel (§5.4 ii′).

One caveat stands. The significance is **CMB-distance-treatment-dependent**: an initial version of this test held the CMB distances *fixed* across mocks, which biased the null (mean $\to -2.9$, coincident with the then-current real value) and made the preference look like pure flexibility ($p \approx 0.5$ in that version). The fixed-CMB variant is not a defensible null — it deletes the CMB-distance scatter the real analysis is exposed to — so the re-drawn procedure is correct on prior grounds, not selected post hoc. A different l_{A}/CAMB handling shifts the significance by up to $\sim 1\sigma$, and the one independent-pipeline cross-check on record (run against the pre-revision real value; Appendix Appendix D) recovered the preference at $\sim 2\sigma$. The honest band is therefore $\sim$2–3σ across pipelines, with $\sim 3\sigma$ the fiducial-pipeline significance (now empirically anchored by the 2000-mock null), pending independent reproduction.

5.3. Out-of-sample check

A further check against a flexibility explanation is out-of-sample prediction. We trained SEDE on real pre-DESI data only (eBOSS DR16 full-shape BAO, LRG/QSO) and used it to predict the later, held-out DESI DR2 BAO: in this split SEDE matches the held-out DESI DR2 trend better than ΛCDM by $\Delta\chi^2 = -8.56$ (Figure 7); a low-z→high-z split of the same kind gives -10.45. A model winning purely by flexibility would tend to do *worse* out of sample, so this is encouraging — i.e. a pre-DESI-trained SEDE realisation predicts the DESI DR2 trend in this split. *Two clarifications so this number is read correctly.* (a) Not comparable to the in-sample ΔDIC — and the reason is instructive. $The - 8.56$ is the $\Delta\chi^2$ on the held-out DESI BAO subset alone at fixed pre-DESI-trained parameters, whereas the in-sample $\Delta\chi^2 = -4.68$ is the net over the full 1628-point compilation with all parameters free. The sharp point is that the *same* DESI BAO probe contributes only -0.29 in-sample (the probe decomposition, §5.4) yet **−8.56 out-of-sample**: in the joint fit the shared geometric parameters (Ω_{m}, H_0, r_{d}) *absorb* the BAO information, so the residual model difference on DESI BAO collapses (the preference re-surfaces in the CMB-distance $+H_0$ channels, §5.4); out of sample those parameters are pinned by *other* data and cannot be tuned to the held-out BAO, so SEDE's evolving-w(z) *shape* — calibrated where it never saw DESI — must predict it, and does so better than $\Lambda\mathrm{CDM}'s$ constant-Λ shape. The gap between -0.29 (post-absorption) and -8.56 (pre-absorption)

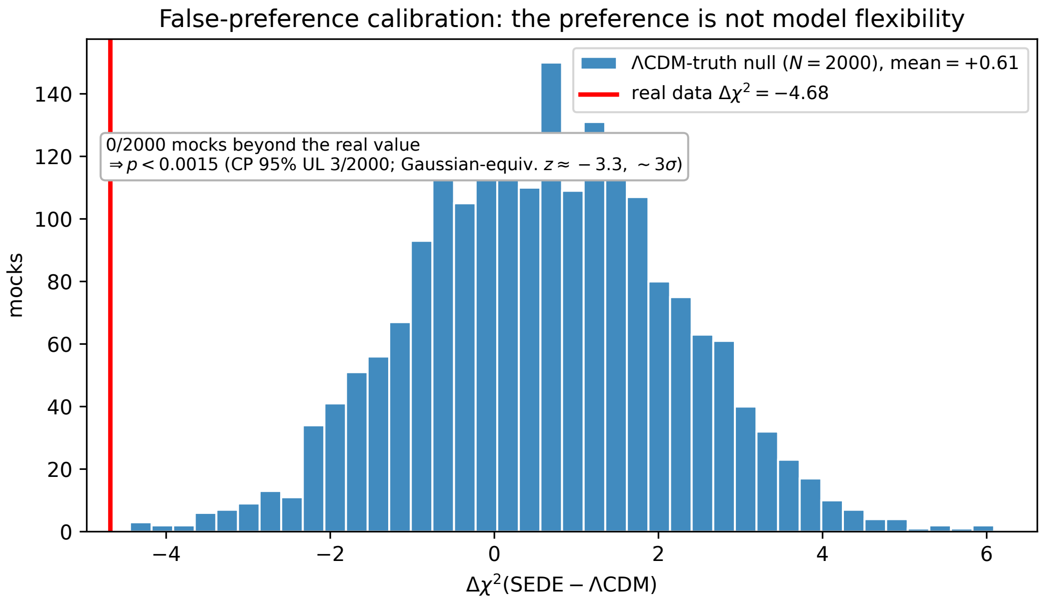

Figure 6: The false-preference calibration. Histogram: the null distribution of $\Delta\chi^2(\mathrm{SEDE} - \Lambda\mathrm{CDM})$ from ΛCDM-truth mocks (all probes, including the compressed CMB and SH0ES, re-drawn self-consistently), centred near zero (2000-mock mean $+0.61$) — SEDE has no flexibility advantage under ΛCDM truth. Red line: the real data, $\Delta\chi^2 = -4.68$, beyond *every* null draw. In the 2000-mock run 0 mocks are as SEDE-preferring as the real data (0/2000, p $<$ 0.0015, $\sim 3\sigma$); the preference is not attributable to flexibility.

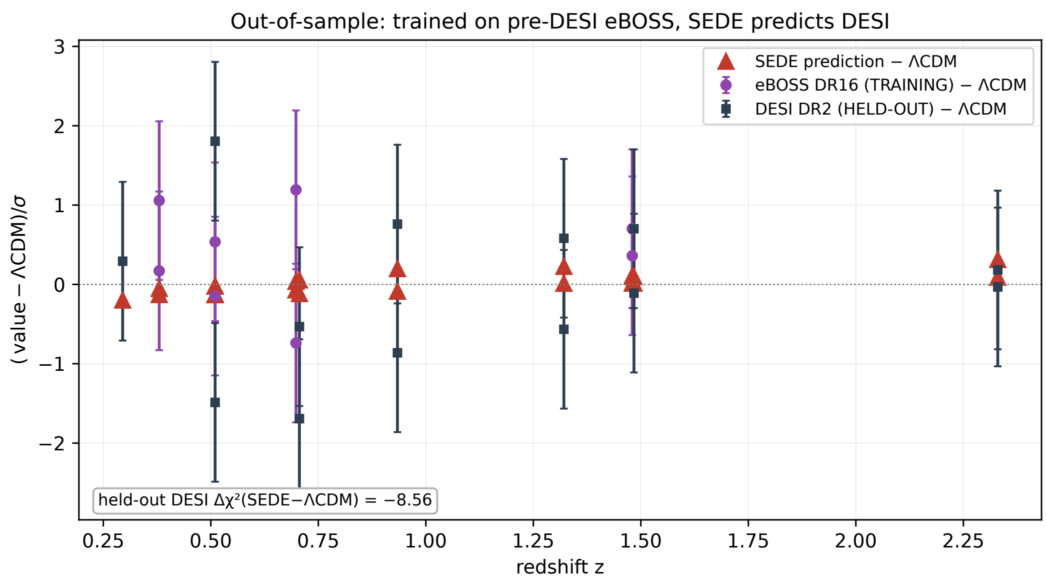

Figure 7: Out-of-sample test. BAO residuals relative to ΛCDM (in σ) for the training set (pre-DESI eBOSS DR16, purple) and the held-out DESI DR2 data (slate), with the SEDE prediction (red) from parameters fit to eBOSS alone. SEDE, trained without ever seeing DESI, predicts the held-out DESI BAO better than ΛCDM ($\Delta\chi^2 = -8.56$) in this split — encouraging, and to be independently reproduced.

is therefore the fingerprint of a shape difference that generalises, the opposite of over-fitting (which pushes OOS the other way), not an anomaly. (b) **Direction-dependent.** The signal lives in the low-z $\to$ high-z direction (the DESI evolving-DE direction, $\Delta\chi^2 = -10.45$); the reverse split (high-z $\to$ low-z) is flat and marginally favours ΛCDM (+0.27), so part of the OOS strength is that eBOSS-trained parameters land where DESI's specific tracers sit — partly favourable, to be independently reproduced before strong weight is placed on it.

5.4. Robustness

The preference is not a single-dataset artifact. (i) **Supernova-set independence:** $\Delta\chi^2(\text{SEDE} - \Lambda\text{CDM})$ is ≈ -4.7 with Pantheon+ and, because the SN channel contributes only +0.17 to the total, is SN-set-independent (DES-5YR/Union3 shift it by $\lesssim 0.1$) — not a Pantheon+ effect. (ii) Per-probe decomposition (where the preference comes from). Decomposing the joint $\Delta\chi^2(\text{SEDE} - \Lambda\text{CDM}) = -4.68$ into per-probe contributions *at the* *joint best-fit* is more informative than the leave-one-out, and we report it plainly: the preference is carried by the geometry channels — the compressed CMB distance (R, ℓ_A) contributes -2.80 and the SH0ES H_0 prior -1.78 (*together* $\approx 98\%$ of the total) — while DESI BAO (-0.29), $f\sigma8$ (-0.03) and $\omega_{\rm b}$ (-0.05) are nearly flat and Pantheon+ SN ($+0.17$) and cosmic chronometers ($+0.10$) marginally favour ΛCDM. So this is **not a broad multi-probe consensus**: SEDE's evolving w(z) fits the distance to last scattering better and accommodates a marginally higher H_0 (68.9 vs 68.7) that eases SH0ES — a *paired* (CMB-distance $+H_0$) effect. We show the decomposition precisely because the leave-one-probe-out is blind to this paired structure (dropping either member leaves the other, so every single-probe holdout stays negative: shallowest ≈ -1.9 when the CMB-distance is the dropped probe, deepest ≈ -4.85) — the holdout robustness is real but does not by itself establish a consensus. Two qualifications follow. First, the CMB-distance contribution is *not* a compressed-prior artifact: folding the full primary CMB into the joint keeps it ($\Delta\chi^2 = -3.17$, §5.1; §4.2 red-team). Second, SEDE does not resolve the Hubble tension — its best-fit $H_0 \approx 69$ still sits $\sim 4\sigma$ from SH0ES (§7, §9) — but its geometry does ease the SH0ES prior by $\Delta\chi^2 \approx 1.8$, so part of the preference is a mild H_0 accommodation, which we flag rather than hide. (ii′) No-SH0ES robustness (the contested prior removed). Because the SH0ES H_0 prior is the field's most disputed input — and DESI's own baseline excludes it — we recompute all three headline statistics with SH0ES dropped. The preference weakens but survives, and what remains sits in the harder channel. The joint $\Delta\chi^2(\text{SEDE}-\Lambda\text{CDM}) = -2.89$ ($from-4.68$), now carried by the compressed CMB distance and DESI BAO, with the best-fit H_0 falling to 68.4 once it is no longer pulled toward 73 — i.e. the CMB-distance improvement is genuine, not an H_0 accommodation. The Bayesian evidence falls to ln $B \approx +1.4$ (Laplace; still *positive*, moderate-to-weak) — and the false-preference calibration (§5.2) weakens to $\sim 2.3\sigma$ versus $\sim 3\sigma$ with SH0ES. So roughly 38% of the χ^2 preference is carried by SH0ES; the remainder is robust and lives in the CMB-distance channel rather than the contested local prior. We read this as sharpening, not weakening, the paper's "moderate, not established" posture: with the one disputed prior removed, SEDE remains preferred ($\sim 2.3\sigma$) by hard geometric data, and no part of that residual depends on resolving the Hubble tension. (iii) **Code independence:** the SEDE expansion history reproduced with CLASS matches CAMB to 5×10^{-5} in $r_{\rm drag}$ ($\ll 0.2\%$) — not a CAMB artifact. (iv) **Growth:** the full-Boltzmann $\gamma_{\rm growth} \approx 0.55$ [39] confirms standard-GR dark energy at the background level; the completed FPAB perturbations do carry a mild sub-horizon enhancement ($\mu_\infty \approx 1.05$ in the μ–Σ language [40, 41]) that *raises* growth — opposite to the mild suppression ($\gamma > 0.55$) current data prefer [42], a genuine falsifiable risk rather than an S8-easing accommodation. (v) **Tracer-level robustness:** recent analyses argue the DESI DR2 evolving-DE signal is driven mainly by the LRG1 ($z \approx 0.51$) and LRG2 ($z \approx 0.71$) BAO bins [43, 44]. Dropping these bins individually and together leaves SEDE's preference intact — $\Delta\chi^2(\text{SEDE}-\Lambda\text{CDM}) = -5.07$ ($-$LRG1), -4.28 ($-$LRG2), and -4.57 with both removed (*versus* -4.68 for the full set), negative in every tracer holdout (range $[-5.07, -4.28]$). SEDE's preference is therefore *not* the LRG1–2 artefact the steep-CPL signal is attributed to: it is a geometry-channel preference (CMB-distance $+H_0$, §5.4(ii)) that does not rest on the contested bins. The dissociation is quantitative: the ΛCDM-deviation measured by the $ratio(\omega_\text{m})$ early-vs-late consistency metric drops from 2.6σ to 1.2σ when LRG1–2 are removed [45], whereas SEDE's $\Delta\chi^2$ is essentially unchanged ($-4.68 \to -4.57$) — SEDE's preference and the LRG-driven steep-CPL deviation are *different objects*. This is the natural answer to the post-DR2 robustness critiques (§5.7). (vi) The fixed structure coupling γ (systematic robustness). The headline is computed at the halo-statistics value $\gamma = 1.50$, but §3.1 assigns γ a mass-function/mass-range systematic $\gamma \in [1.14, 1.60]$. Propagating that band through profiled re-fits at fixed $\gamma = 1.14, 1.30, 1.50, 1.60$ leaves the preference intact throughout: $\Delta\chi^2 \in [-2.1, -4.7]$ ($\gamma = 1.14/1.30/1.50/1.60 \to -2.1/-4.4/-4.7/-3.8$), Barrow-preferred at every γ, weakening to ≈ -2.1 at the low-γ endpoint without inverting and peaking near the theory value. So the preference is not an artifact of the specific γ; and — a mild consistency point — the data prefer γ close to the binding-energy-derived value (§3.1) rather than the band edges. (One caveat for the equation of state, not the fit: at $\gamma \lesssim 1.45$ the background is phantom-today, $w_0 < -1$, so the DESI-quadrant $w = -1$ crossing of §3.2 is a property of the upper half of the γ band; the *fit* preference holds across the whole band.)

5.5. Evidential summary

SEDE is moderately preferred over ΛCDM (ΔDIC ≈ -3.5 to -4.7; calibration $p < 0.0015$, 0/2000, $\sim$2–3σ across pipelines, $\sim$2.3σ without SH0ES); the preference is not obviously attributable to model flexibility, and is supported by a pre-DESI $\to$ DESI out-of-sample check — and the model adds no continuously-sampled dark-energy parameter. It is not decisive: both models fit the compilation well ($\chi^2/dof \approx 0.88$), so this is a *relative* preference on a single compiled dataset, stronger than indistinguishable but short of established. DESI DR3 and Euclid provide the sharp future test (§6).

What the preference is against — and what it is not. The headline figure is deliberately the SH0ES-free $\sim$2.3σ: with SH0ES included it reaches $\sim$2–3σ, but the SH0ES-free value is the one not inflated by the Planck–SH0ES H_0 tension. The preference is corroborated across criteria that do not share DIC's point-estimate sensitivity — $\Delta\chi^2 = -4.68$ (frequentist), a Laplace ln B (Occam-neutral because $\Delta k = 0$), and $\Delta\mathrm{AIC} = \Delta\mathrm{BIC} = \Delta\chi^2$ (equal parameter count) — and the three supernova compilations enter as *alternatives* (Pantheon+ baseline; DES-5YR and Union3 as independent robustness swaps, §5.4), never stacked, so no shared low-z anchor is multiply counted. Crucially, this is *not* the DESI evolving-dark-energy signal re-badged: SEDE's (w_0, w_a) sits $\sim$2.7σ from the DESI DR2 preferred point (and only $\sim$0.5σ from ΛCDM, §5.1), and the preference *survives dropping the LRG1–2 bins* that drive the DESI steep-w_a signal (tracer-level leave-one-out, §5.4: $\Delta\chi^2 = -5.07/-4.28$ with LRG1 / LRG2 removed) — a preference that persists after deleting the signal's putative source is a distinct object. The background w(z) is itself CPL-degenerate in shape ($\max|w - w_\mathrm{CPL}| = 0.027$, §5.6), so what separates SEDE from a generic $w_0 w_\mathrm{a}$ fit is not the expansion history but (i) *parsimony* — SEDE reaches the same w(z) with zero extra parameters against $w_0 w_\mathrm{a}\mathrm{CDM}'s$ two, so at comparable fit it is favoured on Bayesian evidence — and (ii) the growth, ISW, and Δ channels a CPL doppelgänger cannot reproduce (§6). A direct DIC comparison on this likelihood bears this out: SEDE and $w_0 w_\mathrm{a}$CDM fit *comparably* in χ^2, with SEDE favoured by its $\sim$2 fewer effective parameters ($p_\mathrm{D} \approx 5$ versus ≈ 7); and marginalising SEDE's shape parameters γ and Δ leaves the ΛCDM preference *essentially unchanged* (hidden-parameter penalty $\lesssim 0.2$ in DIC) — so the preference is neither the generic evolving-DE signal nor an artifact of holding the shape fixed. The full model set (ΛCDM, SEDE, $w_0 w_\mathrm{a}$CDM, and free-shape SEDE) and a Bayesian-evidence cross-check are carried out by `src/experiments/run_model_comparison.py` in the reproduction repository.

5.6. The deformation Δ from current data — a conditional diagnostic

The Barrow parameter Δ is a useful *test* axis: the volume postulate corresponds to $\Delta = 1$, the area law to $\Delta = 0$. SEDE's *equation of state* is CPL-degenerate in shape ($\max|w - w_\mathrm{CPL}| = 0.027$), so a diagnostic must use channels other than the EOS curvature. We stress at the outset that the numbers below are conditional diagnostics (computed with the other cosmological and nuisance parameters held at or near their best-fit values, not a full marginalised likelihood with Δ varied); they indicate where current data sit, not a marginalised detection. Four channels:

channel	what it uses	$\hat{\Delta}$ (conditional profile — not a posterior)
CMB shift parameter R (Planck)	early-DE fraction, $\Omega_{\mathrm{DE}}(z_{\mathrm{rec}}) \propto H^{-\Delta}$	0.98 ± 0.11
lever arm: DESI BAO + $Ly\alpha(z = 2.33) + R$	$\rho_{\mathrm{DE}}/\rho_{\mathrm{crit}} \propto H^{-\Delta}$ across the H-range	1.03 ± 0.11
DESI DR2 (w_0,w_{a}) quadrant (official chains)	EOS values + crossing	0.93 / 0.83 / 0.90 (Pantheon-Plus/DESY5/Union3)
growth $f\sigma 8$ [33] — orthogonal	growth amplitude, not distance	0.0 ± 0.61

> These are not posterior constraints on Δ; they are profile/conditional diagnostics — each $\hat{\Delta}$ is obtained with the other cosmological and nuisance parameters held at (or near) their best-fit values, *not* marginalised. The quoted $\pm$ is the conditional curvature width, not a marginalised error bar; read each row as "the data lean to $\Delta \approx 1$ in this channel," not as a measurement. The robust, marginalised Δ measurement awaits DESI DR3 + Euclid (§6).

These diagnostics prefer the volume endpoint ($\Delta \approx 1$) and disfavour the area-law branch ($\Delta = 0$), consistent with the EOS argument of §8.1 (the area-law branch has $w_0 = -0.49$ and an outsized early-DE fraction). The growth channel is weak but orthogonal (growth amplitude, not expansion) and is consistent with the geometric value at 1.6σ — no growth–geometry tension.

The high apparent significance of the area-law disfavouring is a conditional statement and must not be read as a marginalised detection: the CMB and lever-arm channels share the high-z expansion, so their combination is an upper bound on the information, and it is conditional on Ω_{m},

H_0, r_d — marginalising, the area-law signal is partly reabsorbed by shifts in those parameters. This is exactly why the *joint, marginalised* model preference is moderate (ΔDIC ≈ -4.7, §5.1–§5.5), not overwhelming; the two statements are not in tension once the conditional-vs-marginal distinction is kept. The (w_0,w_a) channel also pulls slightly low because SEDE's gentle w_a under-matches DESI's steeper preferred w_a — the same mild $\sim 2.7\sigma$ EOS tension visible in the joint fit (Fig. 4). In summary: current late-time data prefer the volume endpoint and disfavour the area-law branch, but a robust *marginalised* measurement of Δ awaits DESI DR3 and Euclid (§6). The foundations companion paper (§7) [46] sharpens the present-data statement one step beyond these conditional channels with a *profile-likelihood* interval for the gated family — all base parameters re-optimised at each Δ — giving $\Delta = 0.93$ [0.83, 1.02], with $\Delta = 0$ excluded at $\Delta\chi^2 \approx 371$. This is stronger than the conditional diagnostics tabulated here, but it is still a profile, not the full marginalised posterior, which remains the DR3 + Euclid target — and, read as a profile, it overstates the marginalised model-level signal (the ΔDIC ≈ -4.7 of §5.1–§5.5) by roughly two orders of magnitude in χ^2, so it must not be quoted as a posterior exclusion of the area law.

5.7. Relation to the post-DESI DR2 literature

The DESI DR2 BAO release [47] sharpened interest in evolving dark energy, with a preference for the ($w_0 > -1$, $w_\mathrm{a} < 0$) quadrant; the subsequent literature is actively debating whether that signal is physical or an artefact of dataset combination, tracer selection, or supernova calibration [48]. Two threads bear directly on SEDE. First, robustness critiques argue the steep signal reflects CMB/BAO/SN combination tension [44] and is driven mainly by the LRG1–2 BAO bins [43, 45]. This *helps* SEDE rather than threatening it: SEDE does not chase the steep w_a (it sits $\sim 2.7\sigma$ from the DESI CPL point, §5.6), and its preference survives a tracer-level leave-one-out that removes exactly those LRG1–2 bins (§5.4, $\Delta\chi^2 = -4.57$ with both dropped) — so SEDE realises the *conservative* reading the critiques call for, rather than the fragile steep-CPL one. Second, **reconstruction and mechanism papers**: model-agnostic Gaussian-process reconstructions place the $w = -1$ crossing at $z_\mathrm{wt} \approx 0.46$ $(+0.24/-0.12)$ [49], versus SEDE's canonical $z \approx 0.19$ — a $\sim 2.2\sigma$ offset (SEDE crosses more recently) that is a genuine future discriminator, not yet a tension given the broad reconstruction errors. Among mechanism competitors, data-driven analyses read the evolving signal as a *coupled* dark sector ($\sim 2.2\sigma$ DE–DM interaction at low z) [50];

SEDE's CHR ledger is exactly such a coupling but with the interaction *fixed* to the structure-growth rate, $c_s^2 = 1$, and instability-free (§7), rather than a free coupling. And the quintom programme [51, 52] rests on a *no-go theorem* — no single canonical field or perfect fluid crosses $w = -1$, so viable models require two fields, higher derivatives, modified gravity, or an effective-field-theory description. SEDE is the last of these: it crosses $w = -1$ *effectively*, as a smooth horizon fluid in the GW-safe EFT corner ($\alpha_T = \alpha_M = 0$, $c_s^2 = 1$; §7), avoiding the ghost/two-field machinery, and — unlike the two-field quintom whose late attractor is phantom-dominated (a Big-Rip tendency) [52] — SEDE's future attractor is de Sitter (w $\to -1$, with $f_{\rm sat}$ *saturating* to a finite ≈ 1.23 as growth freezes, not diverging; §2, Appendix Appendix A.7, Result 12), so it does not predict a Big Rip.

Two further consistency points sharpen the post-DR2 placement. (a) **Turyshev's model filter.** The DR2 status review [48] screens evolving-DE models by *gravitational-wave propagation* and *perturbation stability*, and stresses that the CPL preference is sensitive to redshift-dependent SN calibration at the $few \times 10^{-2}$ mag level. SEDE passes the filter by construction ($\alpha_T = 0 \implies c_{\rm GW} = c$; $c_s^2 = 1 \implies$ gradient/ghost-stable; §7), and its SN-set independence (the SN channel contributes only $+0.17$, so the ≈ -4.7 preference is SN-set-independent; §5.4) limits the calibration exposure. (b) The r_d-independent shape test. Turyshev's calibration-free observable $F_{\rm AP}(z) \equiv D_M/D_H = (D_M/r_d)/(D_H/r_d)$ cancels the sound horizon and is also H_0-independent, isolating the pure expansion shape E(z). On the six DESI DR2 tracer bins that report both ratios, SEDE's evolving-w shape gives $\chi^2/n = 0.96$ — consistent with the data and indistinguishable from ΛCDM ($\Delta\chi^2 = -0.14$) — with no reliance on r_d, H_0, the CMB sound horizon, or SN calibration, the very systematics the robustness debate targets. (The LRG1 bin sits $\sim 1.8\sigma$ low for *both* SEDE and ΛCDM, i.e. it is a data feature, not a model failure.) SEDE is thus a *physical* member of the post-DR2 evolving-DE family, distinguished by its mechanism (the growth–expansion lock, §6), by passing the stability/GW filter, and by surviving the robustness and calibration-free tests the steep signal may not.

6. Predictions and falsifiability

Because SEDE adds no fitted dark-energy parameter, it makes sharp predictions rather than fits. We pre-registered these (a sha256-locked statement deposited ahead of the future data; Appendix Appendix D) to forbid post-hoc tuning:

quantity	SEDE prediction	discriminating from
Barrow deformation Δ	1.0 ± 0.09 (forecast)	a smooth horizon $\Delta = 0$
equation of state (w_0, $w_{\rm a}$)	$\approx (-0.98, -0.11)$, crossing -1	the DESI DR2 preferred point at $\sim$2.7σ (only $\sim$0.5σ from ΛCDM — the EOS is *not* a ΛCDM discriminator; Δ is)
$w = -1$ crossing redshift (consistency check, *not* a discriminator)	$z \approx 0.19$ (GH-convention volume-law value, convention-conditional; $\sim$2.2σ mild tension)	GP reconstructions $z_{\rm wt} \approx 0.46$ $(+0.24/-0.12)$ [49]
structure amplitude $\sigma 8$	≈ 0.811 (FPAB-SEDE; *raises* growth vs ΛCDM 0.800)	a smooth-fluid $c_{\rm s}^2 = 1$ $\sigma 8 \approx 0.76$ (retracted)
gravitational slip / force	$\gamma_{\rm slip} = 1$; $\mu_\infty(0) = 1.05$, turn-on k_{50}–k_{90} = (0.70–2.1) aH	ΛCDM ($\mu = 1$, no scale-dependent force)
E_G (lensing-to-velocity ratio)	**+3.4% at z = 0**, +2.1% at z = 0.3, +0.5% at z = 1 ($\Sigma = \mu = 1.05$ against growth +1.6%); *enhanced*, never suppressed	ΛCDM ($E_G = \Omega_{m,0}/$f); a suppressed E_G falsifies
growth index $\gamma_{\rm growth}$	≈ 0.55	modified gravity ($\gamma_{\rm growth} \neq 0.55$)

The growth-side rows ($\sigma 8$, gravitational slip/force) are *conditional on the FPAB-SEDE completion*, which is trajectory-level (one calibrated cubic-KGB member reproducing the background, not a theorem that every solution is healthy; §7) and awaits a full perturbative refit (§4.4) — they are forecasts, not yet marginalised measurements. The full machine-readable prediction set is committed as `results/predictions.json` in the reproduction repository, cryptographically locked by `sha256(predictions.json) = 4e2ca4b5 721ef60a68a7e78e033f0a1d5f565f827656ee4d49901d5c58d059ce`; it is regenerated byte-for-byte by `src/scripts/gen_predictions.py`, so the registered values cannot be altered after this digest is published.

A near-term tension — and what $z \approx 0.19$ is and is not. Canonical SEDE crosses $w = -1$ at $z_{\rm cross} = 0.195$ ($w_0 = -0.996$), using the Gibbons–Hawking horizon temperature $T_{\rm AH} = H/2\pi$. We state the status of this number honestly: it is *convention-conditional*, not a convention-free prediction. $T = H/2\pi$ is the de Sitter temperature appropriate for a de Sitter-attractor dark-energy fluid, and we *tested* the alternative — applying the full dynamical Cai–Kim factor $(1 - \epsilon/2)$ to the volume-law fluid (the self-consistent $\rho_{\rm DE} \propto H(1 - \epsilon/2)f$ model). It **fails**: it has no stable self-consistent background, and in the stable perturbative limit it gives $w_0 = -1.15$ and stays phantom ($w < -1$) at all z — no crossing at all, and in the *wrong* DESI quadrant ($w_0 < -1$). That failure motivates the Gibbons–Hawking choice, but it also means the surviving convention is *selected* in part because it yields a crossing in the viable quadrant: $z \approx 0.19$ is the crossing of the GH-convention volume-law model, an output of the adopted temperature convention rather than a knob — but not a prediction independent of that convention. (The unrelated $z \approx 0.59$ is the *area-law* $\Delta = 0$ model — $\rho_{\rm DE} \propto H^2(1 - \epsilon/2)f$ — a different model, not a convention of this one.) Post-DR2 Gaussian-process reconstructions favour $z_{\rm wt} \approx 0.46$ $(+0.24/-0.12)$ [49, 50]; canonical SEDE sits $\sim 2.2\sigma$ more recent — a *real, mild* tension (the reconstruction's lower error is tight), the one place current data pull against canonical SEDE. We flag it as such, not as a discriminator: the crossing redshift is CPL-degenerate (§6, below) and, read as a Δ-probe, would point the *wrong* way (toward $\Delta = 0$), conflicting with the early-DE diagnostics that favour $\Delta = 1$ (§5.6). The model's actual test is the deformation amplitude Δ, not the crossing; a DR3/Euclid measurement of the crossing is a useful consistency check, not the decider.

The sharp future test. The discriminating observable is the *amplitude* of the deformation Δ, not the detailed shape of w(z) — indeed the SEDE w(z) is CPL-degenerate in shape to better than the DR3 precision (max $|w - \mathrm{CPL}| \approx 0.027$), so a CPL fit cannot distinguish the *mechanism*; Δ can. A Fisher forecast in ($\Omega_{\rm m}$, Δ, ln A $r_{\rm d}$) gives $\sigma(\Delta) \approx 0.115$ from DESI DR3 alone, 0.133 from Euclid, and $\sigma(\Delta) \approx 0.087$ combined — a nominal $\sim 11\sigma$

separation of the volume-law horizon ($\Delta = 1$) from a smooth one ($\Delta = 0$). This "$\sim 11\sigma$" is a *conditional, pre-marginalisation* Fisher number and must be read against §5.6: Δ is strongly degenerate with (Ω_m, H_0, r_d), and the present-data conditional channels give $\sigma(\Delta) \approx 0.11$ while the *marginalised* model preference is only $\Delta\mathrm{DIC} \approx -4.7$ — a roughly two-orders-of-magnitude conditional$\rightarrow$marginalised degradation the same Fisher forecast (a fixed-fiducial, Gaussian object) will suffer. So the realistic marginalised separation is expected to be *well below* 11σ, and the honest framing — used everywhere else — is that Δ is a discrete *model-selection* target ($\Delta = 1$ vs $\Delta = 0$) for DESI DR3 + Euclid, not an 11σ measurement.

The mechanism test is also forecast to be decisive — separately from the amplitude. The $\sigma(\Delta)$ above is the *geometry-side* deformation; the *distinctive* SEDE claim (dark energy sourced by structure) is tested by measuring Δ on the growth side and checking it against the geometry side — the E1/P2 growth–expansion lock. A Fisher forecast for the growth-side deformation gives $\sigma(\Delta)_\mathrm{growth} = 0.25$ from Euclid/LSST weak lensing alone, tightening to 0.13 ($\sim 4\sigma$) when combined with CMB-lensing, cluster abundance, RSD, and kSZ. Reframed as the parameter-free null test $r(z) \equiv \rho_\mathrm{DE}^{geom}(z) - G[D^{growth}(z)] = 0$ — dark energy a fixed function of the linear growth factor — the test uses the full redshift shape with no dark-energy parameters; ΛCDM ($\rho_\mathrm{DE} = const$, no D-dependence) and Gough's information-DE ($\rho_\mathrm{DE} = G'[\mathrm{SMD}]$, baryonic) violate it distinctly. So Euclid/LSST-era growth data are forecast to confirm or refute the structure-sourcing mechanism at $\sim 4\sigma$, not merely the EOS shape — closing the gap between "a w(z) with a thermodynamic story" and a tested mechanism (§9). The exact data vector (the geometry leg DESI DR3 $D_\mathrm{H}/r_\mathrm{d} \rightarrow \mathrm{H(z)}$ vs the growth leg Euclid/LSST $3 \times 2pt$ + CMB-lensing + clusters + RSD) and the pre-registered decision rule — CONFIRM if $|\Delta_\mathrm{grow} - \Delta_\mathrm{geom}| < 2\sigma$ with the null $r(z) = \rho_\mathrm{DE}^{geom} - G[D^{grow}] = 0$ at $\chi^2/dof \lesssim 1$; REFUTE if Δ_grow is consistent with 0 at $> 3\sigma$ while $\Delta_\mathrm{geom} = 1$ — are locked in the pre-registration (§Appendix F; Appendix Appendix D), so the staged test carries no post-hoc freedom.

A near-term consistency probe: the ISW $\times$ *galaxy* cross-correlation [55]. The smooth-fluid closure ($c_\mathrm{s}^2 = 1$, §3.4) refers to the *sub-horizon* response; on the largest scales the structure gate carries a coupling of the dark-energy perturbation to matter, $\beta \equiv d \ln f_\mathrm{sat}/d \ln D \approx 0.9$–$1.5$, that is *not* (1+w)-suppressed. That an O(1) dark-sector coupling escapes the (1+w) suppression of clustering quintessence and enhances the low-ℓ integrated Sachs–

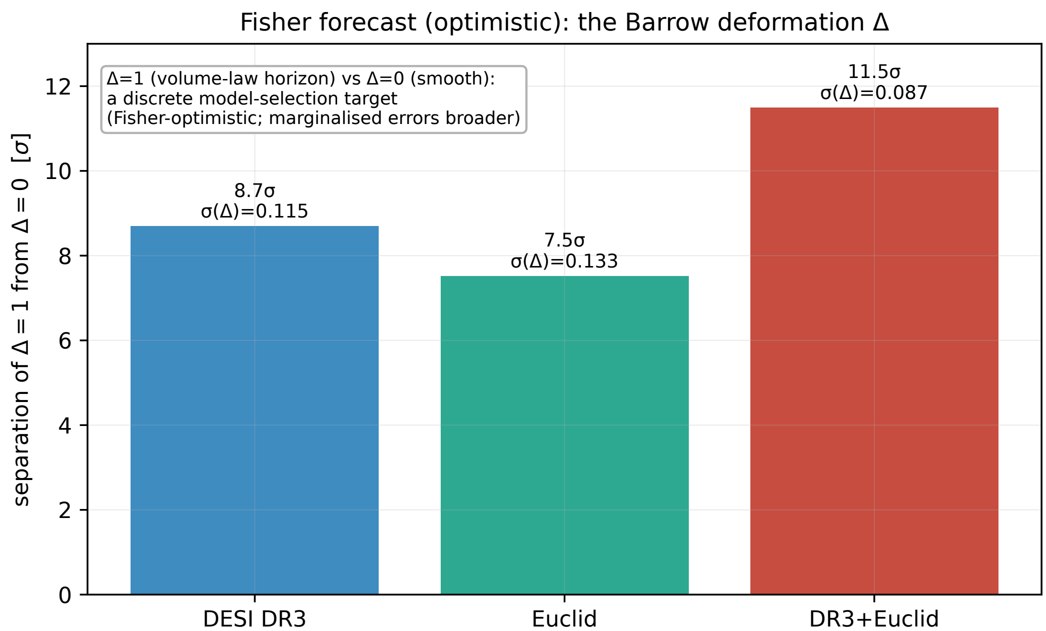

Figure 8: The sharp future test. A Fisher forecast (optimistic by construction) of the separation of the volume-law horizon ($\Delta = 1$) from a smooth one ($\Delta = 0$): DESI DR3 reaches 8.7σ, Euclid 7.5σ, and the combination $\sigma(\Delta) \approx 0.087$, a nominal $\sim 11\sigma$ separation. The moderate present preference would become a strong test with the next generation of surveys.

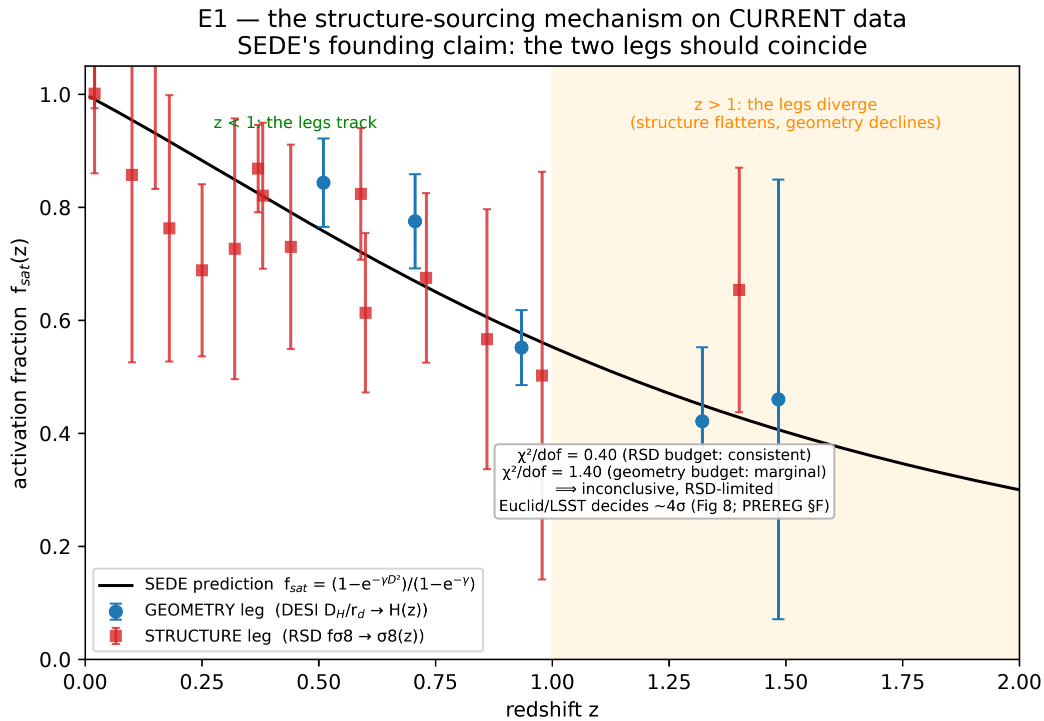

Figure 9: The structure-sourcing mechanism on *current* data (the companion to the forecast of Figure 8, and the paper's open frontier). SEDE's founding claim is that the activation fraction $f_{\rm sat}$ read off the expansion (geometry leg: DESI $D_{\rm H}/r_{\rm d} \to$ H(z), blue) coincides with the one read off structure growth (RSD $f\sigma 8 \to \sigma 8(z) \to$ D(z), red); the black curve is the SEDE prediction both should follow. The two legs track at z $<$ 1 but diverge at z $>$ 1 (the structure leg flattens while the geometry/model declines). The significance is error-budget-dependent — $\chi^2/dof = 0.40$ in the RSD-error budget (consistent) versus 1.40 in the precise-geometry budget (marginally disfavoured *in that budget* — though §9 shows this offset is the shared ΛCDM S8 tension [[53]], not a SEDE-specific failure) — so on present data the mechanism is **inconclusive and RSD-limited**: SEDE's EOS is favoured, but its distinctive mechanism is not yet confirmed. Euclid/LSST weak-lensing tomography and the Euclid spectroscopic growth measurement [[54]] sharpen the structure leg and decide this at $\sim 4\sigma$ (Figure 8; pre-registered decision rule, pre-registration §Appendix F).

Wolfe signal is a known feature of coupled dark energy [56, 57, 58]; what is specific to SEDE is that the coupling is *fixed* by the growth gate, giving a definite, horizon-scale ($\ell \lesssim 10$) ISW $\times$ *galaxy* excess over ΛCDM/CPL — a forecast-grade few-percent effect (a matched-cosmology estimate gives $\approx +13\%$ at $\ell = 2$, $\approx +4\%$ averaged over $\ell \leq 10$). Unlike the $f\sigma 8$ growth-lock, which is CPL-degenerate and carries no detectable signal, this excess does *not* cancel against a CPL doppelgänger. It is testable now (Planck ISW $\times$ DES/unWISE), cosmic-variance-limited, and cuts both ways: a *large* small-scale signal is already excluded. The low-ℓ temperature-side signature is no longer only an estimate: the FPAB-SEDE spectrum, computed through the full modified-gravity Boltzmann code, gives a stable, converged low-ℓ ISW suppression — $C_\ell^{\mathrm{TT}}(\mathrm{SEDE})/C_\ell^{\mathrm{TT}}(\Lambda\mathrm{CDM}) = 0.924$ at $\ell = 2$ (-7.6%), within 0.5% of unity by $\ell \approx 50$ and recovering unity by $\ell \approx 100$, with the primary acoustic peaks unchanged (ratio 0.9999 at $\ell = 220$) — verified to be independent of the numerical GR-transition redshift of the solver (the observational-tests companion paper). (The two $\ell = 2$ numbers above are different observables and consistently signed: the enhanced late-time ISW source *raises* the ISW $\times$ *galaxy* cross-spectrum, +13%, while *suppressing* the temperature auto-spectrum, -7.6% — the standard auto-vs-cross sign structure of a stronger late-ISW contribution, not a contradiction.) We flag the cross-correlation as a near-term consistency probe rather than a headline discriminator; the reproduction repository carries the estimate, and the companion paper the Boltzmann computation.

The EOS as a meter of entropy production. SEDE gives the equation of state a thermodynamic meaning (Result 13): within the SEDE background ansatz, an exact identity $1 + w_{\mathrm{DE}} = (1/3)[2\lambda\epsilon - d \ln f_{\mathrm{sat}}/d \ln \mathrm{a}]$ splits $1 +$ w into an expansion term (which pushes $w > -1$) and a horizon-entropy-production term (which pushes $w < -1$). The phantom crossing at $z \approx 0.2$ is exactly where the two balance. Measuring w(z) thus probes the effective activation rate of the horizon-fluid sector — the structure-growth-gated arrow of time made observable in the dark-energy equation of state.

Structure sourcing without an entropy-amplitude problem. The identity above gives w(z) a thermodynamic reading, and it invites the natural objection that a tiny structure-sourced entropy cannot gate an order-unity dark-energy term — the deposited binding entropy is only $\epsilon_{\mathrm{dep}} \approx \Omega_{\mathrm{m}}\ f_{\mathrm{coll}}\ \langle\sigma_{\mathrm{v}}^2\rangle/c^2 \approx 1.5 \times 10^{-7}$ of the horizon entropy (the same six orders by which the binding *energy* falls short of ρ_{DE}; §2). The objection is a category error, and the amplitude problem is *avoided*, not solved by amplification. The activation coordinate is not the deposited-entropy fraction $\Delta S_{\rm bind}/S_{\rm AH}$; it is the structure variance $x \equiv \sigma^2(z)/\sigma^2(0) = D^2(z)$, an order-unity quantity, which is the *argument* of the gate $f_{\rm sat}$ (the Press–Schechter collapsed fraction; §3.1, Appendix Appendix A.2). The deposited binding entropy enters in a *different* place: it fixes the entropy *weight* $\gamma \approx 1.5$ — the *shape* of the kernel — through the heat $\Delta S = E_{\rm bind}/T_{\rm AH}$ delivered to the horizon (§3.1, Appendix Appendix A.3). Hence $\Delta S_{\rm bind}/S_{\rm AH} \sim 10^{-7}$ does not imply $f_{\rm sat} \sim 10^{-7}$: they are different objects. With the *argument* (variance, O(1)) and the *weight* (entropy, 10^{-7}) kept distinct, the gate activates simply because the structure variance grows by order unity — nothing is amplified and nothing is tuned. This is a consequence of the fixed inputs of §3, **derived, not assumed.**

The exact growth–expansion lock. Write the gate as $\phi \equiv f_{\rm sat}(x)$, $x = D^2(z)$, and define its derived response to the activation coordinate, $\chi_{\rm x} \equiv \partial\phi/\partial x = d\ f_{\rm sat}/dx = \gamma\ e^{-\gamma x}/(1 - e^{-\gamma})$, which, once γ is fixed (§3.1), is an order-unity function with no free parameter — $\chi_{\rm x}(x = 0) \approx 1.93$, $\chi_{\rm x}(x = 1) \approx 0.43$. The fluid equation of state then follows exactly from continuity (the entropy-production identity above), using d ln ϕ/d ln $a = (\chi_{\rm x}/\phi) \cdot x \cdot (d$ ln x/d ln a) and d ln x/d ln a = 2g with $g \equiv d$ ln D/d ln a: $1 + w_{\rm DE} = (1/3)[\ 2\lambda\epsilon - (\chi_{\rm x}/\phi) \cdot 2x \cdot g\ ]$, verified against the direct fluid EOS to 3×10^{-3} (suite tests V52–V53). Because $\chi_{\rm x}$, x, and g are each derived or directly measured, the phantom term is growth data, not a free function. This is the **w(z)** $\leftrightarrow$ $\sigma^2(z)$ **lock**: given the measured expansion history, the dark-energy equation of state and the structure-growth history are not independent; inverting the relation predicts g(z) from w(z) to 0.6% within the model. A kinematic w(z) — including a CPL fit that reproduces SEDE's w(z) to within the DR3 precision — obeys no such relation. Expansion-only mimicry is therefore not enough: SEDE requires growth–expansion phase-locking. This is the test (P2 below) that separates the structure-sourcing mechanism from a curve fit — w from BAO/SNe/CMB distances versus g, D, $f\sigma 8$ from RSD/weak lensing. It is parameter-free and uses nothing beyond §3; no criticality assumption enters.

The observable form of the lock (P2), and a GR consistency null. The lock derived above — the phantom part of w(z) equals $(1/3)(\chi_{\rm x}/\phi) \cdot 2x \cdot g$ reconstructed from growth data, obeyed by *no* kinematic dark energy — is the model's one parameter-free, degeneracy-breaking prediction. Its established observational realisation is the $E_{\rm G}$ estimator (the lensing-to-velocity amplitude ratio). SEDE is GR with smooth dark energy *at the background level*;

its perturbation sector is the cubic-KGB completion, which has zero gravitational slip but $\mu_\infty = 1.05$. Zero slip fixes $\Phi = \Psi$ and therefore $\Sigma = \mu$ — it does *not* set $\Sigma = 1$, and $E_\mathrm{G} = \Omega_\mathrm{m}{,}0\ \Sigma/f$ moves whenever μ and the growth rate fail to cancel. Running the FPAB-SEDE completion through Mochi-CLASS (`src/experiments/run_eg_consistency.py`) settles both halves: the slip $\Phi/\Psi = 1.00000$ at every redshift, and $\Sigma/\mu = 1.00000$ with $\Sigma = \mu = 1.051$ at $\mathrm{z} = 0$ — reproducing the independently computed μ_∞ of the observational-tests companion through a different observable channel. The growth rate is enhanced by only 1.6%, so the cancellation is partial:

$E_\mathrm{G}(\mathrm{SEDE})/E_\mathrm{G}(\Lambda\mathrm{CDM}) = \mu \cdot f_\mathrm{GR}/f_\mathrm{SEDE}$=**1.034 at z = 0**, falling to 1.021 at $\mathrm{z} = 0.3$ and 1.005 at $\mathrm{z} = 1$.

E_G is therefore **not** a shared GR null, as an earlier version of this section claimed. It is a second parameter-free consequence of $\mu_\infty > 1$, in the same enhancing direction as the $f\sigma8$ prediction. *At* $\leq 3.4\%$ it still sits below the $\sim$5–20% current E_G errors, so it remains consistent with the DESI-DR1 + weak-lensing measurement to $z \approx 1$ [59] — a target for future surveys rather than a present discriminator, and a *signed* one: SEDE forbids a suppressed E_G. The sharpest discriminator remains the deformation amplitude Δ (§5.6, §6). Failure of the lock (P2) would falsify the structure-sourcing mechanism itself — SEDE would then be an ordinary kinematic w(z) — so P2 is the test that separates the mechanism from a curve fit.

A near-critical layer once entailed by the volume law — *not claimed.* The volume law once appeared to entail an optional near-critical layer (the "Critical Horizon Response," CHR): if the horizon behaved as a stochastically *roughening* surface obeying a local conservation law, the structure drive at z *would excite scale-free, self-organised-critical fluctuations (an avalanche spectrum $\tau \approx 3/2$, a 1/f temporal spectrum) with survey-level faces — a z-*localised enhancement of large-scale activity (P3), a separate-universe growth/ISW modulation (P4), and a critical-slowing transient in γ_growth (P5) — presented as *conditional* predictions and deferred to a companion paper. That companion (the count paper [60]) has since settled the premise negatively: the classical apparent horizon is constraint-slaved — $\theta_+ = 0$ is elliptic, with no growth equation at any order (its eq. 2.9) — and does not roughen, so there is no height field and no self-organised-critical dynamics of the horizon. **Predictions P3–P5 are therefore withdrawn.** This leaves the core of the present paper untouched: they added no fitted parameter, were never required for the background result, and their retraction leaves the falsifiable predictions P1–P2 and the equation-of-state tests exactly as they stand.

Cross-horizon universality (speculative; count companion). Whether $\Delta = 1$ extends to black-hole horizons — giving $S_{\rm BH} \propto A^{3/2} \propto M^3$ and strongly altered primordial-black-hole evaporation — is a speculative extension logically separate from the core model, which requires the deformation only in the dark-energy sector. We do not adopt it: SEDE is consistent with the standard area law for astrophysical black holes, and on the evidence of this paper alone the black-hole-vs-cosmic-horizon asymmetry remains a genuine conceptual cost (supplementary material, §S2). The count companion paper [60] *offers a resolution* — a *state-dependent* count in which a horizon is area-law when undriven (a black hole, in equilibrium) and volume-law when sustainedly structure-driven (the cosmic horizon), which would make the asymmetry a predicted feature rather than a defect — but that reframing rests on a conjecture (developed there); absent it, the cost stands. The black-hole side is at least derivable: the structure entropy deposited at a black hole is negligible against the entropy it *creates* ($S_{\rm rad}/S_{\rm BH} \sim (M/M_{\rm P})^{-1/2}$), so no black hole — not even a primordial one built entirely by collapse at formation — activates volume. Either way the asymmetry sharpens a falsifier: an isolated *volume-law* black hole would imply a universal deformation. We pursue this in the count companion paper, and the Verlinde dark-matter connection (§7) in the foundations companion paper [46].

6.1. Tensor sector and standard-siren predictions ($\Xi_0 = 1$)

The propagation frame. In any Horndeski theory a tensor mode obeys $h'' + (2+\alpha_M)\,\mathcal{H}\,h' + c_T^2 k^2 h = 0$, so a running Planck mass ($\alpha_M \neq 0$) gives the GW luminosity distance an extra friction relative to the electromagnetic one, $d_L^{\rm GW}(z) = d_L^{\rm EM}(z)\,\exp\big[-\int_0^z \frac{dz'}{1+z'}\,\delta(z')\big]$ with $\delta = -\alpha_M/2$. This is captured by the two-parameter form $\Xi(z) \equiv d_L^{\rm GW}/d_L^{\rm EM} = \Xi_0 + (1-\Xi_0)/(1+z)^n$ [61], the GW-distance analogue of (w_0, w_a); GR is $\Xi_0 = 1$.

The FPAB result (derived). Expanding the FPAB action $\mathcal{L} = M_{\rm P}^2 R/2 + K(X,\phi) - G_3(X,\phi)\,\Box\phi$ to second order in a transverse-traceless tensor perturbation (exponential parametrisation $g_{ij} = a^2(e^{\epsilon h})_{ij}$), three facts are exact to all orders in the perturbation: $\sqrt{-g} = a^3$, $X = \dot\phi^2/2$, and $\Box\phi = -(\ddot\phi + 3H\dot\phi)$. Hence K and G_3 contribute **nothing** at $\mathcal{O}(\epsilon^2)$ for arbitrary functional forms — the entire quadratic tensor action comes from the Einstein–Hilbert term, and (with no integration by parts and, crucially, **no background-EOM input**)

$$\mathcal{L}_2 = \frac{M_{\rm P}^2 a^3}{8}\left[\dot h^2 - \frac{k^2}{a^2}h^2\right]. \tag{9}$$

Reading off: $M_*^2 = M_\mathrm{P}^2$ (constant $\Rightarrow \alpha_M = 0$), $c_T^2 = 1$ ($\alpha_T = 0$), zero graviton mass. This upgrades the $c_T = 1$ statement of §3.4 from an EFT structural fact to a from-scratch derivation, and because it uses no background equation it is independent of the fixed-point closure and survives any refinement of f_sat or γ. Therefore $\delta(z) = 0$ for all z and $\Xi_0 = 1$ **exactly,** n **unconstrained.** The result agrees with the general Horndeski tensor coefficients [28] evaluated at $G_4 = M_\mathrm{P}^2/2$, $G_5 = 0$; both tiers are verified symbolically in `src/experiments/tensor_sector/fpab_tensor_sector.py` (regenerated on build).

Two discriminants. (i) *Versus running-Planck-mass dark energy.* $f(R)$, $G_4(\phi)$ Horndeski and nonlocal RT gravity generically give $\Xi_0 \neq 1$ — the RT minimal model predicts $\Xi_0 = 0.93$, $n = 2.59$ [62]. On the (Ξ_0, w_0) plane ΛCDM is $(1, -1)$ and these theories sit at $(\neq 1, \cdot)$, whereas SEDE occupies $(1, -0.98)$ by force of construction, not tuning: a standard-siren measurement of $\Xi_0 \neq 1$ falsifies SEDE outright, while $\Xi_0 = 1$ with $w_0 \neq -1$ is its signature corner. (ii) *Versus action-level Barrow/deformed-entropy cosmologies.* Implementations of a volume-law entropy as a modification of the gravitational action generically induce $G_\mathrm{eff}(z)$ and $\alpha_M \neq 0$; SEDE's Clausius-flux routing (entropy deformation $\to$ effective source, then braiding completion) is what forces $\alpha_M = 0$. So Ξ_0 separates SEDE from its nearest thermodynamic-cosmology relatives, not only from ΛCDM. Consistency with GW170817 ($\alpha_T = 0$) is automatic, never tuned against the c_T bound.

Screening at the source. The $\sim 5\%$ scalar-force enhancement ($\mu_\infty(0) = 1.05$, §3.4) does not reach the binary: it is strictly a near-horizon effect, turning on only across $k_{50} = 0.70\,aH$ to $k_{90} = 2.1\,aH$ (§3.4) — i.e. at Gpc scales — and is absent at the sub-parsec scales of compact-binary sources, where the cubic braiding additionally Vainshtein-screens ($G_\mathrm{eff} \to G_N$, r_V growing with the G_3 normalisation and far exceeding any binary separation for cosmological-strength braiding). The chirp-mass–amplitude relation at the source is therefore unmodified.

Testability. A BAO-anchored dark-siren analysis forecasts Ξ_0 to $\sigma(\Xi_0) \approx 0.02$ (fixed redshift dependence) rising to ≈ 0.04 (marginalised over n, cosmology, and bias) with ~ 3500 events of $\sim 30\,M_\odot$ to $z = 0.5$ [63], tightening with ET/CE/LISA. So $\Xi_0 \neq 1$ is a sharp near-term kill switch; distinguishing $w_0 = -0.98$ from -1 by sirens alone is sub-percent in distance and a long-horizon confirmation, not a near-term falsification. The whole argument is captured on the (Ξ_0, w_0) plane — ΛCDM at $(-1, 1)$, running-M_* rivals off the $\Xi_0 = 1$ line, SEDE alone at $(-0.98, 1)$ — reproduced by `src/experiments/tensor_sector/fig_xi0_w0.py` in the repository.

Pre-recombination corollary. The same gate that suppresses the late-time source drives $f_{\rm sat} \to \gamma D^2/(1-e^{-\gamma})$ as $D \to 0$; at last scattering the pipeline growth gives $f_{\rm sat}(z=1089.9) = 3.4\times10^{-6}$ and $\rho_{\rm DE}/\rho_{\rm tot} = 1.0\times10^{-10}$ (canonical $E_{\rm SEDE,\,volume}$, $\Omega_m = 0.30$, $\gamma = 1.496$; `src/experiments/tensor_sector/prerecomb_earlyde.py`, consistent with the $\Omega_{\rm DE}(z=3) = 0.029$ of Fig. 5). SEDE therefore carries no early dark energy and the sound horizon r_s is exactly standard — the model bets entirely on the late-time side of the Hubble tension; a confirmed percent-level EDE detection would falsify the gate as the sole mechanism. And since $\alpha_M = 0$ holds at *all* epochs (background-EOM-free), the primordial tensor transfer function is exactly GR: any future feature in CMB B-modes or the primordial stochastic GW background is attributable to sources, never to SEDE propagation.

7. Relation to other sectors

Dark matter. SEDE does not derive dark matter; it assumes the standard cold component. What it *does* provide are falsifiable *relationships*: the equation of state is locked to $\Omega_{\rm m}$ through the fixed-point background — the canonical model giving $w_0 \approx -1$ with a crossing set by the growth gate (§3.2; the closed-form $w_0 \approx -0.85$ enters only as a structural diagnostic, not the relation) — and the coincidence problem is reframed: dark energy's activation gate $f_{\rm sat}(z)$ tracks the (dark-matter-dominated) growth of structure, so "why now" corresponds to $f_{\rm sat}$ crossing ≈ 0.5 near $z \approx 1$, shortly before matter–DE equality. These are testable links, not a unification, and we state them as such.

There is, however, a *principled* route to more, which we flag as a future direction and pursue in the foundations companion paper, not here. The same volume-law dark-sector entropy SEDE uses for dark *energy* is the object from which Verlinde's emergent gravity [64] derives apparent dark *matter* (baryonic displacement of the de Sitter volume-law entropy → the MOND scale $a_0 \sim cH_0$); if SEDE's $\Delta = 1$ horizon entropy and Verlinde's volume-law de Sitter entropy are the same object, one volume-law postulate could in principle source both dark energy and apparent dark matter. We do not adopt this here: Verlinde's proposal is contested (it struggles with clusters and lensing and lacks a covariant CMB/structure completion — the regime SEDE *is* tested in), and our own cross-check finds only an *order-of-magnitude* consistency — the CKN/flatness normalisation sets a_0 within an O(1) factor of observed $((c\sqrt{\Omega_{\rm DE}0}\ H_0)/2\pi \approx 0.73\ a_0$, $\sim$25% low), not a derivation. We record it as a motivated, partially-consistent target for the foundations companion paper; SEDE's data fits use standard cold dark matter throughout.

Quantum and cross-field. SEDE's horizon is fractal/volume-law while its *bulk* is standard GR (holographic-DE scope), so it predicts a deformed horizon without a Lorentz-violating, foamy bulk. Two real datasets are consistent with this restricted-scope picture (they constrain bulk effects SEDE does not predict, so they do not contradict it): the Fermi GRB 090510 31 GeV photon excludes Planck-scale linear Lorentz violation ($E_{\rm QG} >$ *several* $M_{\rm P}$), and the Fermilab Holometer's null result excludes bulk Planck-amplitude "holographic noise." The CKN reading gives a gravitating-vacuum cutoff $\Lambda_{\rm UV} \approx 4$ meV at the dark-energy scale (§8.2). We tested one cross-field extrapolation and it **failed**: the cosmological $\delta = 3/2$ does *not* transfer to galaxy-cluster velocity statistics — a real SDSS/Coma fit gives a Gaussian (Tsallis q = 1.01, not the $q \approx 1.5$ a naïve transfer would predict). We report this negative result plainly; the galaxy-dynamics link is the weakest cross-field claim and the data do not support it.

A contingent tension: cosmic birefringence. Minimal SEDE predicts $\beta = 0$, and this is *derived*, not merely asserted, within the model's stated effective description. Isotropic cosmic birefringence requires a parity-odd coupling of the photon to a dynamical pseudoscalar, $\mathcal{L} \supset -\frac{g}{4}\,\phi\, F_{\mu\nu}\tilde{F}^{\mu\nu}$, giving a rotation $\beta = (g/2)\Delta\phi$ accumulated along the line of sight. SEDE contains no such field: its dark-energy source $\rho_{\rm DE} = T_{\rm AH}\; s_{\rm grav}\; f_{\rm sat}$ is constructed *entirely* from parity-even background scalars — the expansion rate H, the horizon volume $V \propto R^3$, and the linear growth factor D(z) — so there is no pseudoscalar to play the role of ϕ, and no $\phi F\tilde{F}$ operator consistent with the construction, because a parity-even scalar cannot couple to the parity-odd density F$\tilde{F} = E \cdot B$ without an explicit CP-violating coefficient the model does not introduce. (The natural objection — that SEDE's phantom is *effective*, with no fundamental scalar, so an effective $\phi F\tilde{F}$ might still be generated — is answered by the same fact: what forbids the coupling is the *absence of a pseudoscalar degree of freedom*, which holds whether the description is fundamental or effective.) This is the same parity-even, minimally-coupled structure that fixes $\alpha_{\rm T} = 0$ (unmodified gravitational-wave speed) and zero gravitational slip — the three nulls share one origin. It also makes the discriminator physical: a pseudoscalar carrying $\phi F\tilde{F}$ generically also couples to the gravitational Chern–Simons term $\phi R\,\tilde{R}$, so a *confirmed cosmic, dark-energy-sourced* β would predict parity-violating (chiral) gravitational waves — an $\alpha_{\rm T}$-sector partner signal SEDE forbids, and an independent test. A nonzero isotropic rotation is reported (ACT DR6 $0.215^\circ \pm 0.074^\circ$, 2.9σ [65]). We treat this as a *real-but-contingent* threat, not a present falsification, and assess it in the §9 red-team (threat 2): the signal's cosmic, dark-energy-sourced nature is not established (miscalibration- degenerate $\alpha+\beta$; the cleanest $\beta \times$ *galaxy* channel is null at SEDE's zero), but a confirmed cosmic β would favour a single-axion rival that does SEDE's crossing and more.

Where SEDE sits among dark-energy models — and what it borrows. SEDE is a thermodynamic model, but it is not outside the standard effective-field-theory description of dark energy, and locating it there removes its main field-theoretic disadvantage. In the EFT of dark energy (Gleyzes–Gubitosi–Piazza– Vernizzi), SEDE occupies the **safe corner**: the bulk is general relativity (holographic scope), so the Planck-mass run rate $\alpha_{\rm M} = 0$ ($\dot{\rm G}/{\rm G} = 0$, §6/V7) and the tensor speed excess $\alpha_{\rm T} = 0$ — gravitational waves travel at c, so SEDE is GW170817-safe by construction ($|c_{\rm T}/c - 1| = 0$, against the $\sim 10^{-15}$ bound), having never switched on the higher Horndeski operators that GW170817 excludes. In the completed FPAB-SEDE cubic-KGB theory the dark energy is gradient-stable with $c_{\rm s}^2 \in [0.018,\ 0.18] > 0$ (§3.4) — near the sub-horizon $c_{\rm s}^2 = 1$ limit but with a computed scale-dependent force, $\mu_\infty \approx 1.05$, and unit slip — so SEDE sits at the ($\alpha_{\rm T} = \alpha_{\rm M} = 0$) safe corner of the EFT of dark energy. Field-theory representation and stability through the crossing: a *single* minimally-coupled scalar cannot reproduce the $w = -1$ crossing — for any P(X,ϕ), $\rho + p = 2X\ P_{\rm X}$, so the crossing forces $P_{\rm X} \to 0$ (verified: $1 + {\rm w} \to 0$ at $z \approx 0.18$), at which point the perturbation kinetic term $\alpha_{\rm K} \propto P_{\rm X} + 2X\ P_{\rm XX}$ passes through zero (a ghost) and $c_{\rm s}^2$ is ill-defined (the Vikman no-go). A *fundamental* ghost-free crossing nonetheless exists with braiding (Kinetic Gravity Braiding / Horndeski $G_3(\phi,{\rm X})\Box\phi$): the no-ghost determinant $Q_{\rm S} = \alpha_{\rm K} + (3/2)\alpha_{\rm B}^2$ stays positive even when $\alpha_{\rm K} \to 0$, because a nonzero $\alpha_{\rm B}$ carries it (we verify $Q_{\rm S} = 1.15 > 0$ at the crossing for a representative generic-KGB realisation, $\alpha_{\rm B} = 1.5\ \Omega_{\rm DE}$, with $c_{\rm s}^2 > 0$ throughout) [66, 67]; the calibrated FPAB-SEDE braiding is smaller, $\alpha_{\rm B} = b \cdot \Omega_{\rm X}$ with $b \approx 0.206$ (giving $\mu_\infty \approx 1.05$, §3.4), and remains ghost/gradient-stable when solved dynamically.

Status of the covariant completion. The completion is now explicit: **FPAB-SEDE** realises the $\rho_X \propto H\,g$ background as a local, luminal cubic-KGB action $\mathcal{L} = M_{\rm P}^2 R/2 + K(X,\phi) - G_3(X,\phi)\,\Box\phi$ (a finite 16-coefficient reconstruction; §3.4), whose linear perturbations are solved dynamically in Mochi-CLASS — ghost/gradient-stable, $c_T = 1$, $\gamma_{\rm slip} = 1$, $c_s^2 \in [0.018, 0.18]$. The route to this matters and is recorded honestly: identifying $\rho_{\rm DE} \propto H$ with a *single* bulk-viscous fluid **fails** — its principal-symbol speed uses the total enthalpy $(1 + w)\rho_{\rm DE}$, which changes sign at the crossing, giving a gradient instability — so the smooth-fluid (PPF, $c_s^2 = 1$) treatment is only the sub-horizon limit of a theory that *requires* the positive kinetic structure of braiding [68, 69]. This is the same classical $c_s^2 < 0$ instability established long ago for the bare $\rho_{\rm DE} \propto H$ ghost-dark-energy fluid [70]; the braided completion is precisely its known cure, so the pathology motivates the FPAB sector rather than sinking it. The cubic Galileon supplies exactly this with $c_T = 1$ automatic (no G_4/G_5); the shift-symmetric luminal crossing is obstructed and is resolved either by the designer FPAB reconstruction or by a shift-symmetry-breaking potential [71]. Two honest residuals remain: the reconstruction is trajectory-level (not symmetry-derived, and not a theorem that every phase-space solution is healthy), and the microscopic origin of the gate/scale/amplitude is unresolved (§8, §9). Neither affects the background/distance results of §§1–5. The full FPAB analysis and the finite cubic-KGB action are carried out in the companion observational-tests paper [37] (the manuscript repository here reproduces only the background/thermodynamic layer).

The structure-coupling — SEDE's most distinctive and least-confirmed ingredient (the structure-sourced- dark-energy idea itself traces to Gough's information dark energy, §7; what is specific here is the horizon-thermodynamic realisation) — is also free of the two instability classes that afflict coupled dark energy (V55). There is no gradient or ghost instability because $c_{\rm s}^2 = 1 > 0$; and there is no early-time large-scale instability because the interaction fraction $\Omega_{\rm DE}(z)\cdot|\text{d ln } \phi/d \text{ ln a}| \to 0$ at high z (it peaks at ≈ 0.38 near $z \approx 0.5$ and falls to $\sim 10^{-10}$ by recombination, since the dark-energy fraction itself vanishes as $f_{\rm sat} \to 0$). The coupling switches *off* in precisely the radiation/early-matter era where coupled-DE instabilities are seeded; the full perturbation evolution is validated end-to-end by CLASS-PPF ($f\sigma 8$ to 0.1%, §4.4). So two of the apparent disadvantages relative to other programmes in fact *invert*: against Horndeski SEDE is less constrained (GW-safe), and against interacting dark energy it is more stable.

framework	what SEDE shares or borrows	net position
ΛCDM (Planck)	empirical benchmark	**limit:** far less established — only DR3/Euclid closes this
quintessence (Ratra–Peebles)	dynamical DE; tracker/attractor crossing (§8.1, Result 12–13)	structure-locked w(z) with a non-tuned -1 crossing
CKN bound	the UV–IR energy bound fixes the DE scale (§8.2)	SEDE turns it into a cosmology; **limit:** needs the volume-law form on top
Li holographic DE	holographic DE with a horizon cutoff	uses the apparent horizon (no future-horizon causality/circularity); **limit:** volume-law is radical
standard Barrow-HDE [16]	the Barrow entropy $S \propto A^{1+\Delta/2}$	SEDE's Δ enters through the H-coupling $\lambda = 1 - \Delta/2$ *and* the structure gate, not the HDE density directly — so neighbouring Barrow-HDE fits that prefer $\Delta \ll 1$ [72] constrain a *different* model and do not exclude SEDE's $\Delta = 1$ (§9); **limit:** the volume-law endpoint is the radical input
Hu–Sawicki f(R)	viable late-time acceleration	keeps GR ($\alpha_{\mathrm{M}} = \alpha_{\mathrm{T}} = 0$): no screening/GW tension; **limit:** less mature

framework	what SEDE shares or borrows	net position
CPL	the data language; w(z) is CPL-degenerate in shape (§6)	adds the growth–expansion lock CPL lacks; **limit:** model-dependent, not a neutral fit
early dark energy	—	**limit:** does not address H_0 (out of scope, r_{d}-pinned, §9)
Jacobson [6]	Clausius → Friedmann; $\rho_{\mathrm{DE}} = T \cdot s$ (Appendix Appendix A.1)	applies horizon thermodynamics to DE; entropy-law change confined to the DE fluid (§8.3), not gravity
Horndeski / EFT-of-DE	the α-function language	sits at $\alpha_{\mathrm{T}} = \alpha_{\mathrm{M}} = 0$, $c_{\mathrm{s}}^2 = 1$ — GW170817-safe corner; less constrained, not less fundamental
Verlinde [64]	volume-law de Sitter entropy; elastic/memory response (§6, §8.1)	independent motivation for the postulate; $a_0 \sim cH_0$ consistent at O(1) (§7)
interacting DE	the dark-sector energy-exchange ledger (CHR, §6)	coupling fixed to growth, $c_{\mathrm{s}}^2 = 1$, off at high z $\Longrightarrow$ instability-free

The genuinely irreducible disadvantages, after these borrowings, reduce to two, and we state them plainly: SEDE is empirically younger than ΛCDM (the decisive tests are DR3/Euclid, §6), and it rests on the one volume-law postulate (§8.4) that motivation — Barrow, the GSL, CKN, Verlinde — supports but does not derive. Everything else in the table is either shared, borrowed, or a matter of scope.

Symmetry origin of the null predictions. SEDE's many "null" results are not independent assumptions: they follow from a *single* structural condition — *the dark sector adds no symmetry-breaking operator to the GR + Standard-Model effective action; it is a parity-even scalar* ($\rho_{\mathrm{DE}} = T_{\mathrm{AH}} \cdot s_{\mathrm{grav}} \cdot f_{\mathrm{sat}}$) *minimally coupled through* $T_\mu\nu$. Each precision null is then forced by a specific symmetry:

null prediction	symmetry that forces it
$\beta = 0$ (no cosmic birefringence)	parity in electromagnetism $\rightarrow$ no ϕ F $\tilde{F}$ coupling
$\alpha_{\mathrm{T}} = 0$, $c_{\mathrm{T}} = c$ (GW170817-safe)	2-derivative diffeomorphism-invariant gravity $\rightarrow$ no Chern–Simons / Galileon term
$\alpha_{\mathrm{M}} = 0$, $\dot{\mathrm{G}}/\mathrm{G} = 0$, no fifth force, equivalence principle	minimal (metric-only) coupling $\rightarrow$ no non-minimal $\xi\phi^2 R$
no Lorentz violation (Fermi GRB, Holometer)	bulk Lorentz invariance

So $\beta = 0$ and $c_{\mathrm{T}} = c$ are *the same fact* (§7 birefringence; absence of a parity-violating / modified-propagation sector), and the whole cluster — the reason SEDE passes every GW, fifth-force, birefringence, and Lorentz null *at once* — is one principle, not a list. The de Sitter endpoints add a fifth: $f_{\mathrm{sat}} \rightarrow$ const (with H $\rightarrow$ const) $\Longrightarrow$ a maximally-symmetric de Sitter phase $\Longrightarrow w = -1$ by symmetry (the dark-energy bracket of Result 12; Appendix Appendix A.7), with the crossing in between dynamical. We note the limit sharply: this symmetry argument protects SEDE's *safety*, not its *content*. The distinctive, load-bearing inputs — the volume law $\Delta = 1$ (a geometric/thermodynamic postulate, §8.4), the coupling $\gamma \approx 1.5$ (halo binding physics, §3.1), and the crossing redshift (a dynamical balance, Result 13) — are not symmetry-derivable; symmetry provides SEDE nothing there, and the one open postulate is untouched. The symmetry-protected cluster is precisely the part SEDE shares with any minimally-coupled smooth dark energy; what makes it SEDE remains thermodynamic and postulate-based.

Relation to information dark energy. The idea that dark energy is *sourced by the entropy/information produced as structure forms* is not original to SEDE, and we state the priority plainly. It is due to M. P. Gough, in a sustained but under-cited programme that we date — from a NASA-ADS author search — to 2006–2008 ("On the Similarity of Information Energy to Dark Energy," 2006; "Information Equation of State," 2008), developed through *Holographic Dark Information Energy* [13] to a recent observational analysis [14] that fits the stellar-mass-density history to the dark-energy equation of state from Planck/Pantheon+/DES/DESI, finding $\rho_{\rm DE} \propto \mathrm{SMD}(a)^p$ with $p \approx 0.5$ ($R^2 = 0.93$). In Gough's model dark energy is the Landauer energy of the information/entropy of star-heated gas and plasma. SEDE shares his central physics — structure's entropy *is* dark energy, with a self-regulating feedback (acceleration suppresses growth) that addresses the coincidence problem and predicts time-limited acceleration — and we credit it as the priority for the structure-sourced-dark-energy idea.

What distinguishes SEDE is the mechanism and a testable consequence, not the founding idea. (i) *Mechanism:* Gough uses the thermal/information entropy of hot baryons ($\propto$ T, residing in the bulk); SEDE uses the gravitational binding entropy of bound structure deposited on the apparent horizon ($\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav}$, Barrow volume-law $\Delta = 1$). These are physically different entropies in different places. (ii) *Prediction:* the two are not degenerate. SEDE's driver is the gravitational growth factor D(z) (dark matter), Gough's is the baryonic stellar-mass-density SMD(z) (star formation); computing both gives dark-energy histories that agree to $\lesssim 6\%$ for $z \lesssim 0.6$ — where supernova data are strongest, explaining why both fit current data — but diverge by up to $\sim$50% by z = 3, SEDE staying flatter (more high-z dark energy) because gravitational growth persists where star formation declines. The coincidence that both land on a "0.5" exponent is superficial: Gough's is a power of stellar mass density, SEDE's is the H-exponent $\lambda = 1 - \Delta/2$ from Barrow $\Delta = 1$. (iii) SEDE adds a dark sector with no *fitted* parameter and a w-crossing rather than a fitted CPL. So a clean way to separate them observationally is to ask whether dark energy tracks D(z) (gravity) or SMD(z) (stars), with the high-z dark-energy fraction (DESI $Ly\alpha$, future surveys) as the lever.

Other structure-/entropy-linked programmes differ more sharply and are catalogued in the novelty audit: entropic-force [20, 17] and the generalized-entropy holographic-DE family (Barrow/Tsallis/Kaniadakis and their unifications [73, 74, 75]) source the term from the horizon (structure downstream — and standard entropic-force DE in fact *struggles* with structure growth, which SEDE's design inverts; note also the thermodynamic-consistency caveat for pairing $T \propto H$ with a non-area entropy [76], which SEDE's effective-fluid framing evades); backreaction (Buchert, Räsänen [77, 78]) and timescape (Wiltshire [79]) make acceleration an apparent inhomogeneity effect with no real dark energy; matter-creation/CCDM (Lima) replaces dark energy with a particle-creation pressure; and cosmologically-coupled black holes (Farrah et al.) use one special structure via vacuum interiors. A systematic priority pass over both INSPIRE-HEP and NASA ADS (citation-ranked, multiple phrasings; documented in the novelty audit) returns no precedent for SEDE's specific construction — "dark energy = gravitational binding entropy of bound structure on the apparent horizon" (`abs:("dark energy" "binding energy" horizon)` yields zero relevant hits in either database) — nor for its terminology. So while the *general* structure-sourced-dark-energy idea belongs to Gough (since ∼2006–2008), SEDE's *gravitational-horizon mechanism, its gravity-driven (D(z), not SMD) signature, and its no-fitted-parameter (prescription-fixed) dark sector* appear to be new. We also ran Gough's model through our identical CAMB-marginalised pipeline (its $\rho_{\mathrm{DE}} \propto \mathrm{SMD}^{0.5}$ expressed as a w(a) table): both are viable, with SEDE moderately preferred ($\Delta\mathrm{DIC} \approx -4.7$ vs $Gough's \approx -0.5$ relative to ΛCDM), and Gough's fit favouring a higher H_0 — consistent with his Hubble-tension claim. In summary, SEDE is a distinct, more-derived realisation of Gough's idea, decided from ΛCDM and from Gough alike only by the high-z dark-energy and growth data to come (§6).

The most recent independent convergence on the structure-sourced idea is Pandey [80], who shows that the *configuration entropy* of the collapsing matter field, treated as an irreversible backreaction within standard general relativity, supplies an effective energy density that accelerates the late expansion while a dissipative correction to the velocity flow suppresses growth — easing the H_0 and S_8 tensions without touching the CMB sound horizon. This reaches SEDE's *qualitative* conclusion by an independent route, and we welcome it as corroboration of the direction; it differs from SEDE in *where the entropy lives and what it produces.* Pandey's entropy is the matter-sector configuration entropy acting as a GR backreaction — not the gravitational horizon entropy that in SEDE *is* the dark energy ($\rho_{\mathrm{DE}} = T_{\mathrm{AH}} \cdot s_{\mathrm{grav}}$) — and it neither derives a specific equation of state (no w-crossing) nor introduces a horizon deformation Δ. A parallel *generalised mass-to-horizon entropy* programme [81, 82] links a modified horizon entropy to structure growth through a modified-Friedmann route, and so inherits the BBN Δ-bounds (§8.3) that SEDE's holographic-fluid scope evades. SEDE is thus distinguished among these 2026 counterparts by its holographic locus, its *derived* phantom-crossing w(z), and its single geometric input $\Delta = 1$.

One might ask whether SEDE's structure-sourced dark energy is merely a reparametrization of cosmological backreaction. Via the Buchert–Larena–Alimi morphon correspondence [83], the SEDE background maps exactly onto averaged sources (the kinematic backreaction Q_{D} and averaged spatial curvature $\langle \mathrm{R} \rangle$_D); at background level the two are degenerate, as they must be, since the morphon is built to reproduce any $\rho_{\mathrm{DE}}(z)$. They are not the same physics. Reproducing SEDE's $w_0 \approx -1$, $\Omega_{\mathrm{DE}} \approx 0.7$ requires $|\Omega_{\mathrm{Q}}| \approx 0.35$ (and $|\Omega_{\mathrm{R}}| \approx 1.0$), whereas the *perturbative* kinematic backreaction over the SH0ES volume is $\Omega_{\mathrm{Q}} \sim 10^{-5}$ — four to five orders smaller; whether non-perturbative structure can lift this to O(1) is a long-standing open question [77, 79] on which SEDE does not rely. SEDE is degenerate with a morphon whose amplitude no realistic perturbative structure field supplies; its O(1) amplitude derives instead from the horizon conjugate identity $\rho_{\mathrm{DE}} \propto H^{2\lambda}$ f_{sat}, with the structure variance entering only as the O(1) normalised control field shaping $f_{\mathrm{sat}}(z)$.

The two frameworks coincide in trigger (growth of σ^2) and separate in amplitude mechanism, and the resolution of the residual degeneracy is itself amplitude-dependent. The discriminating observable — the scale-dependence a genuine inhomogeneous average imprints on the *locally*-inferred H_0, which the homogeneous SEDE ansatz does not share — bifurcates. A *large* backreaction (the $|\Omega_{\mathrm{Q}}| \approx 0.35$ SEDE would need if it were genuine backreaction, i.e. the timescape regime) is decisively testable today: it implies S/N $\sim 10^2$–10^3 scale-dependence at DESI DR2, and existing bulk-flow / Hubble-bubble analyses already bound any anomalous local-H_0 variation at the $\sim$1% level [84]. That null is precisely what empirically licenses SEDE's homogeneous dark-energy reading over a large-backreaction one — a constraint the data already supply. The *physically expected* backreaction ($\Omega_{\mathrm{Q}} \sim 10^{-5}$), by contrast, gives S/N $\sim 10^{-2}$ and is permanently sub-threshold: LSST gains only $\sim\sqrt{}$volume (a factor $\sim$2), not the $\sim 10^4$ required. SEDE's distinction from small physical backreaction is therefore not a promised future measurement but a stated scope condition — the homogeneity ansatz together with the Q_{D}–$\langle \mathrm{R} \rangle$_D integrability lock. Neither the degeneracy nor its resolution runs through $\rho_{\mathrm{DE}}(z)$ or w(z).

The closest *entropic-acceleration* relative is the General Relativistic Entropic Acceleration (GREA) programme of García-Bellido and Espinosa-Portalés [85, 86], which — like SEDE — removes Λ and sources late-time acceleration from the entropy of the cosmic horizon, with the Helmholtz free energy $F = U -$ TS (not matter alone) as the gravitational source and entropy *production* as the engine. The differences are sharp and observable:

GREA uses the boundary (area) horizon entropy and a single parameter (the horizon-to-curvature ratio), with the growth driven by the horizon's own *expansion*, giving a quintessence-like w(a); SEDE uses the volume entropy ($\Delta = 1$) gated by structure (f_{sat}), which locks w(z) to the growth history and yields a phantom crossing, with a dark sector carrying no *fitted* parameter. The connection is quantitative — SEDE's structure-driven entropy production maps to a positive (second-law) effective bulk viscosity that peaks at the gate-activation epoch, casting SEDE as a *structure-gated* member of the GREA family — and it is the deformation Δ that discriminates them: were SEDE's residue to resolve to the area branch ($\Delta = 0$), its phenomenology would collapse toward GREA's. So SEDE is not "GREA plus a volume postulate": the structure-gating is itself a falsifiable signature — the w(z)$\leftrightarrow$growth phase-lock (§6) that neither GREA's expansion-driven w(a) nor a kinematic CPL fit shares. We develop this relationship, and SEDE's foundational status, in the foundations companion paper [46].

8. The quantum-gravity underpinning

The energy-conservation account of §2–§3 took two horizon properties as given: the magnitude, $\rho_{\mathrm{DE}} \sim \rho_{\mathrm{crit}}$ rather than the QFT vacuum's M_{P}^4, and the H-scaling $\lambda = 1/2$, which follows from the horizon entropy being volume-law ($S_{\mathrm{grav}} \propto V_{\mathrm{AH}}$; §3.3) with *no deformation parameter*. This section supplies their quantum-gravitational basis — it is the support beam under the construction, and a reader content to take the two scales as inputs may treat it as optional. (Its full development — the reduction route map, the driven non-equilibrium-steady-state account, and the closure analysis — is the subject of the forthcoming foundations companion paper [46]; none of the cosmological results of §§1–7 depends on it, and the foundations companion's mechanisms, where invoked above, are flagged as conjectural.) We argue why the horizon entropy is volume-law rather than area-law, anchor the magnitude to the holographic bound and the flatness normalisation, and isolate the single statement that remains a postulate. Throughout, the Barrow parameter Δ appears only to *label* the area-to-volume interpolation ($S \propto A^{1+\Delta/2}$: $\Delta = 0$ area-law, $\Delta = 1 \iff$ volume-law) and as the falsifiable test of the volume postulate (measured in §5.6, forecast in §6); it is not an input to the model.

8.1. Why volume-law, not area-law — consistency motivations (none a derivation)

The dark-sector entropy is the coarse-grained entanglement entropy contained in the apparent-horizon volume, $S_{\text{grav}} \propto V_{\text{AH}} \propto R_{\text{AH}}^3$ — equivalently the Barrow endpoint $\Delta = 1$ ($A^{1+\Delta/2} = A^{3/2} = R^3$). The open physical question is sharp: why the volume law, rather than the Bekenstein–Hawking *area* law? We give three considerations in the main line and collect three further, more speculative ones in the supplementary material. None is a microscopic proof; together they motivate the volume endpoint and make the area-law branch disfavoured in the diagnostics.

1. Geometric ceiling caps the deformation (Result 9). The horizon's fractal dimension is $d_{\text{H}} = 2 + \Delta$, capped at the space-filling value $d_{\text{H}} = 3$ for a two-surface in three-space. Since (for a volume law in embedding dimension d) $\Delta = 2/(d-1)$, the case d = 3 gives $\Delta = 1$. *(This fractal-dimension reading is a heuristic for the bound and is no longer load-bearing: the count companion paper [60] derives the same ceiling* $\Delta \leq 1$ *as a theorem from a finite per-site leg budget — the bond-cut capacity obeys* $C \leq \kappa{\cdot}|\Omega|$ *identically — with no fractal premise, and shows the count is not in fact a Hausdorff dimension. That is consistent with the dof-counting framing of the supplementary material (§S2); the ceiling conclusion below is unchanged.)* We are precise about what this delivers: the ceiling bounds the deformation, $\Delta \leq 1$ — it does *not* select $\Delta = 1$. Landing *on* the endpoint requires, in addition, that the entropy is monotone in Δ (so the cap is saturated); since $\rho_{\text{DE}} = T \cdot s$ makes the free energy $F = E - \text{TS}$ vanish identically, entropy is the governing potential and the generalised second law is consistent with — is a fixed point of — the saturated (volume-law) state. We stress that this is a consistency check, not a selection principle: "total entropy does not decrease" is not "the parameter takes its entropy-maximising value," and the saturation of the cap *is* the volume-law postulate (§8.4) in effective form, not an independent derivation of it. Volume-law scaling is the signature of a thermalised, maximally-scrambled state (Page curve, eigenstate thermalisation).
2. The holographic bound fixes the scale, not the form. The CKN energy bound (§8.2) pins the *magnitude* but not the exponent — its own state count gives a *sub*-area-law ($\delta = 3/4$), the opposite of an enhancement. The volume law is therefore a statement about the horizon's quantum

state (typical / thermalised), which the energy bound cannot supply. This isolates the open content to one sentence: *the horizon is in a typical, volume-law-entangled state.*

3. The area-law branch is disfavoured directly by current data. *Saturating* the CKN bound — the natural "no-deformation" alternative — is algebraically the area-law case $\Delta = 0$, whose equation of state ($w_0 = -0.49$) and large early-dark-energy fraction are disfavoured in the conditional diagnostics of §5.6. On current data the volume branch is favoured over the area-law alternative, not merely an optional choice — though, as §5.6 stresses, the robust marginalised measurement is still to come. The push is *monotone* in Δ: the exact identity $\Omega_{\rm DE}(z) = \Omega_{\rm DE}0 \cdot f_{\rm sat}(z) \cdot E(z)^{-\Delta}$ (from $\rho_{\rm DE} \propto H^{2-\Delta} f_{\rm sat}$) and E $>$ 1 in the past make a larger Δ suppress the early-dark-energy fraction more strongly — $\Omega_{\rm DE}(z = 3)$ falls from 0.127 ($\Delta = 0$) through 0.060 ($\Delta = 0.5$) to 0.029 ($\Delta = 1$, below $\Lambda{\rm CDM}'s$ 0.035). So the geometric ceiling of point 1 ($\Delta \leq 1$) and the early-dark-energy lever here *cooperate*: the ceiling bounds Δ from above and the CMB pushes it to that bound, the volume endpoint. This remains a diagnostic, not a derivation — but it is a third, independent reason the data sit at $\Delta = 1$.

Two further consistency motivations — a *driven non-equilibrium steady state* (the structure gate maintains the volume-law state despite a long scrambling time) and the *independent volume-law entanglement entropy* of Verlinde's emergent-gravity programme — are collected in the supplementary material. Each makes the postulate less *ad hoc*; none derives it. (A third, once listed here — a roughening / Edwards–Wilkinson analogy for the horizon surface — is withdrawn: the count companion paper [60] shows the apparent horizon is constraint-slaved and does not roughen, eq. 2.9.)

8.2. Why $\rho_{\rm DE} \sim \rho_{\rm crit}$ — the CKN bound and the vacuum-energy scale

The deformation fixes the *scaling* of dark energy with H but not its *magnitude*. The magnitude comes from the Cohen–Kaplan–Nelson (CKN) holographic energy bound (Principle 0): an effective field theory in a region of size L cannot hold more energy than a black hole of that size, $\rho\ L^3 \lesssim M_{\rm P}^2$ L, i.e. $\rho \lesssim M_{\rm P}^2/L^2$. With the IR cutoff at the cosmic horizon, L = c/H, this gives

$$\rho_{\rm DE} \lesssim M_P^2 H^2 \sim \rho_{\rm crit} \ \Rightarrow\ \Omega_{\rm DE} \sim \mathcal{O}(1). \tag{10}$$

The naïve QFT vacuum, $\rho \sim M_{\rm P}^4$, overshoots this by $\rho_{\rm P}/\rho_{\rm crit} \sim 10^{122}$ — the cosmological-constant problem. SEDE never invokes the QFT vacuum: dark energy is the horizon's conjugate thermal energy density, and the CKN relation supplies the horizon-scale upper bound $\rho_{\rm DE} \lesssim M_{\rm P}^2 H^2$. This is a bound, not an equality: the present amplitude is fixed by spatial flatness ($\Omega_{\rm DE}0 = 1 - \Omega_{\rm m}$), in the same bookkeeping sense that ΛCDM fixes $\Omega_\Lambda 0$ — once the volume-law branch is chosen, the bound is saturated to O(1) by that normalisation today. We note explicitly that the saturation is a present-day statement: since $\rho_{\rm DE} \propto H$ (volume law) while the bound $\propto H^2$, the ratio $\rho_{\rm DE}/(M_{\rm P}^2 H^2) = \Omega_{\rm DE}(z)$ falls at higher z (0.70 → 0.22 → 0.03 at z = 0, 1, 3), so dark energy sits *below* the CKN bound in the past — consistent, as CKN is an upper bound. So CKN *addresses* the vacuum-energy-scale problem (it explains why the natural gravitational horizon scale is $M_{\rm P}^2 H^2 \sim \rho_{\rm crit}$, not $M_{\rm P}^4$) rather than deriving the magnitude from first principles. The same bound, read as a UV–IR relation, gives a gravitating-vacuum cutoff $\Lambda_{\rm UV} = (M_{\rm P}\, H_0)^{1/2} \approx 4$ meV — numerically the observed dark-energy scale (§7).

8.3. The seam, BBN-safety, and the holographic scope

A naïve *microscopic* volume-law entropy density would be Planckian, so a naïve $\rho_{\rm DE} = T \cdot (S_{\rm vol}/V)$ would overshoot $\rho_{\rm crit}$ by $(M_{\rm P}/H)^\Delta \sim 10^{61}$ — the square root of the 10^{122} hierarchy (Result 11). The same factor appears as a clean *no-go*: the Bekenstein bound on the horizon's energy $E = \rho_{\rm crit}\, V_{\rm H}$ gives $S \leq 2\pi {\rm RE}/\hbar{\rm c} = \frac{1}{4}(A/\ell_{\rm P}^2)$ — exactly the Gibbons–Hawking *area* law (verified to within the O(1) geometric factor) — so the volume law cannot be obtained from any maximum-entropy or energy-bound argument; it *exceeds* the bound by precisely $R/\ell_{\rm P} \sim 10^{61}$. This sharpens rather than weakens the postulate: it proves the missing ingredient is a *state/counting* input (a Planckian-density, volume-law horizon), not an energy bound, and fixes its exact size. The *same* factor controls three apparently separate issues — this seam, the BBN bound on a Barrow-modified Friedmann equation, and the modified-gravity-vs-holographic fork — which are one object. The resolution is the holographic-DE scope adopted in §2: the deformation acts only on the dark-energy fluid (magnitude from CKN, scaling from Barrow), not on the gravitational sector. The early-universe expansion is then exactly standard ($\Delta N_{\rm eff} = \Delta Y_{\rm p} = 0$, verified; BBN-safe), and the naïve $T \cdot (S/V)$ is precisely the modified-gravity path the scope forbids.

This is the load-bearing assumption of the model. In the *modified-gravity* reading — Barrow entropy deforming the Friedmann equations [87] — BBN forces $\Delta \lesssim 1.4 \times 10^{-4}$ [88] and SN+BAO+H(z) give $\Delta \sim 10^{-4}$ (the area law at 2σ [89]), which read literally would exclude SEDE's $\Delta = 1$. Those bounds do *not* apply to SEDE: it keeps standard gravity and lets the volume-law entropy act only on the dark-energy fluid via $\rho_{\rm DE} = T \cdot s$ (§2, Appendix Appendix A.1), so there is no extra Friedmann term, no BBN deviation, and Δ is unconstrained by them. The headline $\Delta = 1$ thus rests on this single scope choice — legitimate and standard (it *is* the holographic-dark-energy framework), but not free: a reader who rejects it recovers the tightly-bounded modified-gravity Barrow cosmology instead.

8.4. The one irreducible postulate

> Postulate (volume-law horizon). The cosmic apparent horizon carries the coarse-grained *entanglement* entropy of its volume, $S_{\rm grav} \propto V_{\rm AH} \propto R^3$ — a constant horizon entropy density, i.e. a thermalised, volume-law (non-extensive) state — rather than the Bekenstein–Hawking area law.

The results in §8.1–§8.3 are bounds, consistency arguments, or motivations *given* this postulate; the postulate itself is not derived from a microscopic theory of quantum gravity. §8.1 *reduces* it (the CKN bound fixes the scale, so the only open content is the horizon's entanglement *state*), *motivates* it (non-extensivity, the GSL, the driven steady state, the Edwards–Wilkinson universality class, and the independent volume-law entanglement entropy of Verlinde's emergent gravity), and shows the *area-law branch is disfavoured by current diagnostics* — but a microscopic state count that delivers the volume law remains the model's single open foundational item (§9). Two points of emphasis. First, the model carries **no fitted deformation parameter**: Δ is not an input but the falsifiable *test* of this postulate — current data already prefer the volume endpoint and disfavour the area-law branch (§5.6, a conditional diagnostic), with the robust marginalised measurement to come from DESI DR3 + Euclid (§6), alongside cross-horizon predictions (a deformed black-hole entropy $S \propto A^{3/2}$; §7, presented as speculative). Second, the open question is now stated plainly — *why is the horizon entropy volume-law (coarse-grained entanglement) rather than area-law?* — rather than hidden in a value of a tunable parameter. We do not claim to have derived quantum gravity; we claim to have reduced its dark-energy content to one testable statement and removed it as a fitted parameter.

On the plausibility of a volume law (genre, not derivation). A volume-law horizon entropy is a strong hypothesis, but volume-law *entanglement* entropy is at least not exotic in cosmological field theory. The entanglement entropy of a field subregion — area-law in the de Sitter adiabatic vacuum — grows and saturates to a *volume law* after a de Sitter→flat transition, ending in a partially thermalised (generalised-Gibbs) state [90]; volume-law entanglement also arises from field–curvature coupling [91]. We cite these only to establish *genre and plausibility* — that volume-law entropy is a real feature of cosmological QFT — and we explicitly resist over-reading them. Two caveats: (i) these results concern the entanglement entropy of field *subregions* under specific transitions, not the cosmic-*horizon* entropy SEDE invokes; and (ii) their assignment (de Sitter = area-law, flat = volume-law) is *not* SEDE's and in fact runs opposite to it (SEDE places the volume law in the de Sitter brackets). So the literature makes a volume-law hypothesis less exotic, but it neither *anticipates* nor *derives* the specific postulate — which remains the model's one open foundational item (§9). (We note for completeness that Padmanabhan's holographic-equipartition route, $dV/dt = N_{\rm sur} - N_{\rm bulk}$, does *not* help here: its surface count is area-law, $N_{\rm sur} = A/\ell_{\rm P}^2$, and it reproduces the *standard* area-law Friedmann equation — it is the baseline SEDE departs from, not independent support for the volume law.)

The postulate, decomposed (after the foundations companion [46]). "Volume-law" is not one claim but four of very different status: (i) *state* — the horizon degrees of freedom are maximally entangled (thermal), not in a low-entanglement ground state; (ii) *form* — the entropy scales as the volume, $S \propto V$; (iii) *scale* — the magnitude is $\rho_{\rm DE} \sim \rho_{\rm crit}$, not $\rho_{\rm Planck}$; and (iv) *count* — the *number* of horizon degrees of freedom grows as the bulk ($N \propto R^{d-1}$) rather than the boundary ($N \propto R^{d-2}$). Because S $\sim$ (state factor) $\times$ N and the Barrow deformation enters as $S \propto A^{1+\Delta/2} \propto R^{(d-2)(1+\Delta/2)}$, the empirical content — the exponent Δ — is carried entirely by the *count* (area $\to \Delta = 0$, volume $\to \Delta = 1$ in d=4); the state only sets whether the prefactor is maximal. The foundations companion argues that the first three are *derivable or reducible* — the state from de Sitter maximal scrambling (a Majorana-SYK diagonalisation gives Page-value saturation, $S/S_{\rm Page} \approx 0.99$), the scale from the CKN bound (§8.2), the form from the driven non-equilibrium steady state — leaving the *count* as the one genuine residue. For *why the count is volume* ($\Delta = 1$) *rather than area* ($\Delta = 0$), that companion advances a driven-NESS selection conjecture: a strongly long-range horizon ($\alpha = 1 \leq d$) driven by structure formation relaxes to a volume-law steady state, and a

Kibble–Zurek scan of the swept, tilted free energy is used to *test* — rather than assume — that the volume basin is the one dynamically selected at horizon "birth". This is a conjecture, not a derivation; it is why $\Delta = 1$ is *ranked above* $\Delta = 0$ (§8.1) while remaining the falsifiable test axis (§5.6, §6). The full route map (five derivation routes A–E, the SYK and causal-set comparisons) is collected in that companion and, in summary, in the supplementary material (§S2–§S3).

9. Discussion: open problems and assessment

The one open item. Every dark-sector input is fixed by a stated choice (§3) given the single postulate that the horizon is volume-law-entangled (§8.4). That postulate is motivated by gravity's non-extensivity and the second law but is not derived from a microscopic state count; it is the model's one irreducible input. The canonical-vs-microcanonical reframing (which gives area-law $\lambda = 1$ or Planckian-volume $\lambda = 1/2$) does not give both for free, but it is not symmetric: the de Sitter horizon has negative specific heat ($C = -2S$) and gravity is strongly long-range ($\alpha = 1 \leq d$; supplementary material, §S3), so by the standard non-additive-systems result [92] the two ensembles are *inequivalent* and the canonical one is ill-defined for the isolated self-gravitating horizon. This *removes the canonical/area-law horn* as ill-defined for a long-range, negative-specific-heat horizon, leaving the *microcanonical accounting* — but it does not by itself select the volume count: within the microcanonical ensemble, whether the *state count* is itself volume is the dS-holography question of §S3 — a genuine open problem, and a *falsifiable* one (§6). The systematic route map of §S3 narrows it: the *state* (maximal entanglement) is first-principles, the *form* ($S \propto V$) reduces to thermalisation, the *scale* is CKN, and the irreducible residue is purely the bulk-vs-boundary count — one sharply-posed driven-NESS conjecture (a strongly-long-range horizon, driven by structure formation, settles in the volume-law steady state rather than the area-law equilibrium) which, if established, would close the residue *and* resolve the black-hole/cosmic-horizon asymmetry (§7). The foundations companion supplies the two microscopic maps this needs (§S3) — connectivity $\to$ cooperative coupling $J \geq J_{\mathrm{c}}$, and deposition $\to$ drive (the growth-*ratio* structure drive *populates* the volume capacity selected at the horizon's birth) — reformulating the static counting postulate into a physical statement: the area$\leftrightarrow$volume free energy is bistable, the volume branch is selected at the birth bifurcation and locked by long-range hysteresis, and the drive populates it. This is a *reformulation, not a reduction* in assumption count, stated as the well-posed successor to the postulate, not a finished proof. (The alternative once entertained — a self-organised-critical z *ratchet with observable CHR signatures — is* excluded*: the count companion [60] shows the apparent horizon is constraint-slaved and does not roughen, §6.)

Background robustness (model-independent H(z)). Against the standard 31-point cosmic-chronometer compilation — the cleanest model-independent expansion-history data, with no sound-horizon calibration — the model fits with $\Delta\chi^2 = +0.04$ relative to flat ΛCDM *at equal parameter count* ($\Omega_\mathrm{m} = 0.325$, $H_0 = 67.9$ vs 0.320, 68.1 km s^{-1} Mpc^{-1}; γ fixed at its halo-statistics value throughout; script `src/experiments/run_deficit_data.py`). The background is indistinguishable from ΛCDM on H(z) alone: the gate's imprint on the $z \geq 0$ expansion shape is absorbed by an (Ω_m, H_0) shift to the ˜0.2% level (an oΛCDM fit to the model's own curve returns Ω_k,eff $= 0.000$) — which is precisely why the model's discriminating power resides in the growth and tensor sectors (§6) rather than in the background.

The deep-future second law selects a saturation endpoint. Extending the background beyond the fitted range exposes a structural constraint invisible to data (script `src/experiments/run_gate_selection.py`). On the accelerating attractor the gate's remaining fill decays at the linear-growth rate $\propto e^{-2N}$ (N = ln a), slower than matter's e^{-3N}, so any gate still strictly filling at the attractor forces H to undershoot H_∞ and the horizon entropy to overshoot its ceiling transiently — a sub-percent ($\approx 0.25\%$), strictly future ($a \gtrsim 6.5$) violation of the generalized second law. No theorem is contradicted: the null energy condition fails there transiently, which is exactly where GSL proofs do not apply, and the entire observable past is monotone. But for a model whose fundamental object *is* horizon entropy, requiring the second law globally is natural, and it bounds the saturation endpoint: the gate must complete its filling at x* $\lesssim 1.8$, where $x = \sigma_8^2(z)/\sigma_8^2(0)$. Every member of the capped family x* $\in [1, 1.8]$ is identical to the uncapped exponential gate over $z \geq 0$ to machine precision — the bound carries no observational cost — and its maximally conservative member (x* $= 1$) places completion at the present epoch, an independent, sharper restatement of the coincidence problem. (The Bousso-bound hypothesis H5, stated in full-saturation units, is automatically satisfied by any CDF gate and is not this bound; the second-law cap is a new, strictly stronger statement.)

Scope and tensions. H_0 is out of SEDE's scope (it is r_d-pinned; the model does not address the distance-ladder tension and we do not claim it does). *S8: the completed FPAB-SEDE theory computes* $\sigma 8(0) = 0.811$*, which* ***raises*** *growth relative to matched* ΛCDM *(0.800) — SEDE therefore sits at the high-*$\sigma 8$ *end and does not ease the weak-lensing S8 tension; the earlier "eases/neutral on S8" reading (based on the retracted smooth-fluid* $\sigma 8 \approx 0.76$*) is superseded, pending the full perturbative refit (§3.4, §4.4).* The structure-sourcing *mechanism* itself — the defining SEDE claim that f_{sat} tracks growth — is only weakly confirmed: a geometry-vs-structure consistency test (E1; the null test $\rho_{\text{DE}} = G[D(z)]$ with f_{sat} read independently off the *expansion*, DESI $D_{\text{H}}/r_{\text{d}} \to$ H(z), and off *structure growth*, RSD $f\sigma 8 \to \sigma 8(z)$) tracks at z $<$ 1 but is RSD-limited at z $>$ 1 and remains inconclusive (current data: χ^2/dof = 0.40 in the RSD-error budget, 1.40 in the precise-geometry budget; Figure 9). A referenced cross-check sharpens this reading and argues *undecided* rather than *disfavoured*: when the low-z (RSD $f\sigma 8$) and high-z (ACT DR6 CMB-lensing) growth amplitudes are each reduced to the $\sigma 8(0)$ they imply *through the same model's own growth*, SEDE and ΛCDM come out indistinguishable — both carry the identical $\approx 1.8\sigma$ low-vs-high-z amplitude offset that is the known S8 tension ($\Delta tension = +0.05\sigma$). The same verdict holds at the full *likelihood* level on real galaxy weak lensing: the DES Y1 cosmic-shear data vector ($\xi\pm$, CAMB-driven cobaya likelihood, SEDE's w(z) via PPF) gives a matched-cosmology $\Delta\chi^2(\text{SEDE} - \Lambda\text{CDM}) = +0.4$ and a shear-preferred $\sigma 8$ that shifts by only $+0.2\sigma$ between the models. So the precise-geometry "1.40" reflects that *shared* S8 offset, not a SEDE-specific failure of the mechanism; the discriminating power lies in the growth-side deformation *shape* (§6), which current growth data do not yet resolve, not in the linear-growth amplitude, which even the best high-z probe available now cannot separate. The pipeline is implemented and staged — it runs now and is ready to apply the moment Euclid/LSST weak-lensing tomography sharpens the structure leg, at which point the forecast of §6 makes it decisive ($\sim 4\sigma$). Its data vector and CONFIRM/REFUTE thresholds are pre-registered (pre-registration §Appendix F) so the staged test cannot be tuned after the data arrive. We have separated this mechanism into a derived, parameter-free part — the growth–expansion lock P2, which follows from $f_{\text{sat}} = f(D^2)$ with no extra assumption (§6), testable with exactly that data. (The near-critical layer once flagged as a further kill test is withdrawn — §6.) But a *derived consistency relation* is not yet a *confirmation*, and the mechanism stays unconfirmed until the lock P2 is measured. We stress this is a *near-term* gap, not a permanent one: a Fisher forecast

(§6) shows the growth-side *deformation-amplitude* test reaches $\sim 4\sigma$ with combined Euclid/LSST weak lensing + CMB-lensing + clusters + RSD — so the distinctive claim is decidable on the same timescale as the amplitude Δ, not deferred indefinitely. Two honest qualifications on that $\sim 4\sigma$. First, it measures the deformation Δ through the growth *shape*; it is a re-measurement of Δ in the growth channel, not by itself a confirmation that $f_{\rm sat}$ is *sourced* by structure (E1). Second, the narrower growth–expansion *lock* — that a matched-geometry model must reproduce SEDE's $f\sigma 8$ — is CPL-degenerate and carries essentially no $f\sigma 8$ signal (a $w_0 w_{\rm a}$ doppelgänger reproduces SEDE's $f\sigma 8$ to $\ll 1\%$), so the mechanism is *not* discriminated in the $f\sigma 8$ amplitude. The one place SEDE departs from ΛCDM/CPL at O(1) is instead the largest-scale ISW/ISW × *galaxy* channel, where the gate's coupling $\beta = d \ln f_{\rm sat}/d \ln D \approx 0.9$–$1.5$ is *not* (1+w)-suppressed (§6). That an O(1) dark-sector coupling escapes the (1+w) suppression of clustering quintessence and enhances the low-ℓ ISW is itself a known feature of coupled dark energy [56, 57, 58]; what is specific to SEDE is the *origin* of that coupling as a structure-growth gate ($\beta = d \ln f_{\rm sat}/d \ln \mathrm{D}$), which fixes its redshift dependence — a near-term ISW × *galaxy* test rather than a claim of a new effect. And the cosmic-birefringence signal (§7) is a live tension. We emphasise the framing: SEDE's *phenomenology* (w crossing −1, the volume-law endpoint) is moderately favoured in the current compiled-data analysis with an encouraging out-of-sample check, but its *distinctive mechanism* (the $f_{\rm sat}$ gate tracking structure growth) is the unconfirmed frontier.

Threats from the recent literature (a red-team). We list the strongest recent results that could damage SEDE, with our assessment of each.

1. *Holographic DE disfavoured by full CMB / the early–late tension.* Wu et al. [38] find that holographic and Ricci DE are disfavoured by full ACT DR6 + DESI through an early-vs-late tension that compressed (R, ℓ_A) priors hide — a direct challenge, since SEDE's headline ΔDIC used compressed CMB. We tested it. Marginalised full primary likelihood: a CAMB-in-the-loop fit on the full Planck *plik*$_{\rm lite}$ TTTEEE+*lowℓ* TT/EE, with the five cosmological parameters and the Planck nuisance *all free* (not θ^*-fixed), gives $\Delta\chi^2(\mathrm{SEDE} - \Lambda\mathrm{CDM}) = -0.43$, the preference sitting in the high-ℓ peak structure (-0.37) — *negative*, i.e. mildly favouring SEDE, not the large *positive* $\Delta\chi^2$ the HDE early–late tension would produce; the best-fits are physical ($\Omega_{\rm m} = 0.295/0.315$ for SEDE/ΛCDM). This is corroborated by two θ^*-matched

single-point proxies — the same $plik_{\rm lite}$ ($\Delta\chi^2_{\rm CMB} = -0.50$) and the official ACT DR6 CMB-lensing likelihood ($\Delta\chi^2_{\rm ACT}$-lens $= -0.32$, validated against the package's fiducial $\chi^2 = 14.06$). ACT DR6 lensing re-evaluated at the marginalised primary best-fit (not a fixed fiducial) gives $\Delta\chi^2_{\rm ACT}$-lens $= -0.31$ (ΛCDM 14.10 $\rightarrow$ SEDE 13.79), and the low-ℓ ISW-sensitive channel is flat (-0.03). All are $|\Delta\chi^2| < 1$: SEDE does not inherit the standard-HDE tension, because it has *less* early dark energy than ΛCDM ($f_{\rm sat} \rightarrow 0$, $\Omega_{\rm DE}(z = 3) \approx 0.029 < 0.035$), not the runaway high-z behaviour that breaks original HDE. Full non-lite plik (foregrounds sampled): we have since run the *complete* high-ℓ plik TT-TEEE likelihood with all 15 foreground/calibration nuisance parameters sampled at their Planck priors (the full $plik_{\rm lite}$ likelihood), the gold-standard primary CMB — it gives $\Delta\chi^2(\text{SEDE} - \Lambda\text{CDM}) = -0.33$ (TTTEEE $-$ 0.27, $lowl$.TT $-$ 0.23, lowl.EE $+0.17$), confirming the foreground-marginalised $plik_{\rm lite}$ result (-0.43) to within the minimiser noise. So letting the foregrounds float does not reveal a hidden tension — the standard-HDE early–late wall is simply absent for SEDE. *Residual scope:* only the ACT DR6 *primary* multifrequency likelihood remains not run (ACT *lensing*, the DE-sensitive channel, is; §above) — formally deferred under the pre-registered trigger (pre-registration §Appendix E; `act_dr6_mflike` + multi-GB products not installed), with the falsifiable committed expectation $|\Delta\chi^2| < 1$ given three concordant CMB datasets already at that level; a full MCMC for ΔDIC is deferred (*expected* $\approx \Delta\chi^2$, $p_{\rm D}$ shared). The full primary likelihood is also folded *into the joint* (§5.1).

2. *Cosmic birefringence — a contingent threat, with a rival waiting if it is confirmed.* SEDE predicts $\beta = 0$; a nonzero *rotation* is reported (ACT DR6 2.9σ [65]; Planck $\sim$0.3°), and *if* it is cosmic and dark-energy-sourced, a single axion explains β + *acceleration* [93] and an interacting axion sector reproduces the phantom crossing $+\beta$ + *dark* matter [94] — a parsimony threat, since the rival does SEDE's crossing and more with one field. But the cosmic, DE- sourced nature is *not established*: the measurement is of $\alpha + \beta$ (miscalibration-degenerate), Planck and ACT disagree by $> 3\sigma$, SPIDER is incompatible with the coupling [95], and the $\beta \times$ LSS cross-correlation is null ($p \approx 37\%$, [96]) — at SEDE's predicted zero, not the axion (§7). So this is a *contingent* threat, not a present falsification, and the cleanest current channels favour $\beta = 0$. The discriminator is the pair {w-crossing, β}; $\beta = 0$ is

itself symmetry-derived (§7), tied to GW-safety, so a confirmed cosmic β would also predict GW parity violation.

3. *The DDE signal may be a low-z SN systematic.* Huang, Cai & Wang [97] show the DESI evolving-DE preference is biased by an external low-z SN sample in DESY5 and drops to $< 2\sigma$ once corrected. *Defence (partial):* the bias is DESY5-specific ("in contrast to the uniform behaviour of PantheonPlus"), and SEDE's headline uses Pantheon+; more decisively, SEDE's leave-one-probe-out drops supernovae entirely and stays preferred ($\Delta\chi^2 \approx -4.85$), and $F_{\rm AP}$/*tracer*-LOO are SN-free. SEDE is not SN-driven — but if the field's evolving-DE evidence softens, SEDE's *context* erodes, and its "moderate, not established" framing must hold.
4. *The nearest Barrow-HDE fit prefers* $\Delta \ll 1$. Luciano, Paliathanasis & Saridakis [72] fit standard Barrow-HDE to DESI DR2 + SN + CC and obtain $\Delta < 0.47$–0.54 ($\Delta = 0$ within 1σ, mild positive tendency) — which, read literally, disfavours SEDE's $\Delta = 1$. *Defence (shown, not asserted; Fig. 10):* both models share the scaling $\rho_{\rm DE} \propto H^{2-\Delta}$; the *only* difference is SEDE's structure gate $f_{\rm sat}(z)$, and the gate is exactly what decouples Δ from the observable the Barrow-HDE bound rests on. At the same $\Delta = 1$ the two models differ sharply: SEDE's gate ($f_{\rm sat} \to 0$ at high z) holds the early-dark-energy fraction to $\Omega_{\rm DE}(z = 3) = 0.029$ — CMB-safe — whereas un-gated Barrow-HDE has $\Omega_{\rm DE}(z = 3) = 0.147$ ($5\times$ larger) and $w_0 = -0.77$ with no $w = -1$ crossing, versus SEDE's $w_0 = -1.0$ in DESI's crossing quadrant. So the Δ-constraint derived from *Barrow-HDE's* early-DE/EOS does not transfer: SEDE's $\Delta = 1$ produces viable observables precisely where Barrow-HDE's $\Delta = 1$ does not, and in SEDE's own construction $\Delta = 1$ is data-preferred over $\Delta = 0$ (§5.6). The bound constrains a related model, not SEDE — and Fig. 10 shows the orthogonality rather than asserting it.

Net: the two threats that targeted the model itself (full-CMB tension; the Barrow-Δ bound) are answered by direct tests or a stated model distinction; the two that target the *context* (birefringence; the low-z SN systematic) are real and partially outside SEDE's control — birefringence especially.

Evidential status. SEDE is a falsifiable, fixed-parameter, Λ-free alternative to ΛCDM, moderately preferred (calibrated p < 0.0015, 0/2000; $\sim$2–3σ across pipelines) with an encouraging out-of-sample check, resting on one open postulate and a few stated theoretical choices. It is stronger than

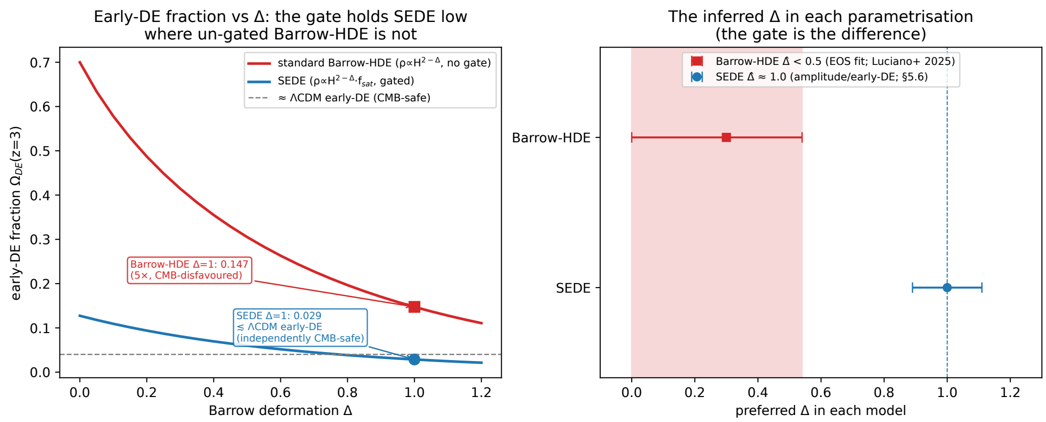

Figure 10: Why the standard Barrow-HDE bound $\Delta \ll 1$ does not constrain SEDE's $\Delta = 1$. Both models share the holographic scaling $\rho_{\rm DE} \propto H^{2-\Delta}$; the only difference is SEDE's structure gate $f_{\rm sat}(z)$. *Left:* the early-dark-energy fraction $\Omega_{\rm DE}(z = 3)$ — the quantity the CMB and the standard-HDE EOS fit actually constrain — versus Δ. Ungated Barrow-HDE (red) carries large, strongly Δ-dependent early DE ($\Omega_{\rm DE}(z = 3) = 0.15$ even at $\Delta = 1$, rising to 0.70 at $\Delta = 0$), so its fits pin $\Delta < 0.5$; SEDE's gate (blue) drives $f_{\rm sat} \to 0$ at high z, holding $\Omega_{\rm DE}(z = 3) \approx 0.03$ — at or below the ΛCDM/CMB-safe level (grey dashed) — for *all* Δ. At $\Delta = 1$ the two models differ five-fold (0.029 vs 0.147), and SEDE additionally crosses $w = -1$ ($w_0 = -1.0$) where un-gated Barrow-HDE does not ($w_0 = -0.77$). *Right:* consequently the data prefer orthogonal Δ in the two parametrisations — $\hat{\Delta} < 0.5$ for Barrow-HDE (EOS/early-DE; [[72]]) and $\hat{\Delta} \approx 1.0$ for SEDE (amplitude/early-DE; §5.6) — with no contradiction. The gate is the difference.

"indistinguishable from ΛCDM" and weaker than "established." Its value is twofold: a *physical story* (no fine-tuned Λ, w(z) from Ω_{m}, the coincidence reframed) and a *decidable prediction* (the volume law, $\Delta = 1$). The data, not the theory, will settle it.

10. Conclusions

We have presented Structural Entropy Dark Energy: a fixed-parameter dark-energy model in which dark energy is identified with the conjugate thermal energy density of a volume-law-entangled cosmic horizon and activated through a structure-growth gate. The model is a single Friedmann fixed-point equation with no fitted dark-sector parameter — λ, w(z), γ, and c_{s}^2 are each fixed by a stated theoretical choice rather than tuned — that replaces the cosmological constant by a structure-gated, horizon-coupled term. It addresses the vacuum-energy-scale problem (the scale is the CKN holographic bound, normalised by flatness, not the QFT vacuum) and reframes the *timing* of the coincidence problem (the activation gate tracks the late-time maturation of the structure-growth history); it does not explain the present-day dark-energy *magnitude*, which — as $\rho_{\mathrm{DE}} \sim M_{\mathrm{P}}^2 H_0^2$ is numerically the same coincidence — it shares with Λ. In a marginalised, CAMB-in-the-loop joint analysis it is moderately preferred over ΛCDM (ΔDIC ≈ -3.5 to -4.7); the preference is calibrated at $p < 0.0015$ (0/2000; $\sim$2–3σ across pipelines; not obviously flexibility) with an encouraging pre-DESI → DESI out-of-sample check, and its equation of state crosses -1 in DESI's evolving-DE quadrant ($\sim 0.5\sigma$ from ΛCDM, so disfavoured alongside it if the steep DESI signal firms up). On the growth side the completed perturbation sector *raises* late-time structure ($\sigma 8 \approx 0.811$ vs 0.800), so we make no claim on the S8 tension. Its quantum-gravitational content reduces to one falsifiable postulate — a volume-law horizon entropy. The sharp future test is near: under the stated Fisher assumptions, DESI DR3 and Euclid could constrain the effective deformation Δ to $\sigma \approx 0.09$, separating SEDE's volume-law horizon from a smooth one at a nominal $\sim 11\sigma$ — a Fisher-optimistic figure; the honest framing is a discrete $\Delta = 1$ versus $\Delta = 0$ model-selection test. SEDE is a moderate preference, not established — and, crucially, it is decidable.

Data availability statement

The code, data vectors, and analysis pipelines that support the findings of this study are openly available at `https://github.com/spsingularity/`

`sede-cosmology`, with a tagged release archived at Zenodo, DOI 10.5281/zenodo.21525524. The repository ships the model package (the background E(z), w(z), the structure gate $f_{\rm sat}$, and the halo-binding γ derivation), the public data vectors, the figure generators, and the **CAMB-in-the-loop inferential pipeline** — the canonical ($\lambda = 1/2$) background $\rightarrow r_{\rm drag}/r_{\rm star} \rightarrow$ BAO + compressed-CMB joint likelihood, the marginalised MCMC (ΔDIC, $p_{\rm D}$), the per-probe decomposition, the false-preference mock calibration (Fig. 6), the out-of-sample split, the *tracer*/γ robustness scans, and the sha256-locked prediction set (`results/predictions.json`, verified byte-for-byte by `src/scripts/gen_predictions.py`). A single entry point (`reproduce.py`) regenerates the figures, and the inference drivers under `src/experiments/` regenerate the headline model-comparison numbers; an automated verification suite checks the background layer numerically. Two components are *not* regenerated locally and are reported from the companion observational-tests analysis [37]: the full-primary-CMB (plik) joint ($\Delta\text{DIC} = -3.5$, robust part $\Delta\langle\chi^2\rangle \approx -3.0$, eroded to -2.3 by the companion's direct Planck-lensing $\phi\phi$ test; this paper's own fold-in $\Delta\chi^2 = -3.17$), which needs the Planck likelihood data, and the FPAB-SEDE cubic-KGB / Mochi-CLASS growth sector ($\sigma 8$, μ_∞, the ISW ratio, gravitational slip); the nested-sampling evidence is reported via a Laplace estimate here (full dynesty run in the companion). The observational inputs are third-party public datasets (DESI DR2, Pantheon+/DES-5YR/Union3, Moresco cosmic chronometers, Gold-2018 $f\sigma 8$, compressed Planck, eBOSS DR16), available at their cited original sources and retrieved via the documented loaders.

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Appendix A. Key results, prescriptions, and assumptions

This appendix documents the six load-bearing items, labelled by their actual epistemic status — one defining ansatz (A.1), supporting derivations (A.2, A.4, A.5), a fixed prescription (A.3), and a motivated postulate (A.6). The headings give that status; the parenthetical "Result/Prescription/ Motivation N" numbering corresponds to the repository's theorem labels (retained there for code stability, *not* a claim that each is a mathematical theorem). We work in units $c = \hbar = k_B = 1$ and write $8\pi G = 1/M_P^2$. The source statements are Results 1, 3, 4C, 5, 8, 9; the remaining items (2, 4, 4B, 6, 6B, 10–13, 5D) are catalogued at the end with one-line statements.

Appendix A.1. The conjugate horizon-fluid ansatz: $\rho_{DE} = T_{AH} \cdot s_{grav}$

Claim. SEDE *identifies* the dark-energy density with the conjugate energy of the horizon entropy, $\rho_{DE} = T_{AH} \cdot s_{grav} \cdot f_{sat}$, with s_{grav} the (volume-law) horizon entropy density and $f_{sat} \in [0,1]$. The Clausius relation supplies the *dynamics*; the absolute density $\rho = T \cdot s$ is the model's defining ansatz, motivated by horizon thermodynamics — not a theorem.

Setup. At the flat-FRW apparent horizon $R_{AH} = 1/H$, the dynamical Cai–Kim temperature is $T_{AH} = (H/2\pi)(1 - \epsilon/2)$, $\epsilon {\equiv} {-}\dot{H}/H^2$. We adopt the Gibbons–Hawking limit $T_{AH} = H/2\pi$ ($\epsilon \to 0$), the leading temperature appropriate for the de Sitter-attractor dark-energy sector; the dynamical $(1-\epsilon/2)$ correction is sub-leading where dark energy matters and is disfavoured in full (§2, §6), so it does not enter the calibrated E(z). The gravitational entropy is the coarse-grained entanglement entropy in the horizon volume, $S_{grav} \propto V_{AH} \propto R_{AH}^3$ (§3.3, §8.1), i.e. a *constant* entropy density $s_{grav} = s_0$ (in contrast to the Bekenstein–Hawking area-law density $s_{AH}{=}\frac{3}{4}M_P^2 H \propto H$, which would instead give the area-law model $\rho_{DE} \propto H^2$ — the branch disfavoured in the conditional diagnostics of §5.6).

Construction. *Step 1 (Clausius — the rigorous, dynamical content).* Hayward's first law for a dynamical horizon, $\delta E = T_{AH}\, dS_{AH} + W\, \delta V$ with work density $W = (\rho - p)/2$, applied to a shell of matter crossing the horizon in time dt with flux $-dE = (\rho + p)H$ V dt, gives the Clausius relation $(\rho + p)H\ V = T_{AH}\ \dot{S}$_AH. This relates *changes* and is standard horizon thermodynamics.

Step 2 (Bousso bound as motivation for a bounded f_{sat}). The covariant entropy bound gives $S_{matter} \le A/(4l_P^2) = S_{AH}$ on any light-sheet bounded by the horizon. This *motivates* interpreting $f_{\rm sat}$ as a bounded effective saturation fraction; in SEDE we impose $0 \leq f_{\rm sat} \leq 1$ and $f_{\rm sat}(0) = 1$ with the redshift dependence supplied by the halo prescription (A.2/A.3), and we do not identify $f_{\rm sat}$ literally with $S_{\rm matter}/S_{\rm AH}$ (consistent with the small binding-entropy budget, Appendix Appendix F).

Step 3 (the SEDE identification — the ansatz). SEDE posits the dark-energy density to be the conjugate energy of the volume-law horizon entropy,

$$\rho_{\rm DE} \equiv T_{AH}\, s_{\rm grav}\, f_{\rm sat} = \frac{H}{2\pi}\, s_0\, f_{\rm sat} \;\propto\; H\, f_{\rm sat}. \tag{A.1}$$

Fixing the one constant s_0 by spatial flatness [$\rho_{\rm DE}(0) = \Omega_{\rm DE}0\, \rho_{\rm crit},0$, $f_{\rm sat}(0) = 1$] gives

$$\rho_{\rm DE}(z) = \Omega_{\rm DE0}\, \rho_{\rm crit,0}\, \frac{H}{H_0}\, f_{\rm sat}(z) = \Omega_{\rm DE0}\, \rho_{\rm crit,0}\, E(z)\, f_{\rm sat}(z), \tag{A.2}$$

i.e. the fixed-point term $\Omega_{\rm DE}0 \cdot E \cdot f_{\rm sat}$ of the boxed closure equation (§2) — the volume-law ($\lambda = 1/2$) scaling, *not* the area-law H^2. This is an identification motivated by horizon thermodynamics, not a derivation: the Clausius relation (Step 1) fixes the dynamics, while the absolute density is the postulate (§8.4).

Step 4 (scale / coincidence). $\rho_{\rm DE} \sim M_{\rm P}^2 H_0^2 \sim (10^{-3}\ eV)^4$ is small because $H_0/M_{\rm P} \sim 10^{-61}$. The CKN bound (§8.2) *explains* why this horizon scale is $\sim\rho_{\rm crit}$ — as an upper bound, taken here to be saturated to O(1) by the flatness normalisation above — replacing the QFT-vacuum conflation behind the 10^{120} problem rather than deriving the magnitude from scratch. ■

Appendix A.2. Result 3 — the structure gate $f_{\rm sat}(z)$

Claim. $f_{\rm sat}(z) = [1-e^{-\gamma x(z)}]/[1-e^{-\gamma}]$, where $x \equiv \sigma 8^2(z)/\sigma 8^2(0) = D^2(z)$ (D the linear growth factor, D(0) = 1).

Setup. We model the cumulative activation gate directly (dimensionless, avoiding any mix of area- and volume-law densities) through a normalised growth-weighted deposition kernel $K(t) \propto (d\sigma 8^2/dt)\, e^{-\gamma\sigma 8^2/\sigma 8^2(0)}$ — the structure-growth rate $d\sigma 8^2/dt > 0$ times a Boltzmann factor $e^{-\gamma x}$ for the exponential rarity of massive halos (γ from A.3) — taking $f_{\rm sat}$ to be its normalised cumulative from early times to epoch z (cosmic time t),

$$f_{\rm sat}(z) = \frac{\int_0^{t(z)} K\, dt}{\int_0^{t_0} K\, dt}, \tag{A.3}$$

so that $f_{\rm sat}(\infty) = 0$ (early, t→0) and $f_{\rm sat}(0) = 1$ (today, $t = t_0$) by construction (the closed-form gate below is this cumulative-kernel form, not a relaxation ODE).

Proof. The kernel integrated over cosmic time is K dt = $(d\sigma8^2/dt)$ $e^{-\gamma\sigma8^2/\sigma8^2(0)}$ dt = $e^{-\gamma u}$ $d\sigma8^2$ with $u \equiv \sigma8^2/\sigma8^2(0)$; the Hubble factors cancel and the integral reduces to one over u. With u running from 0 (z′→∞, no early structure) to $x(z) \equiv \sigma8^2(z)/\sigma8^2(0) = D^2(z)$,

$$f_{\rm sat}(z) = \frac{\int_0^{x(z)} e^{-\gamma u}\, du}{\int_0^1 e^{-\gamma u}\, du} = \frac{1 - e^{-\gamma x(z)}}{1 - e^{-\gamma}}, \qquad x(z) = D^2(z), \tag{A.4}$$

where the denominator (x = 1, i.e. z = 0) enforces $f_{\rm sat}(0) = 1$. Checks: $f_{\rm sat}(0) = 1$; $f_{\rm sat}(z \gg 1) \to 0$ as x → 0; $f_{\rm sat} \in$ [0,1] for $x \in$ [0,1]. The form is the CDF of a truncated exponential of rate γ on $x \in$ [0,1] — the normalised activation fraction reached by epoch z. ■

Appendix A.3. Prescription 4C — the fixed structure coupling $\gamma \approx 1.496$

Claim. The entropy weight exponent is p = 5/3, giving $\gamma = d$ ln $\Sigma_{\rm S}^{(5/3)}/d$ ln $\sigma8^2 = 1.496$ (the derivative is w.r.t. the variance $\sigma8^2$, the gate argument $x = D^2 = \sigma8^2/\sigma8^2(0)$ of A.2; equivalently $\frac{1}{2}$ d ln $\Sigma_{\rm S}/d$ ln $\sigma8 = 1.496$, i.e. d ln $\Sigma_{\rm S}/d$ ln $\sigma8 = 2.993$). A self-similar reduction gives the closed form $\gamma = (p-1)\langle 1/\alpha\rangle$_Σ with $\alpha \equiv -d$ ln σ/d ln M the effective spectral slope over the entropy-weighted mass range; it reproduces the exact mass-function integral to 0.1% (the binding-energy companion note, §7). Entropy is deposited by the $\nu_{\rm S} \approx 2.4$ (∼2σ) clusters at $M_{\rm S} \sim 10^{14.5}\ M\odot/h$, giving an independent cluster-abundance handle; p = 1 gives ≈ 0.27 (∼5× below p = 5/3), a weak gate versus the real one.

Proof. In the SEDE horizon-fluid ansatz (A.1), $f_{\rm sat}$ is the effective fraction of the cosmic-horizon entropy capacity that is gravitationally activated, and $\rho_{\rm DE} = T_{\rm AH} \cdot s_{\rm grav} \cdot f_{\rm sat}$ — so the entropy that defines the $f_{\rm sat}$ weighting is accounted in the *horizon* reservoir at the *horizon* temperature $T_{\rm AH} = H/2\pi$. By Clausius, heat Q deposited into a reservoir at temperature T adds entropy Q/T. When a halo of mass M virialises it releases gravitational binding energy $E_{\rm bind} \propto M^{5/3}$ (NFW; $R_{\rm vir} \propto M^{1/3}$; verified slope 1.64). Under the SEDE prescription this released binding energy is accounted in the horizon entropy ledger at the horizon temperature $T_{\rm AH}$ (we do not claim a demonstrated transport mechanism), so the entropy assigned in this ledger is

$$\Delta S_{AH} = \frac{E_{\rm bind}}{T_{AH}} \propto \frac{M^{5/3}}{H}. \tag{A.5}$$

Because $T_{\mathrm{AH}} \propto H$ is identical for every halo at a given epoch (mass-independent), the per-halo mass weight is $M^{5/3}$: p = 5/3. With Σ_{S}=∫ $M^{5/3}(dn/d$ ln M)d ln M and a Sheth–Tormen mass function (COLOSSUS, EH98 transfer, $M \in [10^{10}, 10^{16}]\ M_{\odot}$),

$$\gamma = \frac{d \ln \Sigma_S^{(5/3)}}{d \ln \sigma_8^2} = \tfrac{1}{2}\,\frac{d \ln \Sigma_S^{(5/3)}}{d \ln \sigma_8} = 1.4964 \approx 1.50 \tag{A.6}$$

(the derivative is w.r.t. the variance $\sigma 8^2$ because the gate's argument is $x = D^2 = \sigma 8^2/\sigma 8^2(0)$; equivalently d ln Σ_{S}/d ln $\sigma 8$ = 2.993). The earlier p = 1 (Result 4B) divided E_{bind} by the *halo's* virial temperature $T_{\mathrm{vir}} \propto M^{2/3}$, which is the entropy of the halo's own internal state — correct for the halo, wrong for the entropy added to the cosmic horizon (where f_{sat} lives). The heat is the same; the reservoir, and hence the temperature, is the horizon's. ■

Appendix A.4. Result 5 — the structural-EOS diagnostic (the w_0 side-note, not canonical w(z))

Claim (structural EOS). $w_{\mathrm{struct}}(0) = -1+\gamma/[3(e^{\gamma}-1)]$, well-approximated by the closed-form algebraic expression $w_{\mathrm{alg}} = (4\Omega_{\mathrm{m}}/3 - 1)/(1 - \Omega_{\mathrm{m}})$. (This is the side-note structural estimate of §3.2, not the model's prediction, which is the canonical crossing w(z).)

Proof. From $\rho_{\mathrm{DE}}(z) = \rho_{\mathrm{crit}}$,0 Ω_{DE}0 $f_{\mathrm{sat}}(z)$ (A.1) with $f_{\mathrm{sat}}(x)$, $x = D^2$, the structural EOS in the variance variable is $w_{\mathrm{struct}}(0) = -1 + (1/3)$ $df_{\mathrm{sat}}/dx|$_{x=1}. Differentiating the Result-3 form,

$$\left.\frac{df_{\mathrm{sat}}}{dx}\right|_{x=1} = \frac{\gamma e^{-\gamma}}{1 - e^{-\gamma}} = \frac{\gamma}{e^{\gamma} - 1}, \tag{A.7}$$

so $w_{\mathrm{struct}}(0) = -1 + \gamma/[3(e^{\gamma} - 1)]$. At $\gamma = 1.4964$ this is 0.432, giving $w_{\mathrm{struct}}(0) = -0.856$ (0.33σ from DESI DR2). Numerically $\gamma/(e^{\gamma} - 1) = 0.432 \approx \Omega_{\mathrm{m}}/(1 - \Omega_{\mathrm{m}}) = 0.451$ at the observed parameters, so the structural value is approximated to ∼4% by the closed-form expression $w_{\mathrm{alg}} = (4\Omega_{\mathrm{m}}/3 - 1)/(1 - \Omega_{\mathrm{m}}) = -0.849$ (0.21σ from DESI). This is a numerical approximation in the $x = D^2$ convention, not an exact identity. (The interacting-fluid effective EOS, from Raychaudhuri + Friedmann at z = 0, is phantom, $w_{\mathrm{fluid}}(0) \approx -1.15$, because f_{sat} is still rising today; the canonical $\lambda = 1/2$ background realises this as a crossing of -1, §5.1.) ■

Appendix A.5. Result 8 — the H-coupling $\lambda = 1 - \Delta/2$

Claim. With the Barrow entropy $S = (A/A_0)^{1+\Delta/2}$, the conjugate horizon-fluid ansatz gives $\rho_{\rm DE} \propto H^{2\lambda}\, f_{\rm sat}$ with $\lambda = 1 - \Delta/2$.

Proof. A quantum-gravitationally rough horizon carries the Barrow entropy $S = (A/A_0)^{1+\Delta/2}$, $\Delta \in [0,1]$, in place of the area law. With $A = 4\pi/H^2$ and $V_{\rm AH} \propto H^{-3}$, the Barrow-deformed entropy density (written s_Δ to distinguish it from the Bekenstein–Hawking area-law density $s_{\rm AH} \propto H$) is

$$s_\Delta = \frac{S}{V_{AH}} \propto \frac{(H^{-2})^{1+\Delta/2}}{H^{-3}} = H^{1-\Delta}, \tag{A.8}$$

$$\rho_{\rm DE} = T_{AH}\, s_\Delta\, f_{\rm sat} \propto H \cdot H^{1-\Delta} \cdot f_{\rm sat} = H^{2-\Delta} f_{\rm sat} \equiv H^{2\lambda} f_{\rm sat}, \qquad \boxed{\lambda = 1 - \tfrac{\Delta}{2}.} \tag{A.9}$$

This is Barrow holographic dark energy with the Hubble IR cutoff ($\rho_{\rm DE} \propto L^{\Delta-2}$, L = 1/H). The endpoints are $\Delta = 0 \Longrightarrow \lambda = 1$ (the area-law branch, disfavoured) and $\Delta = 1 \Longrightarrow \lambda = 1/2$ (the volume-law postulate, A.6). λ is therefore not a free parameter; it is set by the Barrow deformation. **Scope:** the Barrow exponent multiplies only the $f_{\rm sat}$-gated $\rho_{\rm DE}$, *not* the gravitational sector, so H(z) at BBN equals the standard matter+radiation value to ~8 figures ($\Omega_{\rm DE}/\rho_{\rm tot} \sim 10^{-16}$ at z $\sim 4 \times 10^9$); the modified-gravity BBN bound $\Delta \lesssim 10^{-4}$ does not apply, and constant $\Delta = 1$ is BBN-safe. ■

Appendix A.6. Motivation 9 — $\Delta = 1$ *from a geometric ceiling and the GSL*

Claim. $\Delta = 1$ (the volume-law endpoint) is *strongly motivated* — not fitted to cosmological data — by two independent principles; it is the central postulate (§8.4), not a microscopic derivation.

Pillar 1 — geometric ceiling. The Barrow deformation is the fractal (Hausdorff) dimension of the horizon surface, $d_{\rm H} = 2 + \Delta$. A 2-surface embedded in 3-space cannot exceed the space-filling dimension $d_{\rm H} = 3$, so $\Delta \leq 1$, with $\Delta = 1$ the extremum where the area-measure becomes a volume-measure ($S \propto A^{3/2} \propto R^3$). This is Barrow's defining range. *(Superseded, not withdrawn: the count companion paper [60] derives the same ceiling* $\Delta \leq 1$ *as a theorem from a finite per-site leg budget,* $C \leq \kappa{\cdot}|\Omega|$*, and shows the count is not literally a Hausdorff dimension — so the fractal picture here is a heuristic for a bound that now has a stronger, premise-free proof.)*

Pillar 2 — thermodynamic attractor. Three premises: (P1) the DE sector is purely entropic — SEDE posits $\rho_{\rm DE} = T_{\rm AH}\, s_{\rm grav}$, so the free-energy density $F = \rho_{\rm DE} - T_{\rm AH}\, s_{\rm grav} = 0$ identically; the governing potential is therefore the *entropy*, not $F = E - \mathrm{TS}$ (this is the load-bearing step: with a nonzero Δ-dependent free energy, equilibrium would extremise F and $\Delta = 1$ would not follow). (P2) the generalised second law: $S_{\text{total}} = S_\Delta + S_{\text{matter}}$ is non-decreasing, S_{matter} independent of Δ. (P3) Δ is a thermalising horizon degree of freedom (the standard Jacobson–Padmanabhan assumption). By (P1)–(P2), maximising S_{total} over Δ reduces to maximising $S_\Delta = (A/A_0)^{1+\Delta/2}$, whose derivative is

$$\frac{dS_\Delta}{d\Delta} = \frac{S_\Delta}{2} \ln \frac{A}{A_0} + [\text{back-reaction through } A(\Delta)]. \tag{A.10}$$

The explicit term carries $ln(A/A_0) \approx ln(10^{122}) \approx 280 > 0$ (a vastly super-Planckian horizon). The back-reaction (Δ shifts $\rho_{\text{DE}} \propto H^{2-\Delta}$, hence H, hence A) cannot flip the sign: it *vanishes* at $z = 0$ ($H = H_0 \Longrightarrow \ln \mathrm{H} = 0 \Longrightarrow \partial_\Delta H^{2-\Delta} = 0$) and is $\sim 1/280$-suppressed elsewhere. Hence $dS_\Delta/d\Delta > 0$ on all of [0,1]; a strictly increasing function on a closed interval attains its maximum at the right endpoint. With the ceiling $\Delta \leq 1$ (Pillar 1) the constrained maximum is uniquely $\Delta = 1$, and by (P3) a thermalising horizon relaxes to it.

Conclusion. Under the additional assumptions that Δ is a thermalising horizon degree of freedom (P3) and that entropy maximisation over the Barrow family is the correct variational principle (P1), the GSL argument is a consistency check — the endpoint $\Delta = 1$ is a GSL *fixed point* — rather than a selection principle: the geometric ceiling $d_{\text{H}} = 3$ supplies the bound $\Delta \leq 1$, and saturation of that cap *is* the volume-law postulate in effective form (via $\lambda = 1 - \Delta/2$ this gives $\lambda = 1/2$). These assumptions motivate but do not microscopically derive the endpoint (§8.4). Because S_Δ is monotone in Δ, the same argument favours $\Delta = 1$ at all epochs (the adopted constant Δ). The current conditional diagnostics prefer values near $\Delta(0) \approx 1.0$–1.1 (§5.6), consistent with the endpoint. This is a strong motivation, not a derivation: it reduces $\Delta = 1$ from a fitted coupling to a GSL-equilibrium extremum under the stated premises, leaving the microscopic origin open. ■

Appendix A.7. The remaining repository items (catalogue)

Result 2 (reheating master equation): the same logistic $df_{\text{sat}}/dt = H f_{\text{sat}}(1 - f_{\text{sat}}) - \Gamma\, f_{\text{sat}}$ governs the *early* f_{sat} (inflaton→radiation), giving the U-shape of Result 12. **Result 4 / 4B** [superseded by 4C]: the $M^{5/3}$ energy weight, and the $p = 1$ virial-temperature mis-step. **Result 5D**: the $(1-\epsilon/2)$ "$\lambda = 1$ variant" background ($w_0 = -0.85$, no crossing) vs the canonical

$\lambda = 1/2$ crossing background. **Result 6 / 6B**: the smooth-DE growth validation and the $c_s^2 = 1$ closure ($f_{\rm sat}$ a background functional). **Result 10**: the QG chain and the volume-law postulate. **Result 11**: the seam = BBN bound = modgrav/holographic fork. **Result 12**: the $f_{\rm sat}$ U-shape (inflation and dark energy as the two de Sitter brackets). **Result 13**: $1+w = (1/3)[2\lambda\epsilon - d \ln f_{\rm sat}/d \ln a]$, the EOS as horizon-entropy production. Full statements are catalogued here; each is exercised by a numbered check in Appendix Appendix B.

Appendix B. The verification suite

The accompanying code ships a suite of 63 numerical checks (V1–V63) covering: the fixed-point solution (closed-loop H(z) to 10^{-13} over $z \in [0, 10^6]$), BBN-safety ($\Delta N_{\rm eff} = \Delta Y_{\rm p} = 0$), the CAMB $\equiv$ CLASS $r_{\rm drag}$ agreement (5×10^{-5}), the CLASS perturbation validation (0.1%), the GSL along cosmic history, the fixed (derived, Prescription 4C) γ and the smooth-DE $c_s^2 = 1$ closure, the horizon-dimension scaling $\lambda(d_{\rm H}) = 2 - d_{\rm H}/2$ (V59), the state/counting diagnostics of the supplementary material (§S2; SYK scrambling V60; causal-set dof counting in Minkowski + de Sitter V61; the entropy-bound audit V62; the tensor-network realisability test V63), and the cross-horizon and quantum-sector consistency checks. The verification suite and the figures regenerate from the repository, and the CAMB-in-the-loop inference drivers under `src/experiments/` regenerate the headline model-comparison numbers (ΔDIC, the per-probe decomposition, the false-preference calibration, the out-of-sample split, and the *tracer*/γ robustness scans). Only the full-primary-CMB (plik) joint and the FPAB/Mochi-CLASS growth sector are run by the companion observational-tests analysis [37] (they need the Planck likelihood data and the modified Boltzmann code respectively; see the Data-availability statement).

Appendix C. Data vectors, covariances, and the $f\sigma 8$ audit

The full data vectors and covariance matrices, the positive-definiteness checks, the benign Pantheon+ duplicate-CID treatment, and the $f\sigma 8$ data audit (the small set of mis-entered growth points corrected to their published Gold-2018 [33] values, and the SH0ES value correction) — all reproduced from the public data vectors and documented in the data audit. One approximation is disclosed: the Moresco cosmic-chronometer covariance is applied as a diagonal + 3% correlated-systematic floor (the full published covariance is cached separately); the CC channel contributes only +0.10 to $\Delta\chi^2$ and removing it entirely shifts the headline by ≤ 0.1, so the approximation is immaterial to the model comparison.

Appendix D. False-preference calibration methodology

The parametric-bootstrap procedure of §5.2: generation of ΛCDM-truth mocks with all probes (including the compressed CMB R/l_{A} and SH0ES) re-drawn self-consistently; the refit of both models to each mock; the null distribution of $\Delta\chi^2(\mathrm{SEDE} - \Lambda\mathrm{CDM})$ (2000-mock run: mean +0.61, s.d. 1.59); the real $\Delta\chi^2 = -4.68$ beyond all draws, 0/2000, p < 0.0015 ($\sim 3\sigma$); the documented bug (CMB held fixed $\rightarrow$ biased null -2.87, $p \approx 0.5$) and its fix; and the independent-pipeline cross-check ($\sim 2\sigma$, CMB-distance-engine-dependent; run against the pre-revision real value -2.96 and not yet repeated at -4.68). The pre-registration sha256 and contents are listed. The pre-registration additionally records the triggered CMB follow-ups and their status (§Appendix E: full non-lite plik run, $\Delta\chi^2 = -0.33$; ACT DR6 primary formally deferred) and the locked E1 structure-sourcing decision rule (§Appendix F: data vector + CONFIRM/REFUTE thresholds). The no-SH0ES robustness (§5.4 ii′; $\Delta\chi^2 = -2.89$, $\ln B \approx +1.4$, $\sim 2.3\sigma$) is reproduced by the no-SH0ES robustness check.

Appendix E. Status of the QG derivation chain

The full chain from §8, with the status of each link: - Jacobson–Cai–Kim (motivational ansatz): Clausius horizon thermodynamics motivates the horizon-fluid construction and supplies the conservation/dynamical logic ($\delta Q = T$ dS $\Longrightarrow$ Friedmann); the *absolute* identification $\rho_{\mathrm{DE}} = T \cdot s$ is the SEDE ansatz (A.1), not a theorem. - Non-extensivity $\Longrightarrow$ Tsallis–Cirto/Barrow power law (motivated): the weakest step. - Second-law / GSL $\Longrightarrow$ volume-law endpoint $\Delta = 1$ (motivated, given the volume-law thermalisation premise): the geometric ceiling caps $\Delta \leq 1$ and the GSL is *consistent with* (a fixed point at) the saturated endpoint — a consistency check, not a selection principle; a microscopic derivation remains open. - CKN scale (a rigorous bound, normalised by flatness): the holographic energy bound gives the horizon-scale upper bound; the amplitude is normalised by flatness (§8.2). - The one irreducible postulate (open): a microscopic state count delivering the volume law.

The perturbative-QG (log-correction) vs thermalised (volume-law) regimes are reconciled as different *states*, not a contradiction. This summary matches §8.4 and Appendix Appendix F.

Appendix F. The energy and entropy budget

This appendix records the order-of-magnitude estimate that the energy-conservation framing of §1–§2 turns on, and the resulting summary of what is derived, normalised, and open.

Energy budget. Dark energy today is $\rho_{\rm DE}\ c^2 \approx 0.7\ \rho_{\rm crit}\ c^2 \approx 6 \times 10^{-10}$ J/m^3. The gravitational binding energy released by structure is bounded by $|E_{\rm bind}|/Mc^2 \sim \sigma^2/c^2$ per virialised system, with $\sigma \sim 10^3$ km/s for clusters ($\sigma^2/c^2 \sim 10^{-5}$) and ~200 km/s for galaxies (~10^{-7}); even taking all of $\Omega_{\rm m}$ collapsed, $\rho_{\rm bind}\ c^2 \lesssim 10^{-5} \cdot 0.3 \cdot 8.5 \times 10^{-10} \sim 10^{-15}\ J/m^3$. Hence $\rho_{\rm bind}/\rho_{\rm DE}$ $\sim 10^{-6}$: binding energy is six orders of magnitude too small to *be* the dark energy. A literal matter→dark-energy transfer is excluded independently — moving $O(\rho_{\rm crit})$ out of matter over a Hubble time would spoil $\rho_{\rm m} \propto a^{-3}$ and with it BBN and the CMB.

Entropy budget. Deposited at the cold horizon temperature, the binding entropy is $\Delta S_{\rm AH}= E_{\rm bind}/T_{\rm AH}$; with $T_{\rm AH} = \hbar H/2\pi$ this comes to $\Delta S_{\rm AH} \sim 10^{-7}\ S_{\rm AH}$, i.e. ~$10^{-7}$ of the horizon entropy. So binding-entropy deposition does not, by itself, fill $f_{\rm sat}$ to O(1); it sets the *relative* z-dependence (the weight p $= 5/3 \to \gamma$), while the normalisation $f_{\rm sat}(0) = 1$ is set by the horizon capacity. This is the same family as the $(M_{\rm P}/H)^{\Delta} \sim 10^{61}$ "seam" (Result 11), resolved by the holographic-DE scope (Barrow acts only on the $f_{\rm sat}$-gated dark-energy fluid).

Summary.

quantity	status	set by
magnitude $\rho_{\rm DE} \sim \rho_{\rm crit}$	bounded + normalised	CKN holographic upper bound (§8.2), amplitude fixed by flatness (O(1) saturation of the bound assumed)

quantity	status	set by
H-scaling $\lambda = 1/2$	postulate	volume-law horizon entropy $S_{\rm grav} \propto V_{\rm AH}$ (§8.1)
shape $\gamma \approx 1.5$	derived (Prescription 4C)	halo-binding-energy weighting $E_{\rm bind} \propto M^{5/3}$ $\to$ p $= 5/3$ (§3.1)
sound speed $c_{\rm s}^2 = 1$	closure	minimal smooth-DE treatment, $f_{\rm sat}$ a background functional (§3.4)
normalisation $f_{\rm sat}(0) = 1$	imposed	present-day horizon-entropy normalisation
volume-law horizon ($\Delta = 1$ microscopically)	open	the one irreducible postulate (§8.4)

None of these is a parameter *fitted to cosmological data*, but neither are they all "derived from first principles" — the magnitude is a holographic bound normalised by flatness (as ΛCDM normalises $\Omega_\Lambda 0$), the H-scaling is the central volume-law postulate, γ is a fixed halo prescription, and $c_{\rm s}^2 = 1$ is a modelling closure. Binding energy enters SEDE entropically, fixing the *shape* of the gate; the dark energy's *magnitude* is the horizon's own thermal scale, not energy borrowed from structure. The single irreducible foundational input is the microscopic volume-law postulate.

References

[1] A. G. Adame, et al., DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations, JCAP 02 (2025) 021. arXiv:2404.03002, doi:10.1088/1475-7516/2025/02/021.

[2] K. Lodha, et al., DESI 2024: Constraints on physics-focused aspects of dark energy using DESI DR1 BAO data, Phys. Rev. D 111 (2) (2025) 023532. arXiv:2405.13588, doi:10.1103/PhysRevD.111.023532.

[3] K. Lodha, et al., Extended dark energy analysis using DESI DR2 BAO measurements, Phys. Rev. D 112 (8) (2025) 083511. arXiv:2503.14743, doi:10.1103/w4c6-1r5j.

[4] T.-N. Li, Y.-H. Li, G.-H. Du, P.-J. Wu, L. Feng, J.-F. Zhang, X. Zhang, Revisiting holographic dark energy after DESI 2024, Eur. Phys. J. C 85 (6) (2025) 608. arXiv:2411.08639, doi:10.1140/epjc/s10052-025-14279-7.

[5] S. Bhattacharjee, S. Halder, J. de Haro, S. Pan, E. N. Saridakis, Is dark energy an effective manifestation of non-equilibrium thermodynamics? extendash insights from DESI, Class. Quant. Grav. 43 (11) (2026) 115005. arXiv:2507.15575, doi:10.1088/1361-6382/ae6f65.

[6] T. Jacobson, Thermodynamics of space-time: The Einstein equation of state, Phys. Rev. Lett. 75 (1995) 1260–1263. arXiv:gr-qc/9504004, doi:10.1103/PhysRevLett.75.1260.

[7] R.-G. Cai, S. P. Kim, First law of thermodynamics and Friedmann equations of Friedmann-Robertson-Walker universe, JHEP 02 (2005) 050. arXiv:hep-th/0501055, doi:10.1088/1126-6708/2005/02/050.

[8] S. Rasanen, Accelerated expansion from structure formation, JCAP 11 (2006) 003. arXiv:astro-ph/0607626, doi:10.1088/1475-7516/2006/11/003.

[9] D. L. Wiltshire, Gravitational energy as dark energy: Cosmic structure and apparent acceleration, in: International Conference on Two Cosmological Models, 2011. arXiv:1102.2045.

[10] E. W. Kolb, S. Matarrese, A. Riotto, On cosmic acceleration without dark energy, New J. Phys. 8 (2006) 322. arXiv:astro-ph/0506534, doi:10.1088/1367-2630/8/12/322.

[11] I. Zlatev, P. J. Steinhardt, A Tracker solution to the cold dark matter cosmic coincidence problem, Phys. Lett. B 459 (1999) 570–574. arXiv:astro-ph/9906481, doi:10.1016/S0370-2693(99)00707-8.

[12] M. P. Gough, Information equation of state, Entropy 10 (2008) 150.

[13] M. P. Gough, Holographic Dark Information Energy, Entropy 13 (2011) 924–935. arXiv:1105.4461, doi:10.3390/e13040924.

[14] M. P. Gough, Evidence for dark energy driven by star formation: Information dark energy?, Entropy 27 (2) (2025) 110. doi:10.3390/e27020110.

[15] A. K. Çamlıbel, Observational Constraints on the Structure-Induced Dark Energy Model (3 2026). arXiv:2603.03999.

[16] E. N. Saridakis, Barrow holographic dark energy, Phys. Rev. D 102 (12) (2020) 123525. arXiv:2005.04115, doi:10.1103/PhysRevD.102.123525.

[17] S. Basilakos, D. Polarski, J. Sola, Generalizing the running vacuum energy model and comparing with the entropic-force models, Phys. Rev. D 86 (2012) 043010. arXiv:1204.4806, doi:10.1103/PhysRevD.86.043010.

[18] N. Komatsu, S. Kimura, General form of entropy on the horizon of the universe in entropic cosmology, Phys. Rev. D 93 (4) (2016) 043530. arXiv:1511.04364, doi:10.1103/PhysRevD.93.043530.

[19] N. Komatsu, S. Kimura, Non-adiabatic-like accelerated expansion of the late universe in entropic cosmology, Phys. Rev. D 87 (4) (2013) 043531. arXiv:1208.2482, doi:10.1103/PhysRevD.87.043531.

[20] D. A. Easson, P. H. Frampton, G. F. Smoot, Entropic Accelerating Universe, Phys. Lett. B 696 (2011) 273–277. arXiv:1002.4278, doi:10.1016/j.physletb.2010.12.025.

[21] F. R. Urban, A. R. Zhitnitsky, The cosmological constant from the QCD Veneziano ghost, Phys. Lett. B 688 (2010) 9–12. arXiv:0906.2162, doi:10.1016/j.physletb.2010.03.080.

[22] F. R. Urban, A. R. Zhitnitsky, The QCD nature of Dark Energy, Nucl. Phys. B 835 (2010) 135–173. arXiv:0909.2684, doi:10.1016/j.nuclphysb.2010.04.001.

[23] N. Ohta, Dark Energy and QCD Ghost, Phys. Lett. B 695 (2011) 41–44. arXiv:1010.1339, doi:10.1016/j.physletb.2010.11.044.

[24] M. Li, A Model of holographic dark energy, Phys. Lett. B 603 (2004) 1–5. arXiv:hep-th/0403127, doi:10.1016/j.physletb.2004.10.014.

[25] J. Kennedy, L. Lombriser, A. Taylor, Reconstructing Horndeski models from the effective field theory of dark energy, Phys. Rev. D 96 (8) (2017) 084051. arXiv:1705.09290, doi:10.1103/PhysRevD.96.084051.

[26] G. Gubitosi, F. Piazza, F. Vernizzi, The Effective Field Theory of Dark Energy, JCAP 02 (2013) 032. arXiv:1210.0201, doi:10.1088/1475-7516/2013/02/032.

[27] J. Gleyzes, D. Langlois, F. Piazza, F. Vernizzi, Healthy theories beyond Horndeski, Phys. Rev. Lett. 114 (21) (2015) 211101. arXiv:1404.6495, doi:10.1103/PhysRevLett.114.211101.

[28] E. Bellini, I. Sawicki, Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity, JCAP 07 (2014) 050. arXiv:1404.3713, doi:10.1088/1475-7516/2014/07/050.

[29] M. Cataneo, E. Bellini, mochi_class: Modelling Optimisation to Compute Horndeski In class (7 2024). arXiv:2407.11968, doi:10.33232/001c.123470.

[30] M. Zumalacárregui, E. Bellini, I. Sawicki, J. Lesgourgues, P. G. Ferreira, hi_class: Horndeski in the Cosmic Linear Anisotropy Solving System, JCAP 08 (2017) 019. arXiv:1605.06102, doi:10.1088/1475-7516/2017/08/019.

[31] C. Deffayet, X. Gao, D. A. Steer, G. Zahariade, From k-essence to generalised Galileons, Phys. Rev. D 84 (2011) 064039. arXiv:1103.3260, doi:10.1103/PhysRevD.84.064039.

[32] T. Kobayashi, M. Yamaguchi, J. Yokoyama, Generalized G-inflation: Inflation with the most general second-order field equations, Prog. Theor. Phys. 126 (2011) 511–529. arXiv:1105.5723, doi:10.1143/PTP.126.511.

[33] B. Sagredo, S. Nesseris, D. Sapone, Internal Robustness of Growth Rate data, Phys. Rev. D 98 (2018) 083543. arXiv:1806.10822, doi:10.1103/PhysRevD.98.083543.

[34] N. Aghanim, et al., Planck 2018 results. VIII. Gravitational lensing, Astron. Astrophys. 641 (2020) A8. arXiv:1807.06210, doi:10.1051/0004-6361/201833886.

[35] P. Motloch, W. Hu, Tensions between direct measurements of the lens power spectrum from Planck data, Phys. Rev. D 97 (10) (2018) 103536. arXiv:1803.11526, doi:10.1103/PhysRevD.97.103536.

[36] G. E. Addison, C. L. Bennett, M. Halpern, G. Hinshaw, J. L. Weiland, Revisiting the A $_L$ Lensing Anomaly in Planck 2018 Temperature Data, Astrophys. J. 974 (2) (2024) 187. arXiv:2310.03127, doi:10.3847/1538-4357/ad6d61.

[37] S. Pandev, Structural Entropy Dark Energy against DESI DR2, Pantheon+ and Planck 2018: a zero-extra-parameter test, companion paper (observational tests); preprint doi:10.5281/zenodo.21652163 (2026).

[38] P.-J. Wu, T.-N. Li, G.-H. Du, X. Zhang, Observational challenges to holographic and Ricci dark energy paradigms: Insights from ACT DR6 and DESI DR2*, Chin. Phys. C 50 (4) (2026) 045105. arXiv:2509.02945, doi:10.1088/1674-1137/ae3be2.

[39] E. V. Linder, R. N. Cahn, Parameterized Beyond-Einstein Growth, Astropart. Phys. 28 (2007) 481–488. arXiv:astro-ph/0701317, doi:10.1016/j.astropartphys.2007.09.003.

[40] L. Pogosian, A. Silvestri, What can cosmology tell us about gravity? Constraining Horndeski gravity with Σ and μ, Phys. Rev. D 94 (10) (2016) 104014. arXiv:1606.05339, doi:10.1103/PhysRevD.94.104014.

[41] A. Silvestri, L. Pogosian, R. V. Buniy, Practical approach to cosmological perturbations in modified gravity, Phys. Rev. D 87 (10) (2013) 104015. arXiv:1302.1193, doi:10.1103/PhysRevD.87.104015.

[42] N.-M. Nguyen, D. Huterer, Y. Wen, Evidence for Suppression of Structure Growth in the Concordance Cosmological Model, Phys. Rev. Lett. 131 (11) (2023) 111001. arXiv:2302.01331, doi:10.1103/PhysRevLett.131.111001.

[43] H. Chaudhary, S. Capozziello, V. K. Sharma, I. Gómez-Vargas, G. Mustafa, Evidence for evolving dark energy from DESI DR2 BAO and Pantheon$^+$, DES-Dovekie, and Union3, Eur. Phys. J. C 86 (5) (2026) 564. arXiv:2508.10514, doi:10.1140/epjc/s10052-026-15806-w.

[44] D. Wang, D. Mota, Did DESI DR2 truly reveal dynamical dark energy?, Eur. Phys. J. C 85 (11) (2025) 1356. arXiv:2504.15222, doi:10.1140/epjc/s10052-025-15076-y.

[45] G. Liu, Y. Wang, W. Zhao, Impact of LRG1 and LRG2 in DESI 2024 BAO data on dark energy evolution (7 2024). arXiv:2407.04385.

[46] S. Pandev, Is the volume-law horizon a postulate or a consequence? Birth-selection and a driven non-equilibrium reformulation of Structural Entropy Dark Energy, companion paper (foundations); preprint doi:10.5281/zenodo.21651612 (2026). doi:10.5281/zenodo.21651612.

[47] M. Abdul Karim, et al., DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints, Phys. Rev. D 112 (8) (2025) 083515. arXiv:2503.14738, doi:10.1103/tr6y-kpc6.

[48] S. G. Turyshev, Dark energy after DESI DR2: Observational status, reconstructions, and physical models, Phys. Rev. D 113 (10) (2026) 103540. arXiv:2602.05368, doi:10.1103/dqxw-yp1j.

[49] X. Zhang, Y.-H. Xu, Y. Sang, Reconstruction of dark energy using DESI DR2, Commun. Theor. Phys. 78 (3) (2026) 035404. arXiv:2511.02220, doi:10.1088/1572-9494/ae1a5b.

[50] C. You, D. Wang, T. Yang, Dynamical dark energy implies a coupled dark sector: Insights from DESI DR2 via a data-driven approach, Phys. Rev. D 112 (4) (2025) 043503. arXiv:2504.00985, doi:10.1103/f6v7-n9fr.

[51] Y. Cai, X. Ren, T. Qiu, M. Li, X. Zhang, The quintom theory of dark energy after DESI DR2, Natl. Sci. Rev. 13 (10) (2026) nwag115. arXiv:2505.24732, doi:10.1093/nsr/nwag115.

[52] P. Thanankullaphong, P. Sahoo, P. Hassan Puttasiddappa, N. Roy, Quintom dark energy: Future attractor and phantom crossing in light

of DESI DR2 observations, Phys. Rev. D 113 (8) (2026) 084069. arXiv:2601.02284, doi:10.1103/4b1x-6pcx.

[53] I. Pantos, L. Perivolaropoulos, Status of the S_8 Tension: A 2026 Review of Probe Discrepancies, Phys. Dark Univ. 52 (2026) 102286. arXiv:2602.12238, doi:10.1016/j.dark.2026.102286.

[54] E. Majerotto, et al., Probing deviations from General Relativity with the Euclid spectroscopic survey, Mon. Not. Roy. Astron. Soc. 424 (2012) 1392–1408. arXiv:1205.6215, doi:10.1111/j.1365-2966.2012.21323.x.

[55] N. Afshordi, Integrated Sachs-Wolfe effect in cross - correlation: The Observer's manual, Phys. Rev. D 70 (2004) 083536. arXiv:astro-ph/0401166, doi:10.1103/PhysRevD.70.083536.

[56] B. M. Schäfer, The integrated Sachs-Wolfe effect in cosmologies with coupled dark matter and dark energy, Mon. Not. Roy. Astron. Soc. (2008). arXiv:0803.2239.

[57] G. Olivares, F. Atrio-Barandela, D. Pavon, The Integrated Sachs-Wolfe Effect in Interacting Dark Energy Models, Phys. Rev. D 77 (2008) 103520. arXiv:0801.4517, doi:10.1103/PhysRevD.77.103520.

[58] W. Hu, R. Scranton, Measuring dark energy clustering with CMB-galaxy correlations, Phys. Rev. D 70 (2004) 123002. arXiv:astro-ph/0408456, doi:10.1103/PhysRevD.70.123002.

[59] S. J. Rauhut, et al., Testing gravitational physics by combining DESI DR1 and weak lensing datasets using the E_G estimator, Open J. Astrophys. 8 (2025) 145197. arXiv:2507.16098, doi:10.33232/001c.145197.

[60] S. Pandev, Barrow $\Delta \leq 1$ from a long-range entanglement network: a capacity theorem and an area–volume dichotomy, companion paper (count); preprint doi:10.5281/zenodo.21652161 (2026).

[61] E. Belgacem, et al., Testing modified gravity at cosmological distances with LISA standard sirens, JCAP 07 (2019) 024. arXiv:1906.01593, doi:10.1088/1475-7516/2019/07/024.

[62] E. Belgacem, Y. Dirian, A. Finke, S. Foffa, M. Maggiore, Nonlocal gravity and gravitational-wave observations, JCAP (2019). arXiv:1907.02047.

[63] S. Mukherjee, B. D. Wandelt, J. Silk, Testing the general theory of relativity using gravitational wave propagation from dark standard sirens, Mon. Not. Roy. Astron. Soc. 502 (1) (2021) 1136–1144. doi:10.1093/mnras/stab001.

[64] E. P. Verlinde, Emergent Gravity and the Dark Universe, SciPost Phys. 2 (3) (2017) 016. arXiv:1611.02269, doi:10.21468/SciPostPhys.2.3.016.

[65] P. Diego-Palazuelos, E. Komatsu, Cosmic birefringence from the Atacama Cosmology Telescope Data Release 6, Phys. Rev. D 113 (10) (2026) L101302. arXiv:2509.13654, doi:10.1103/pbc3-t52s.

[66] C. Deffayet, O. Pujolas, I. Sawicki, A. Vikman, Imperfect Dark Energy from Kinetic Gravity Braiding, JCAP 10 (2010) 026. arXiv:1008.0048, doi:10.1088/1475-7516/2010/10/026.

[67] R. Kimura, K. Yamamoto, Large Scale Structures in Kinetic Gravity Braiding Model That Can Be Unbraided, JCAP 04 (2011) 025. arXiv:1011.2006, doi:10.1088/1475-7516/2011/04/025.

[68] M. Crossley, P. Glorioso, H. Liu, Effective field theory of dissipative fluids, JHEP 09 (2017) 095. arXiv:1511.03646, doi:10.1007/JHEP09(2017)095.

[69] S. Agüí Salcedo, T. Colas, C. Dufner, E. Pajer, Phenomenology of an Open Effective Field Theory of Dark Energy (2026). arXiv:2603.12321.

[70] E. Ebrahimi, A. Sheykhi, Instability of QCD ghost dark energy model, Int. J. Mod. Phys. D 20 (2011) 2369–2381. arXiv:1106.3504, doi:10.1142/S0218271811020354.

[71] S. Tsujikawa, Crossing the phantom divide in scalar-tensor and vector-tensor theories (2025). arXiv:2508.17231.

[72] G. G. Luciano, A. Paliathanasis, E. N. Saridakis, Constraints on Barrow and Tsallis holographic dark energy from DESI DR2 BAO data, JHEAp

49 (2026) 100427. arXiv:2506.03019, doi:10.1016/j.jheap.2025.100427.

[73] S. Nojiri, S. D. Odintsov, T. Paul, Early and late universe holographic cosmology from a new generalized entropy, Phys. Lett. B 831 (2022) 137189. arXiv:2205.08876, doi:10.1016/j.physletb.2022.137189.

[74] F. K. Anagnostopoulos, S. Basilakos, E. N. Saridakis, Observational constraints on Barrow holographic dark energy, Eur. Phys. J. C 80 (9) (2020) 826. arXiv:2005.10302, doi:10.1140/epjc/s10052-020-8360-5.

[75] E. N. Saridakis, Modified cosmology through spacetime thermodynamics and Barrow horizon entropy, JCAP 07 (2020) 031. arXiv:2006.01105, doi:10.1088/1475-7516/2020/07/031.

[76] H. Gohar, V. Salzano, On the foundations of entropic cosmologies: Inconsistencies, possible solutions and dead end signs, Phys. Lett. B 855 (2024) 138781. arXiv:2307.01768, doi:10.1016/j.physletb.2024.138781.

[77] T. Buchert, Dark Energy from structure: a status report, Gen. Rel. Grav. 40 (2008) 467–527. arXiv:0707.2153, doi:10.1007/s10714-007-0554-8.

[78] S. Räsänen, Dark energy from backreaction, JCAP 02 (2004) 003. arXiv:astro-ph/0311257, doi:10.1088/1475-7516/2004/02/003.

[79] D. L. Wiltshire, Exact Solution to the Averaging Problem in Cosmology, Phys. Rev. Lett. 99 (2007) 251101. arXiv:0709.0732, doi:10.1103/PhysRevLett.99.251101.

[80] B. Pandey, Entropic backreaction from cosmic structure formation: a thermodynamic approach to the late-time cosmological tensions, arXiv e-printsDistinct author from S. Pandev (SEDE Paper I); matter configuration-entropy backreaction in GR (2026). arXiv:2605.28797.

[81] S. S. Shameeem, K. V. Priyesh, P. B. Krishna, T. K. Mathew, Generalized Mass-to-Horizon Entropy and Horizon Thermodynamics, arXiv e-prints (2026). arXiv:2605.12003.

[82] S. Mondal, A. Choudhuri, Cosmological dynamics and structure formation in a generalized mass-to-horizon entropy-inspired modified gravity, arXiv e-prints (2026). `arXiv:2604.15399`.

[83] T. Buchert, J. Larena, J.-M. Alimi, Correspondence between kinematical backreaction and scalar field cosmologies: the 'morphon field', Class. Quant. Grav. 23 (2006) 6379–6408. `arXiv:gr-qc/0606020`, `doi:10.1088/0264-9381/23/22/012`.

[84] W. D. Kenworthy, D. Scolnic, A. Riess, The Local Perspective on the Hubble Tension: Local Structure Does Not Impact Measurement of the Hubble Constant, Astrophys. J. 875 (2) (2019) 145. `arXiv:1901.08681`, `doi:10.3847/1538-4357/ab0ebf`.

[85] J. García-Bellido, L. Espinosa-Portalés, Cosmic acceleration from first principles, Phys. Dark Univ. 34 (2021) 100892. `arXiv:2106.16014`, `doi:10.1016/j.dark.2021.100892`.

[86] L. Espinosa-Portalés, J. García-Bellido, Covariant formulation of non-equilibrium thermodynamics in General Relativity, Phys. Dark Univ. 34 (2021) 100893. `arXiv:2106.16012`, `doi:10.1016/j.dark.2021.100893`.

[87] M. Asghari, A. Sheykhi, Observational constraints of the modified cosmology through Barrow entropy, Eur. Phys. J. C 82 (5) (2022) 388. `arXiv:2110.00059`, `doi:10.1140/epjc/s10052-022-10262-8`.

[88] J. D. Barrow, S. Basilakos, E. N. Saridakis, Big Bang Nucleosynthesis constraints on Barrow entropy, Phys. Lett. B 815 (2021) 136134. `arXiv:2010.00986`, `doi:10.1016/j.physletb.2021.136134`.

[89] G. Leon, J. Magaña, A. Hernández-Almada, M. A. García-Aspeitia, T. Verdugo, V. Motta, Barrow Entropy Cosmology: an observational approach with a hint of stability analysis, JCAP 12 (12) (2021) 032. `arXiv:2108.10998`, `doi:10.1088/1475-7516/2021/12/032`.

[90] S. Khlebnikov, A. Sheoran, The first heat: production of entanglement entropy in the early universe, JHEP 11 (2019) 157. `arXiv:1907.00487`, `doi:10.1007/JHEP11(2019)157`.

[91] A. Belfiglio, O. Luongo, S. Mancini, Entanglement area law violation from field-curvature coupling, Phys. Lett. B 848 (2024) 138398. arXiv:2306.08357, doi:10.1016/j.physletb.2023.138398.

[92] A. Campa, T. Dauxois, S. Ruffo, Statistical mechanics and dynamics of solvable models with long-range interactions, Phys. Rept. 480 (2009) 57–159. arXiv:0907.0323, doi:10.1016/j.physrep.2009.07.001.

[93] W. Lin, L. Visinelli, T. T. Yanagida, Testing quintessence axion dark energy with recent cosmological results, JCAP 10 (2025) 023. arXiv:2504.17638, doi:10.1088/1475-7516/2025/10/023.

[94] B. Barman, S. Girmohanta, Implications of DESI data for dark matter and cosmic birefringence, Phys. Rev. D 113 (1) (2026) 015023. arXiv:2506.12589, doi:10.1103/169s-4g17.

[95] L. Yin, S. Xiong, J. Kochappan, B.-H. Lee, T. Ghosh, Constraints on Cosmic Birefringence from SPIDER, Planck, and ACT observations (10 2025). arXiv:2510.25489.

[96] S. Arcari, N. Bartolo, G. Fabbian, A. Greco, A. Gruppuso, M. Lattanzi, P. Natoli, L. Pagano, G. Zagatti, Stairway to axions: the cross-correlation of birefringence and galaxies from NPIPE and Quaia, JCAP 06 (2026) 015. arXiv:2509.22273, doi:10.1088/1475-7516/2026/06/015.

[97] L. Huang, R.-G. Cai, S.-J. Wang, The DESI DR1/DR2 evidence for dynamical dark energy is biased by low-redshift supernovae, Sci. China Phys. Mech. Astron. 68 (10) (2025) 100413. arXiv:2502.04212, doi:10.1007/s11433-025-2754-5.